

Antibiogram Prediction Through Genome Analysis

SHAP Feature Importance – Antibiotic Resistance Prediction

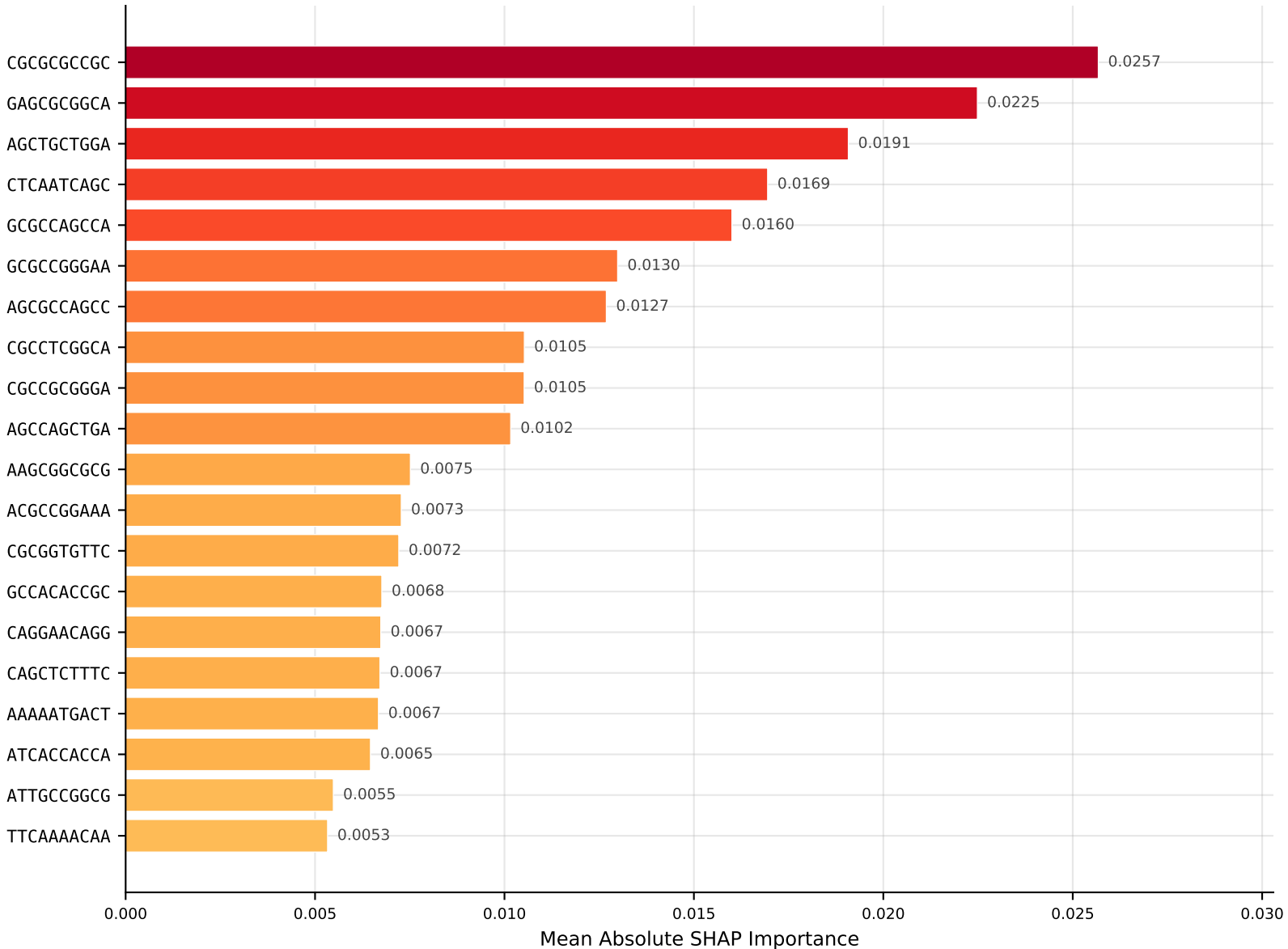
Model: XGBoost (xgboost_f0849bdd)
Training set: 3,204 bacterial genomes
K-mer size: 10-bp
Features: 155,211 unique k-mers
Classes: Susceptible (S) / Intermediate (I) / Resistant (R)
Antibiotics: 78 with SHAP data

Each page shows the Top-20 k-mer features for one antibiotic, ranked by mean absolute SHAP importance score.

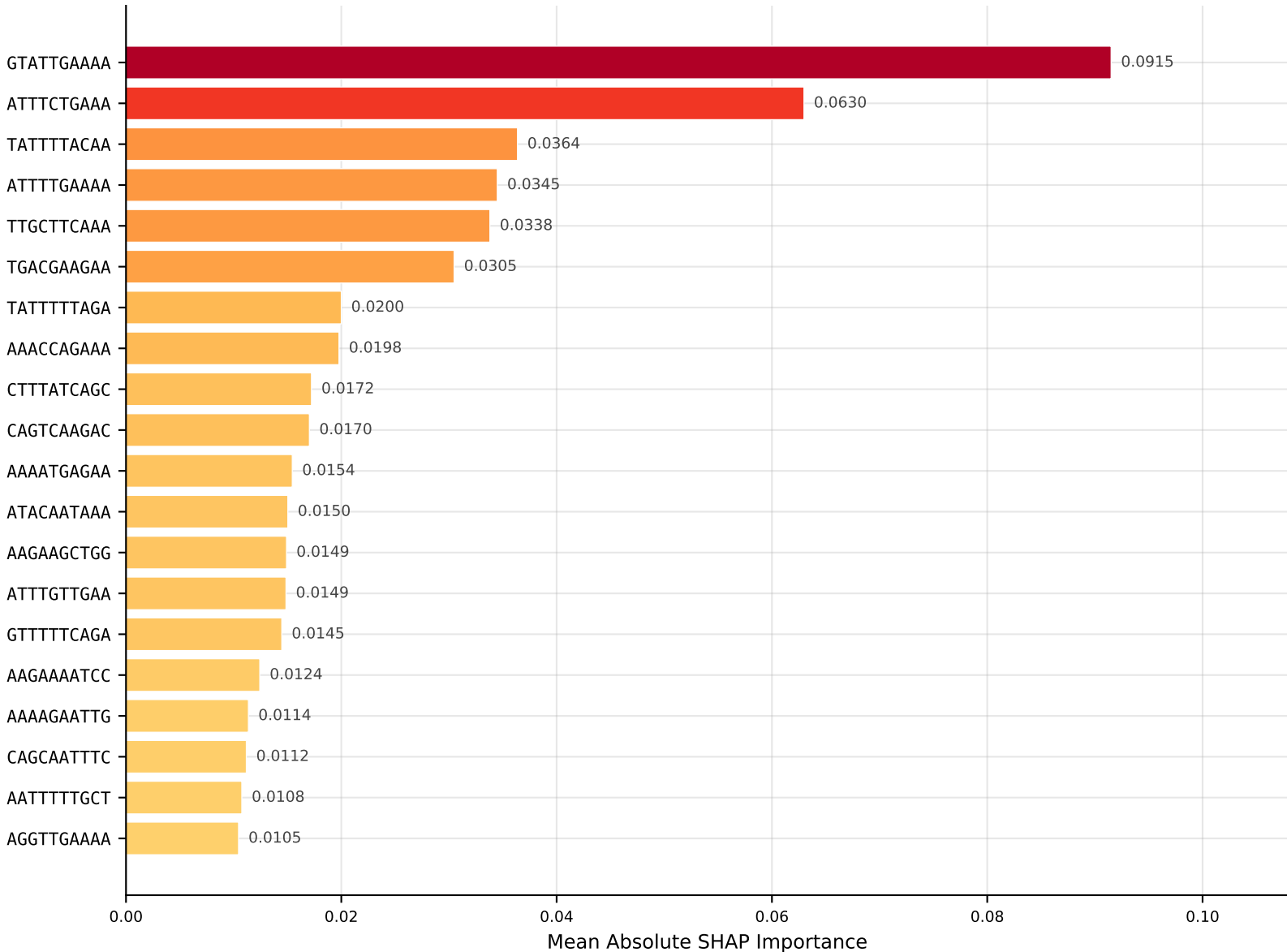
Higher importance → k-mer is more predictive of resistance.

SHAP Feature Importance — Amikacin

Top 20 k-mer predictors

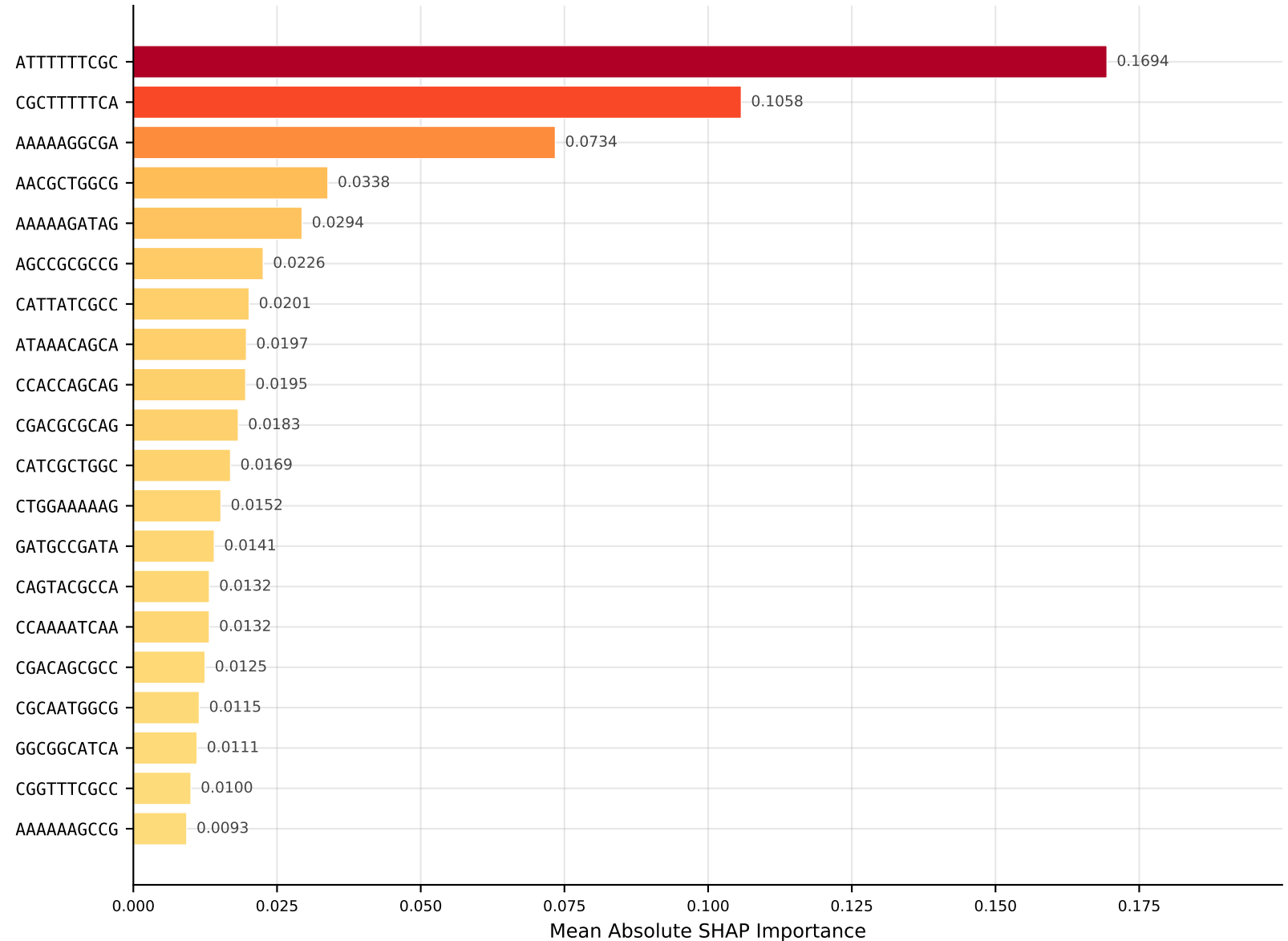


SHAP Feature Importance — Amoxicillin Top 20 k-mer predictors

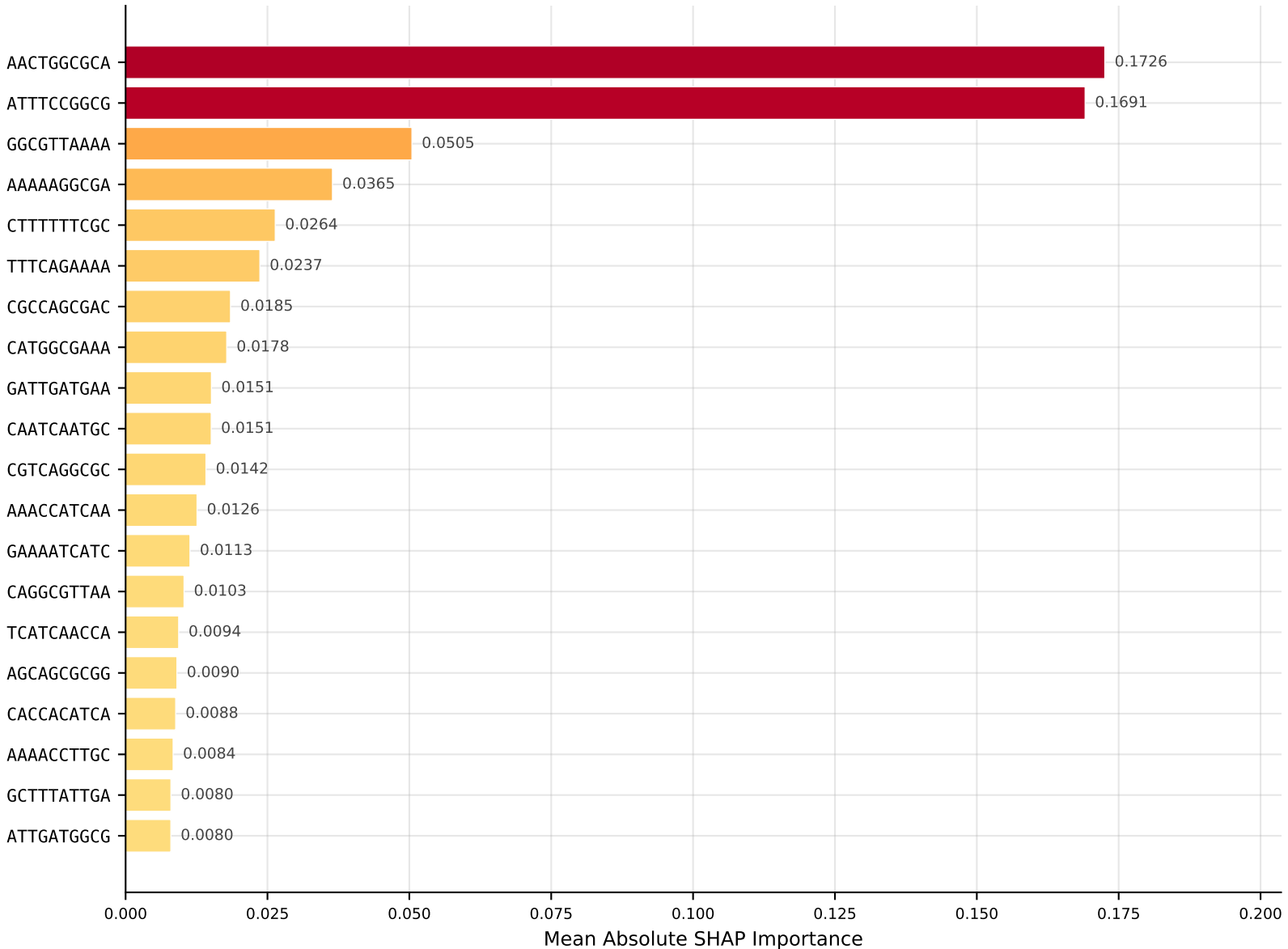


SHAP Feature Importance — Amoxicillin/Clavulanic Acid

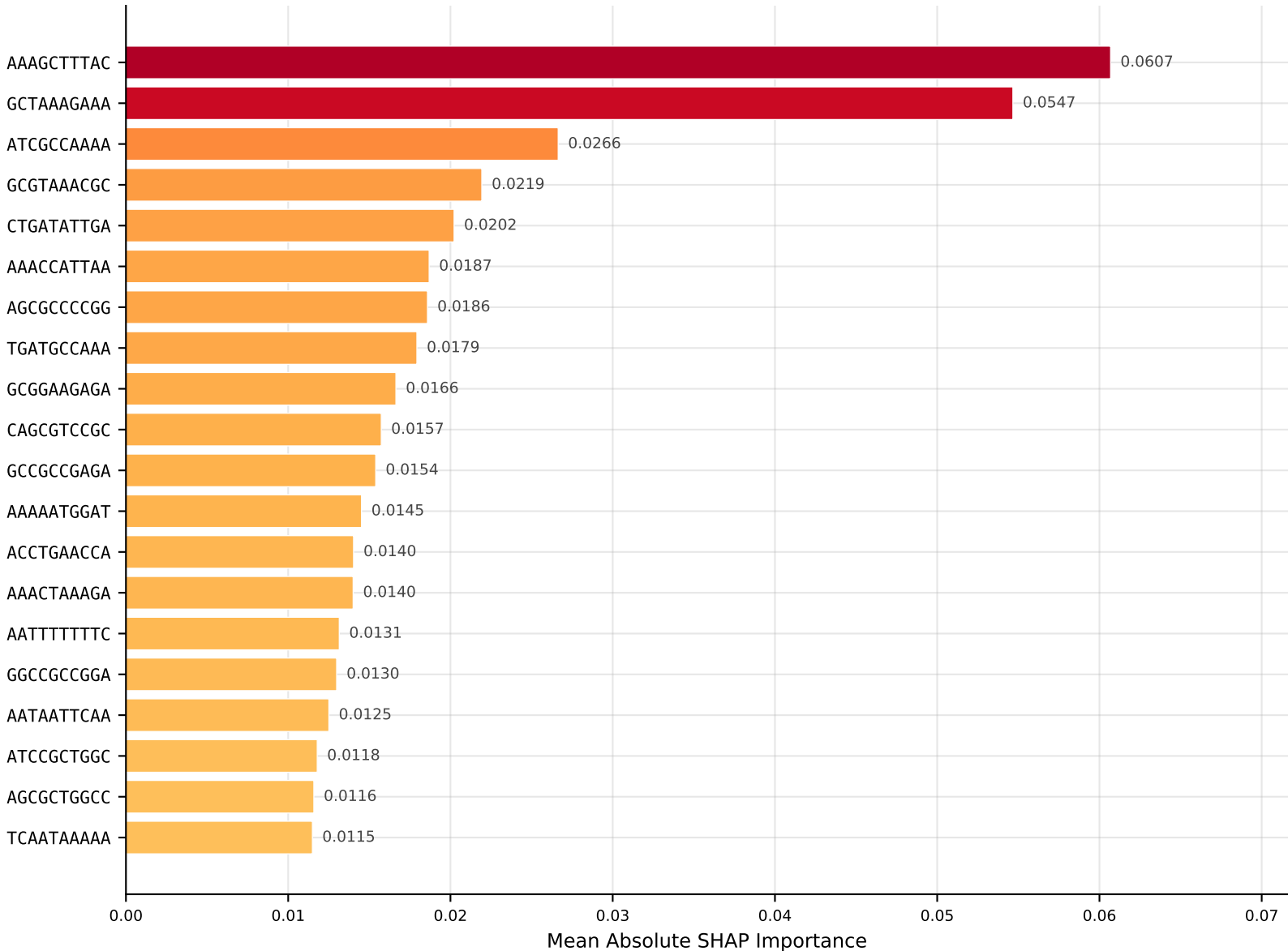
Top 20 k-mer predictors



SHAP Feature Importance — Ampicillin Top 20 k-mer predictors

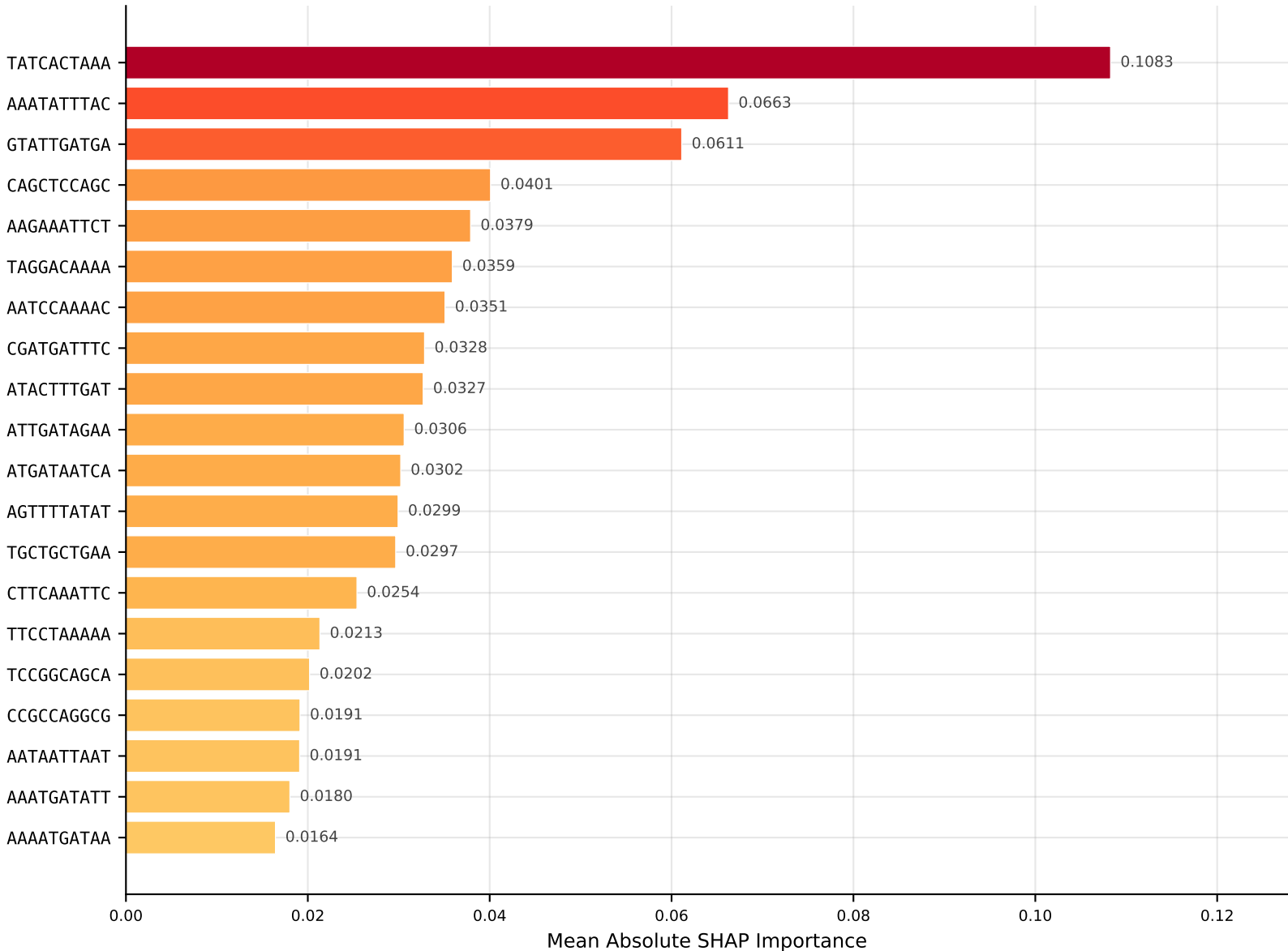


SHAP Feature Importance — Ampicillin/Sulbactam Top 20 k-mer predictors



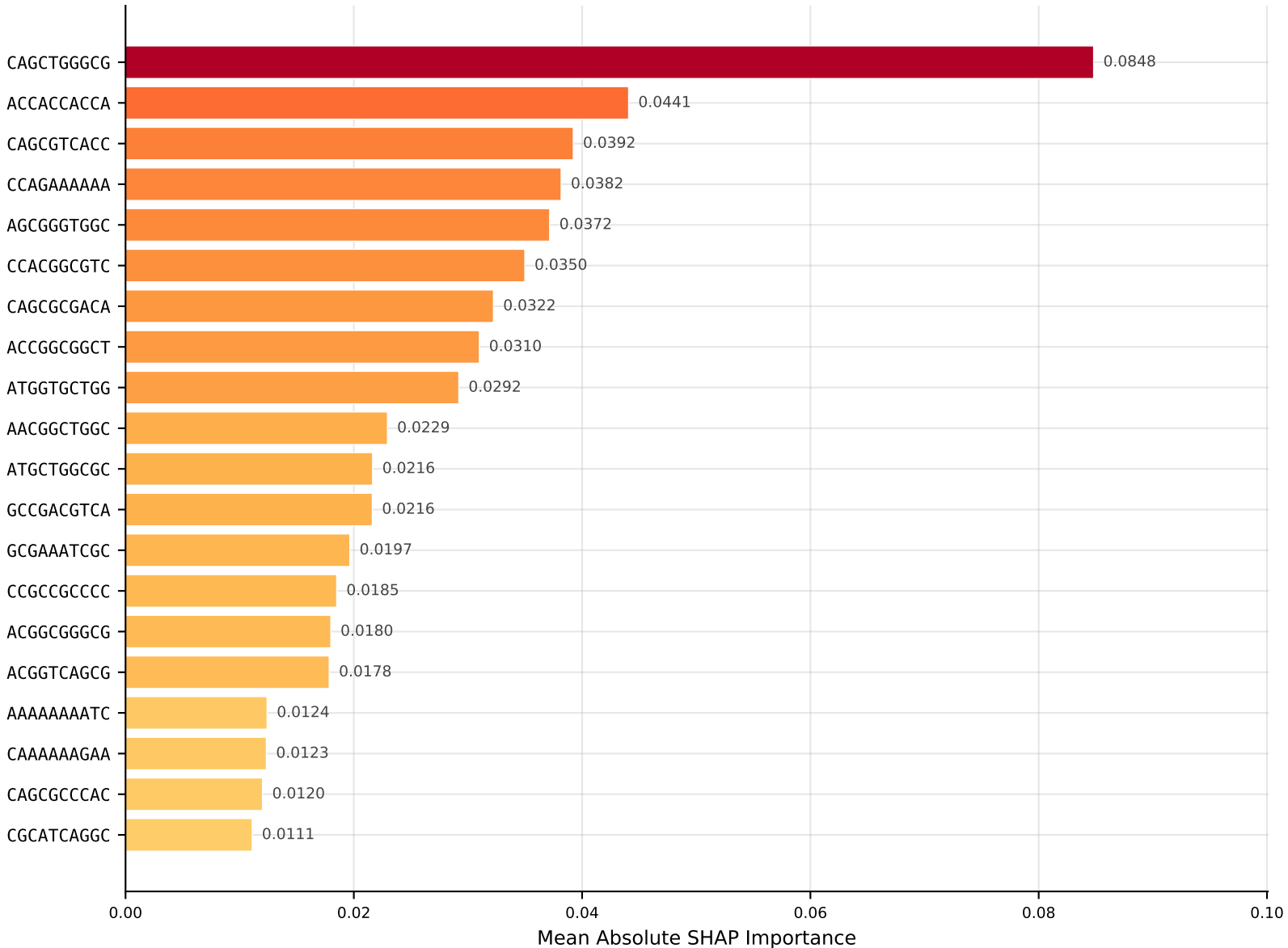
SHAP Feature Importance — Azithromycin

Top 20 k-mer predictors

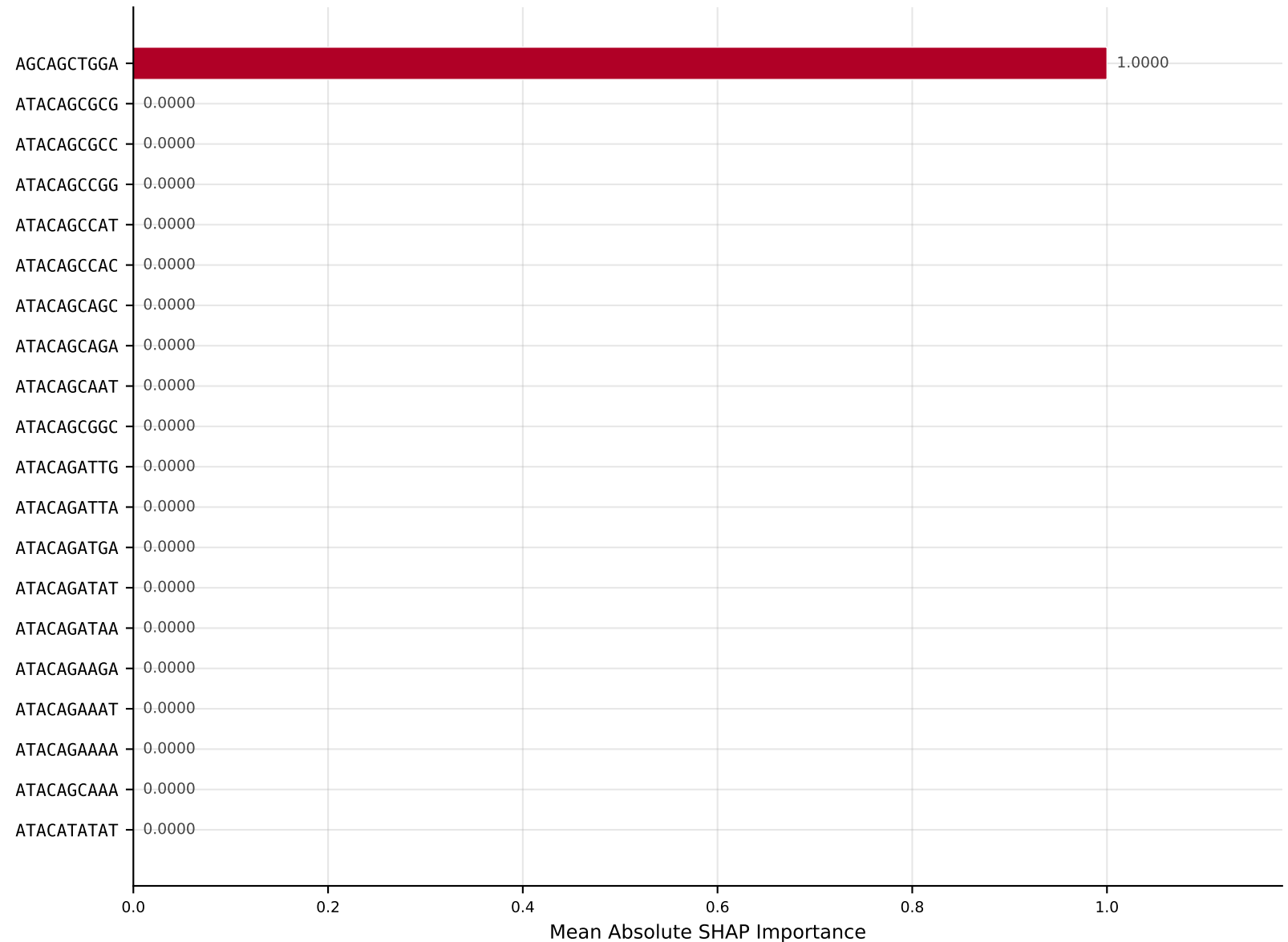


SHAP Feature Importance — Aztreonam

Top 20 k-mer predictors

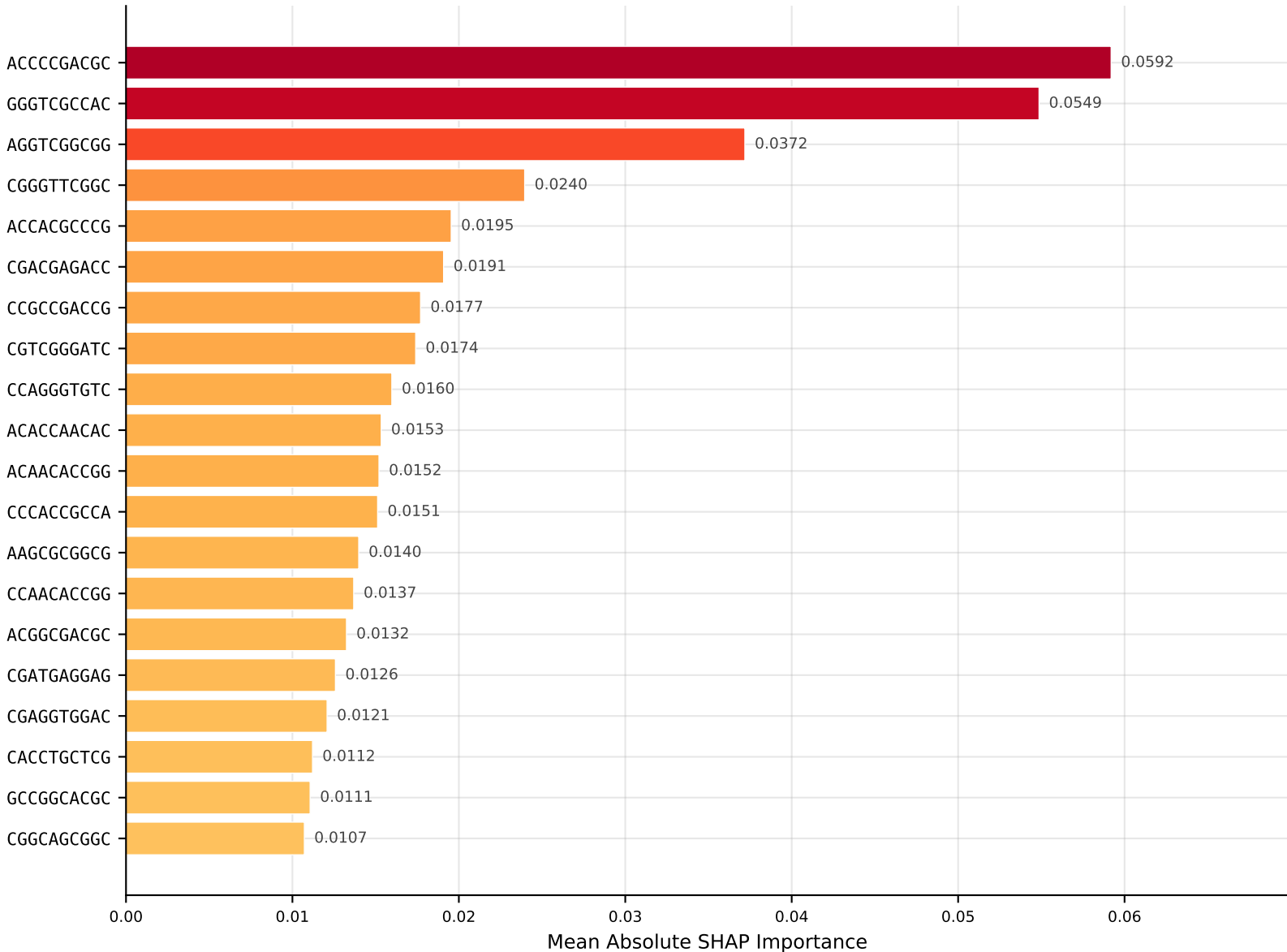


SHAP Feature Importance — Beta-Lactam Top 20 k-mer predictors

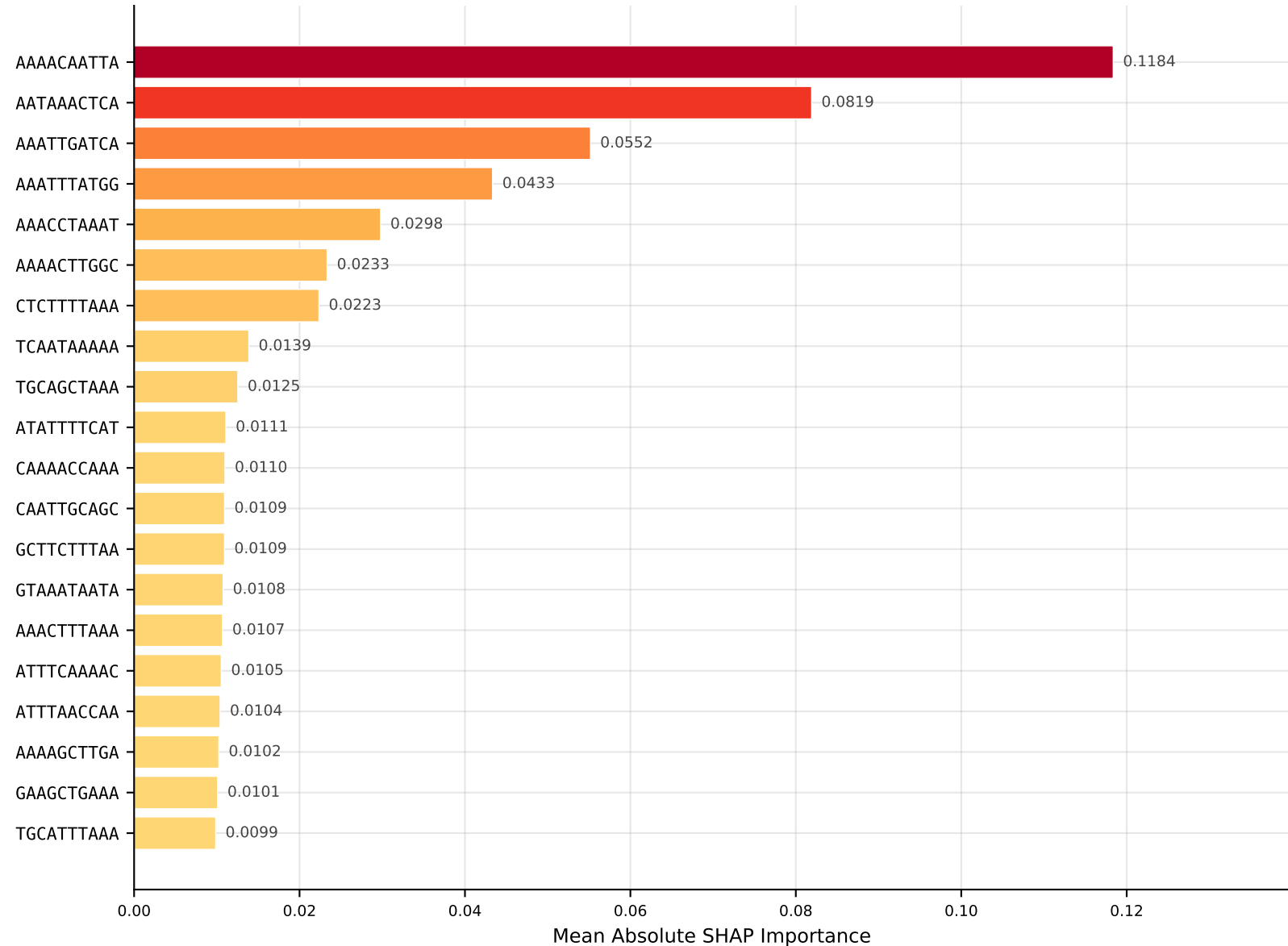


SHAP Feature Importance — Capreomycin

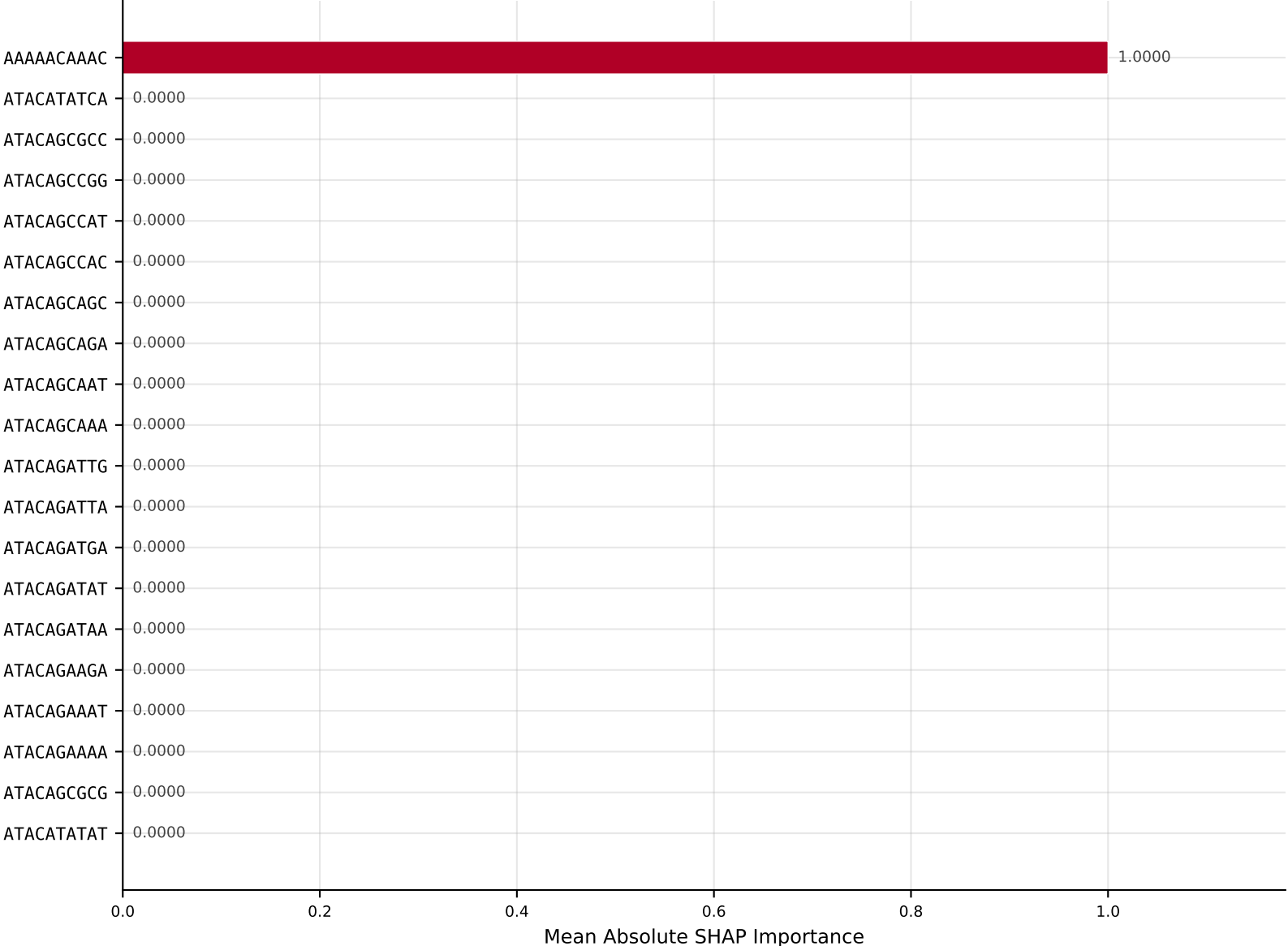
Top 20 k-mer predictors



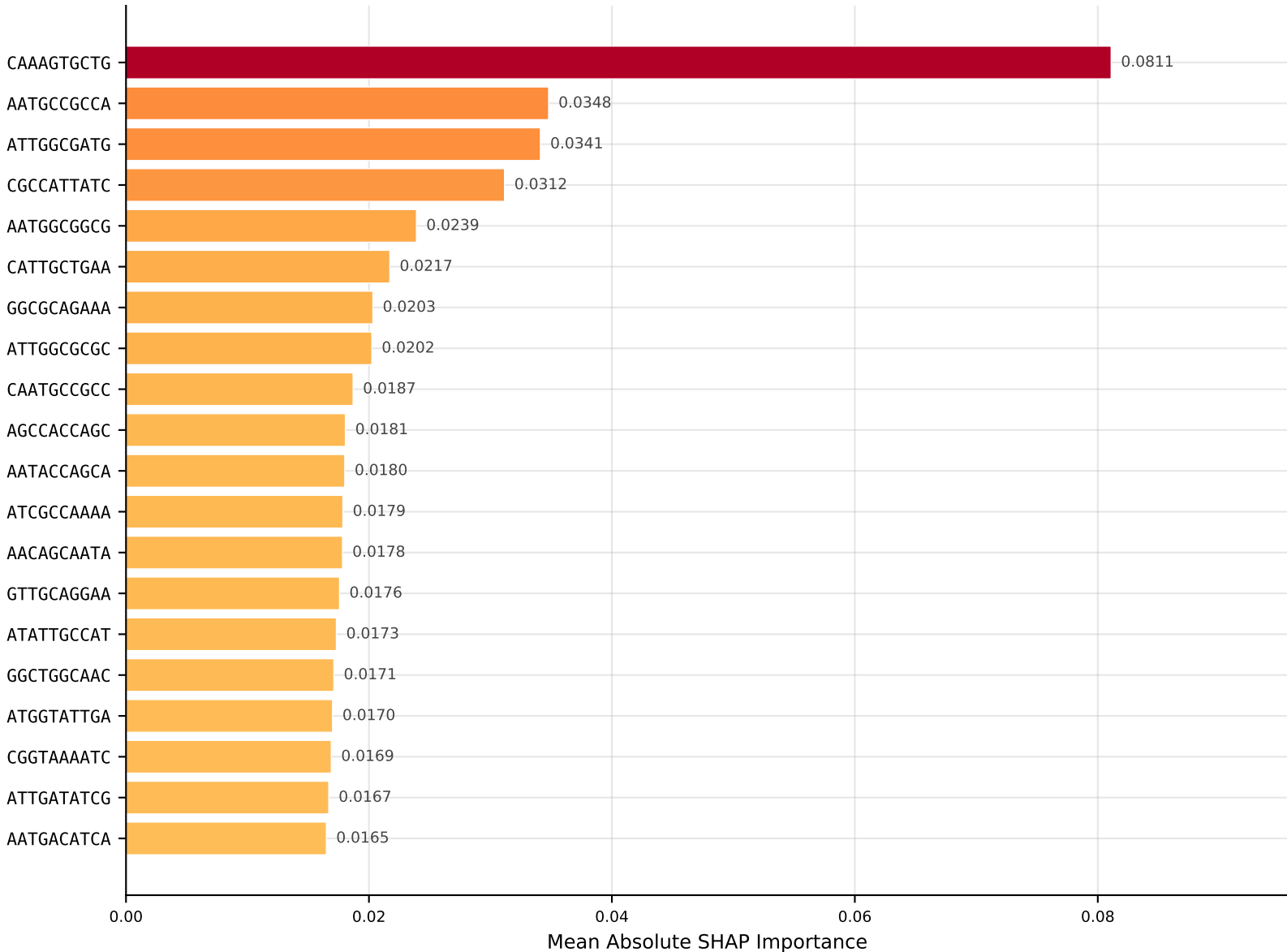
SHAP Feature Importance — Carbapenem Top 20 k-mer predictors



SHAP Feature Importance — Carbenicillin
Top 20 k-mer predictors

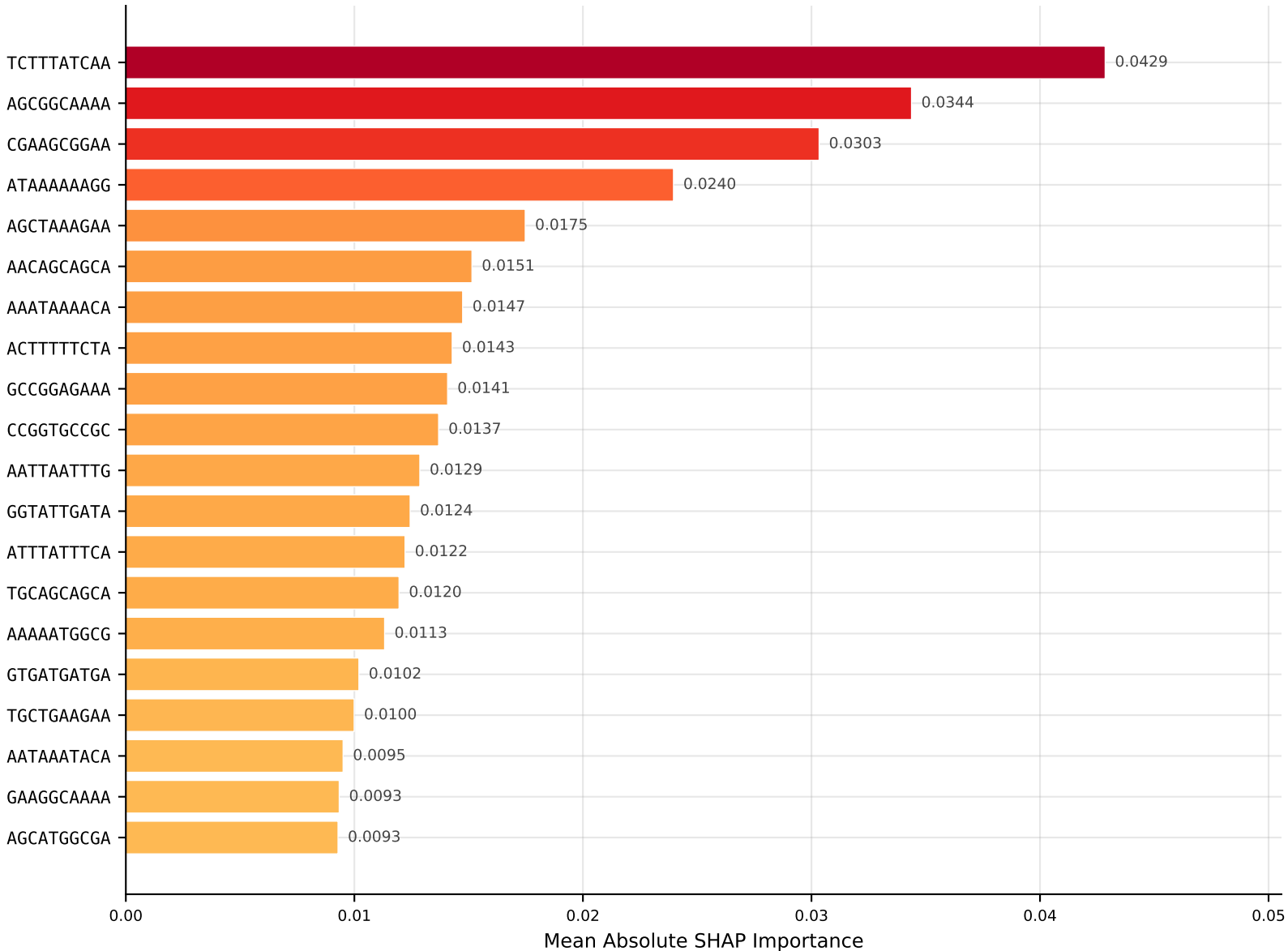


SHAP Feature Importance — Cefazolin Top 20 k-mer predictors



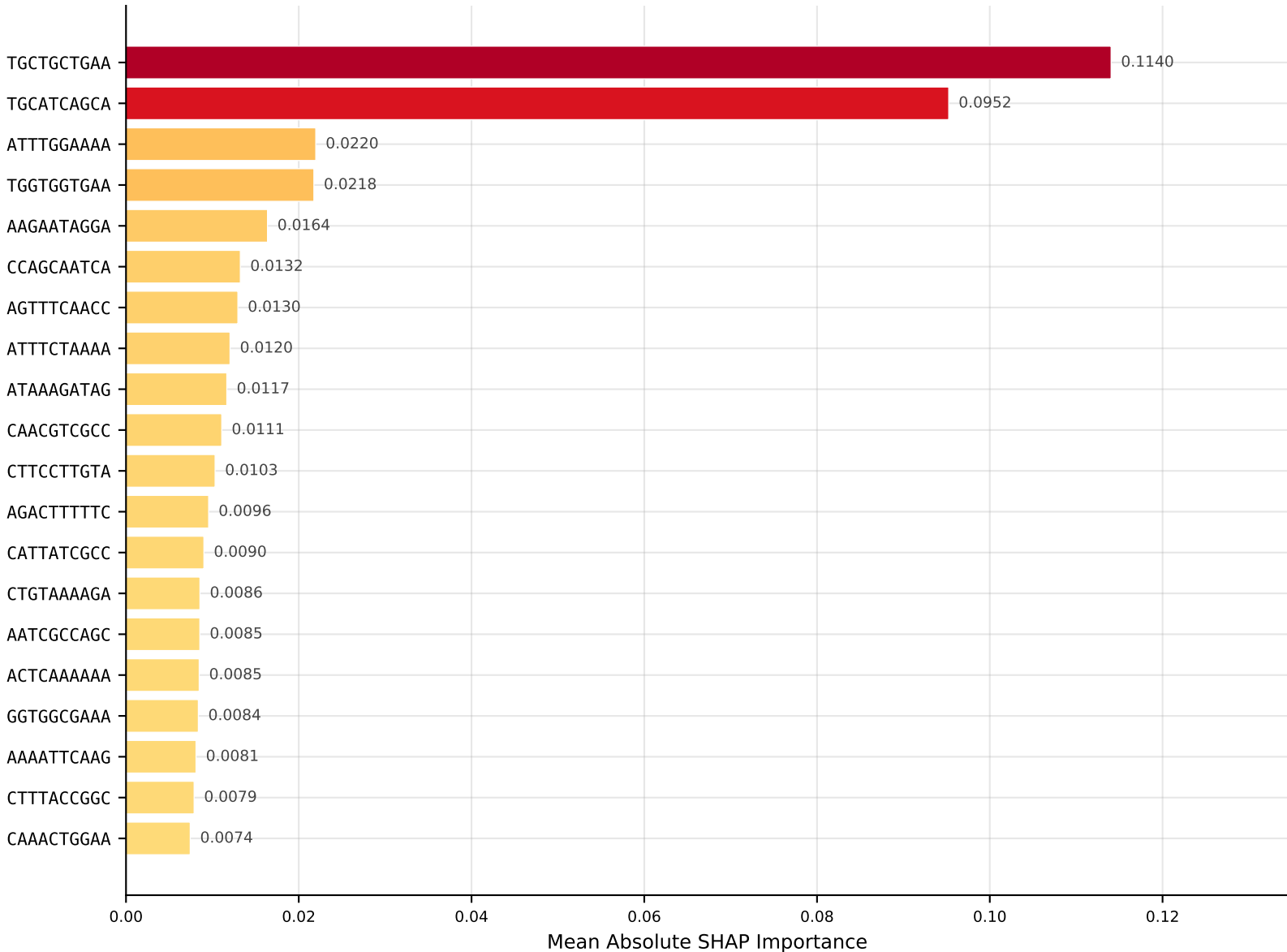
SHAP Feature Importance — Cefepime

Top 20 k-mer predictors



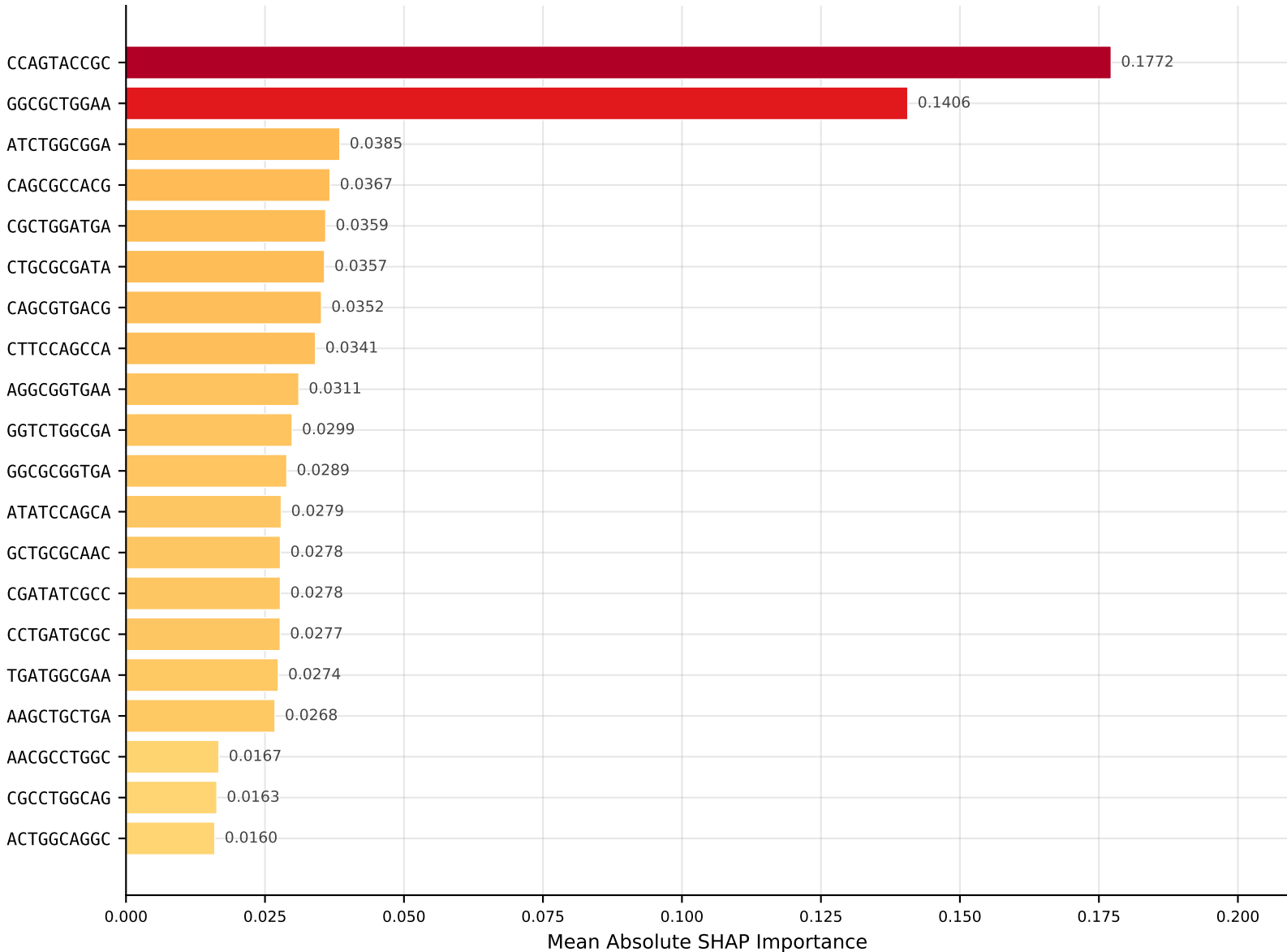
SHAP Feature Importance — Cefotaxime

Top 20 k-mer predictors



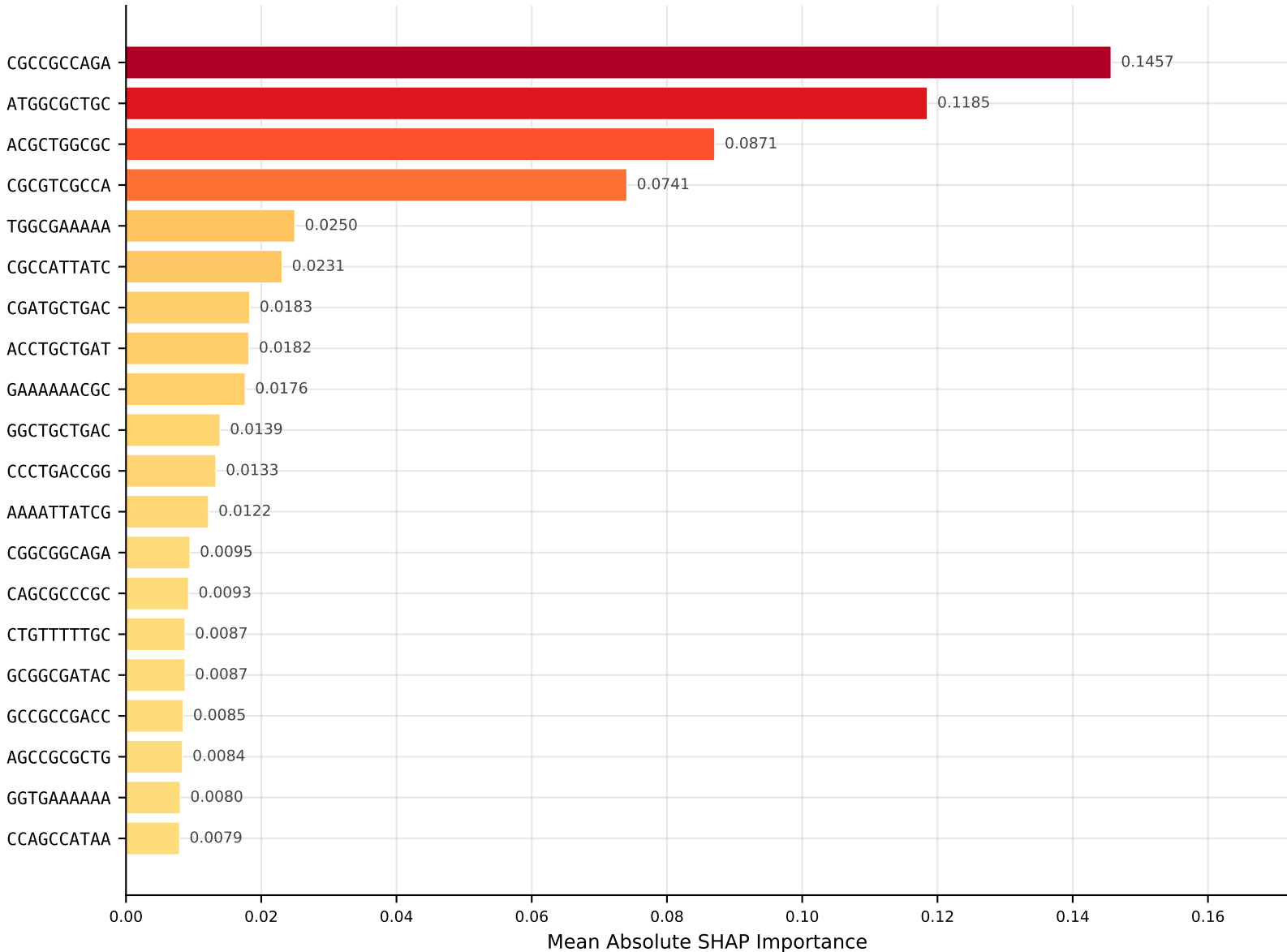
SHAP Feature Importance — Cefotetan

Top 20 k-mer predictors



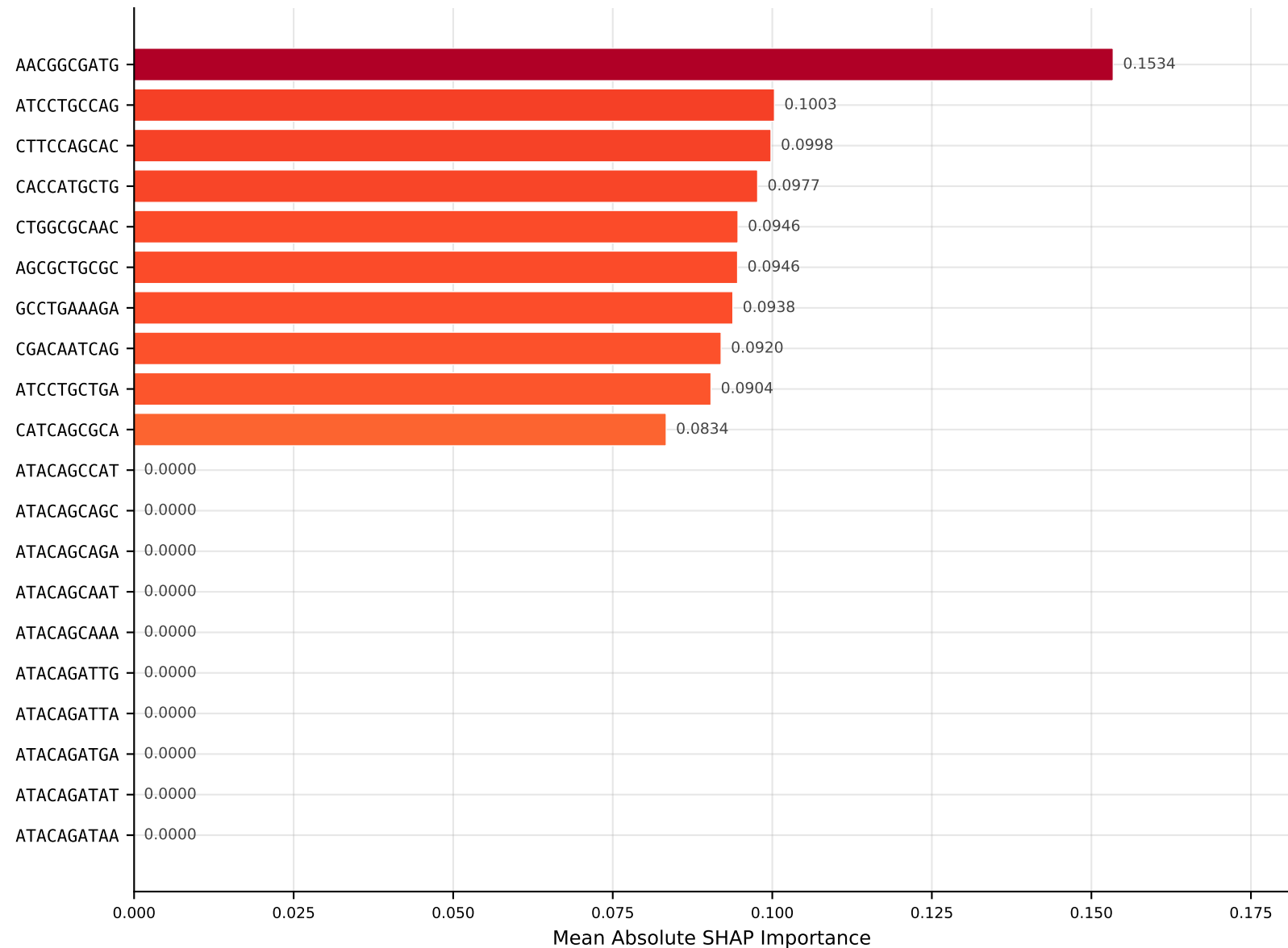
SHAP Feature Importance — Cefoxitin

Top 20 k-mer predictors



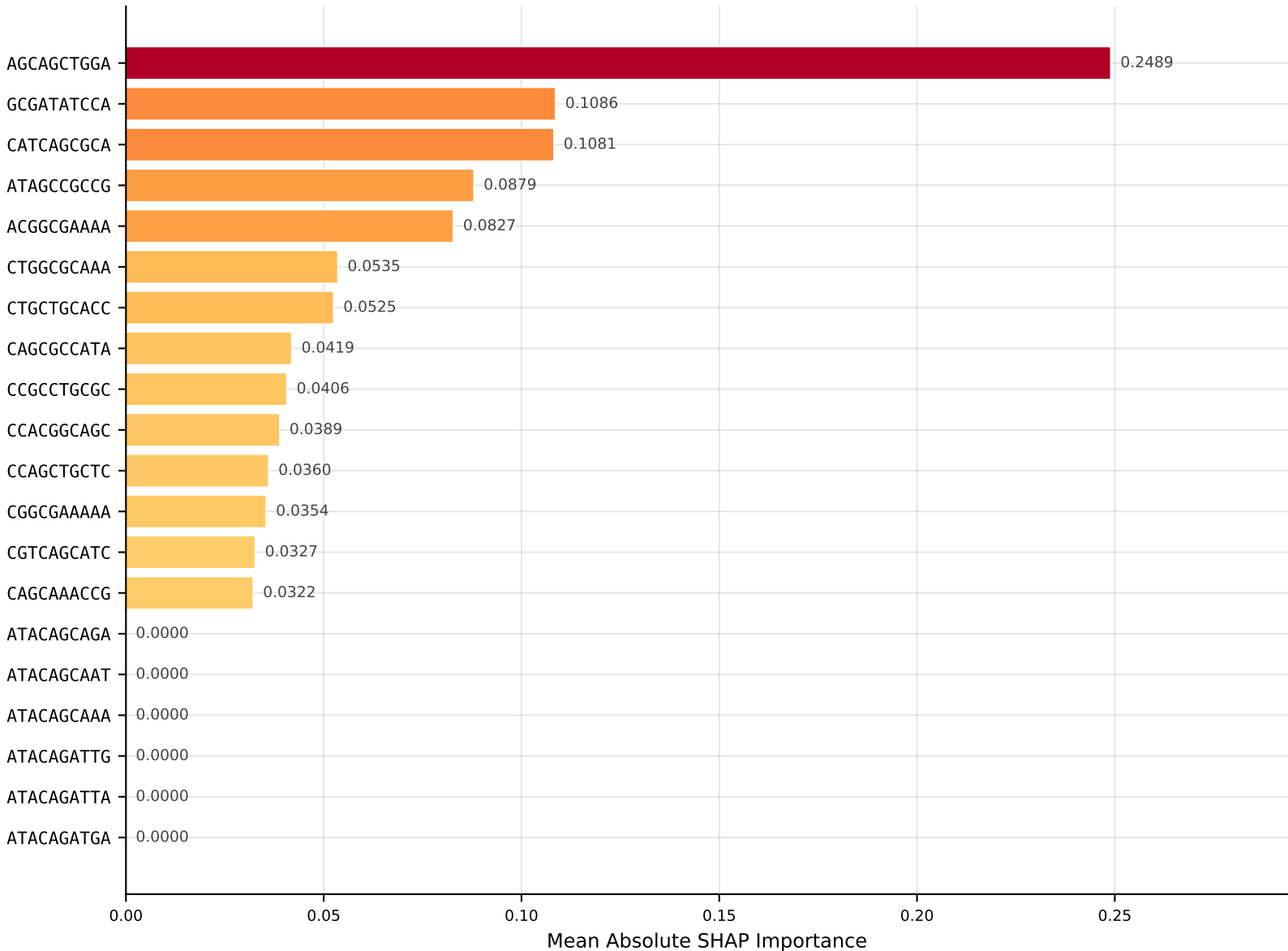
SHAP Feature Importance — Cefpodoxime

Top 20 k-mer predictors



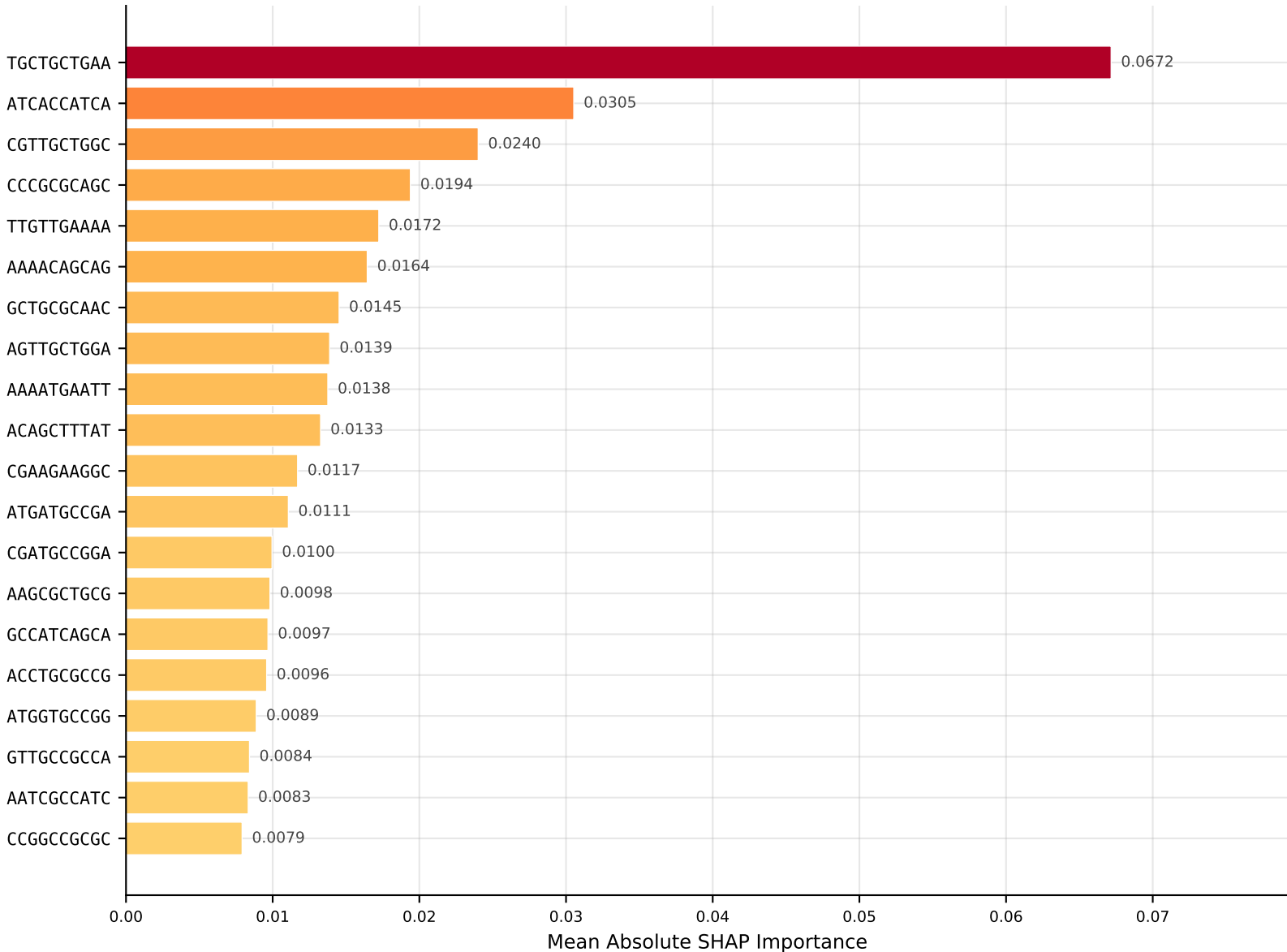
SHAP Feature Importance — Cefpodoxime_Clavulanic_Acid

Top 20 k-mer predictors

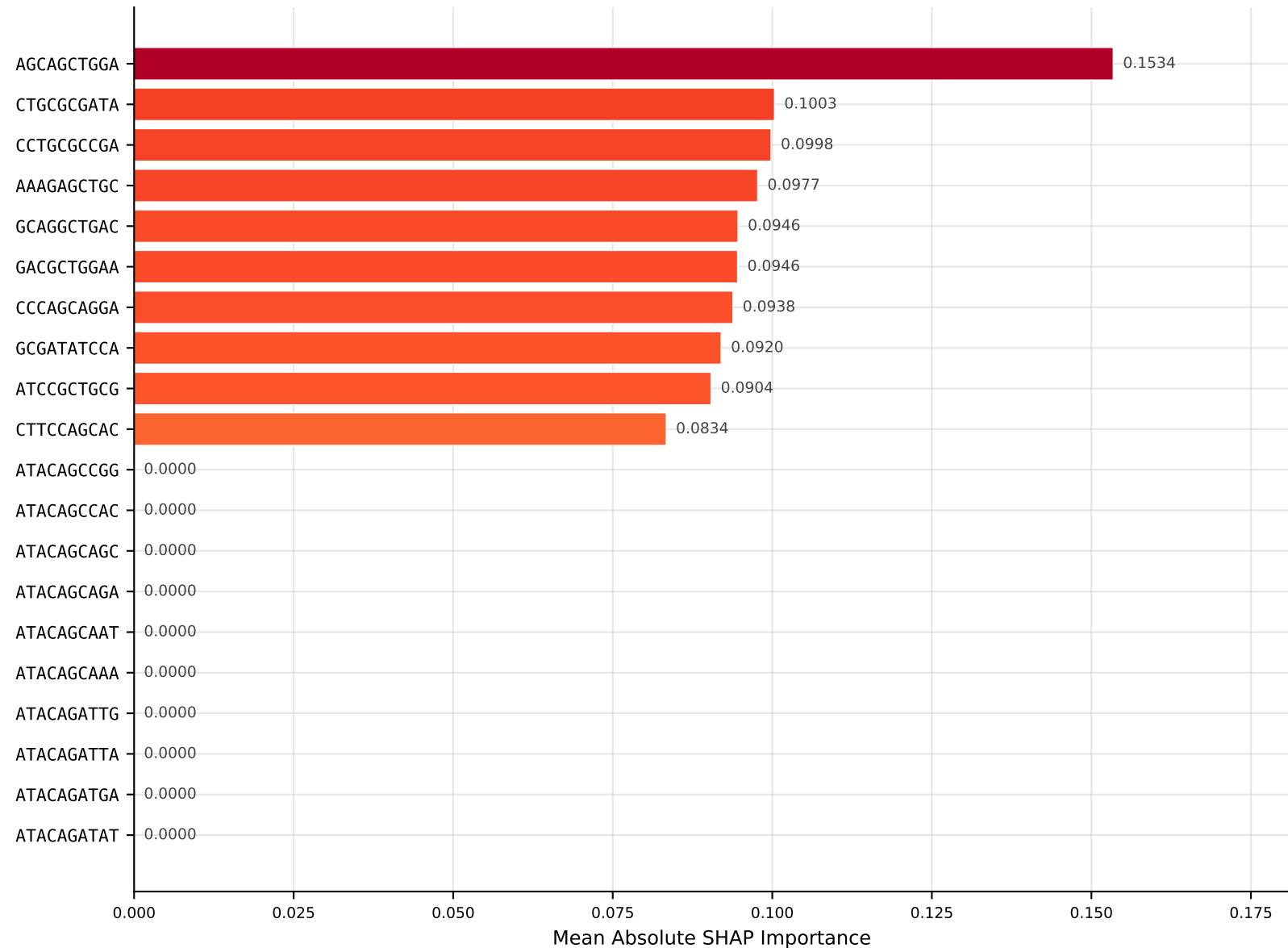


SHAP Feature Importance — Ceftazidime

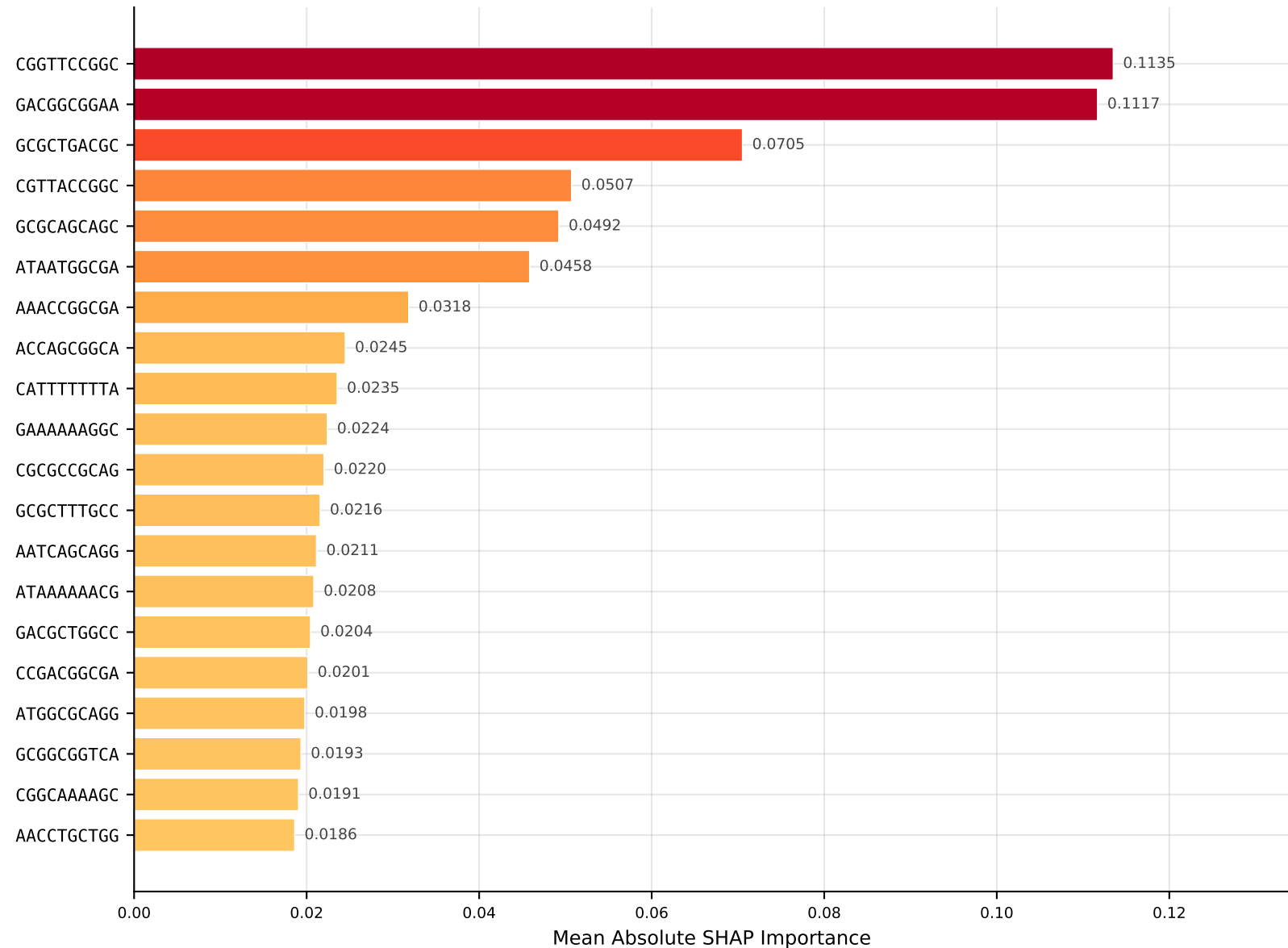
Top 20 k-mer predictors



SHAP Feature Importance — Ceftazidime/Avibactam Top 20 k-mer predictors

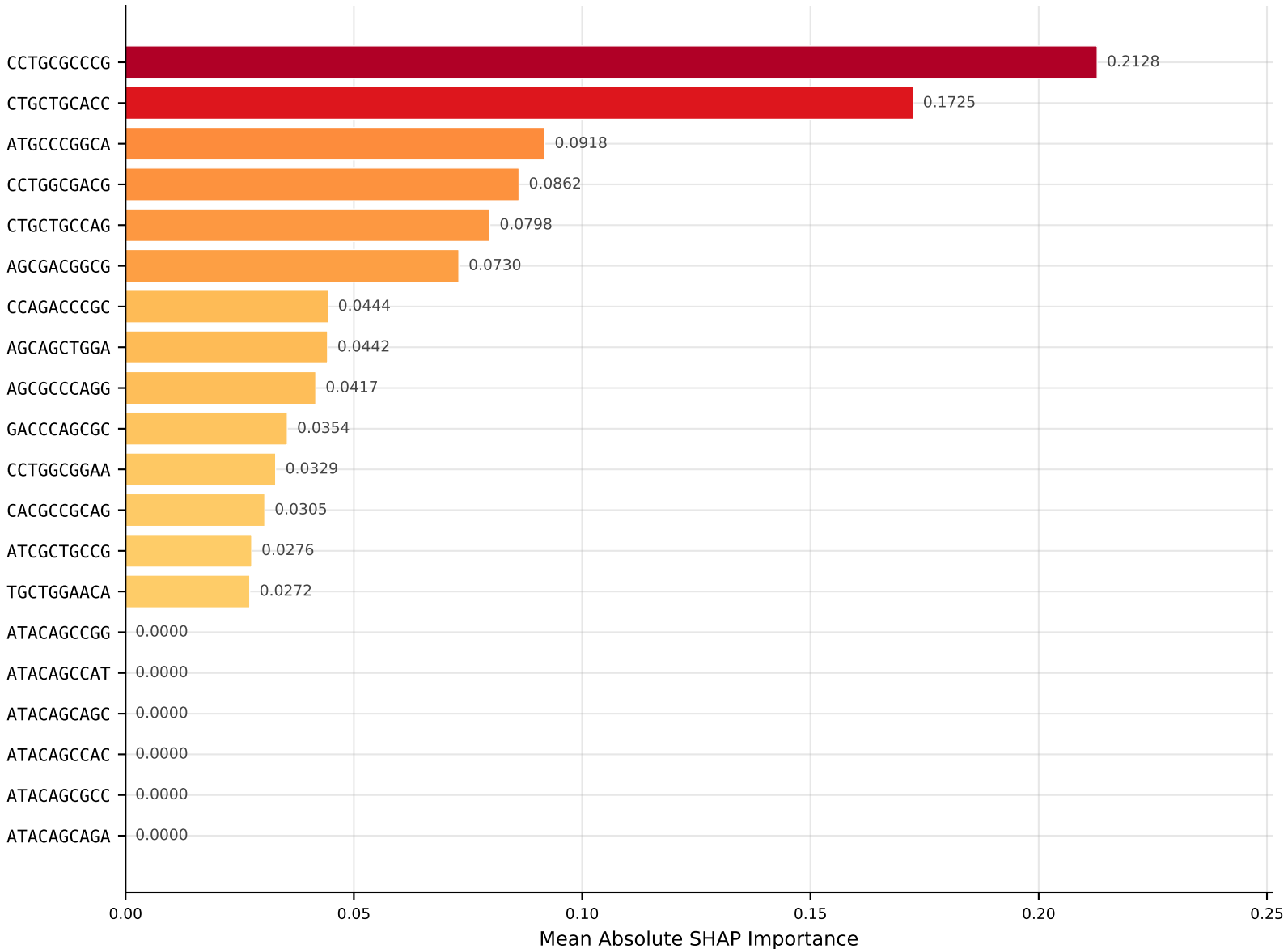


SHAP Feature Importance — Ceftiofur Top 20 k-mer predictors



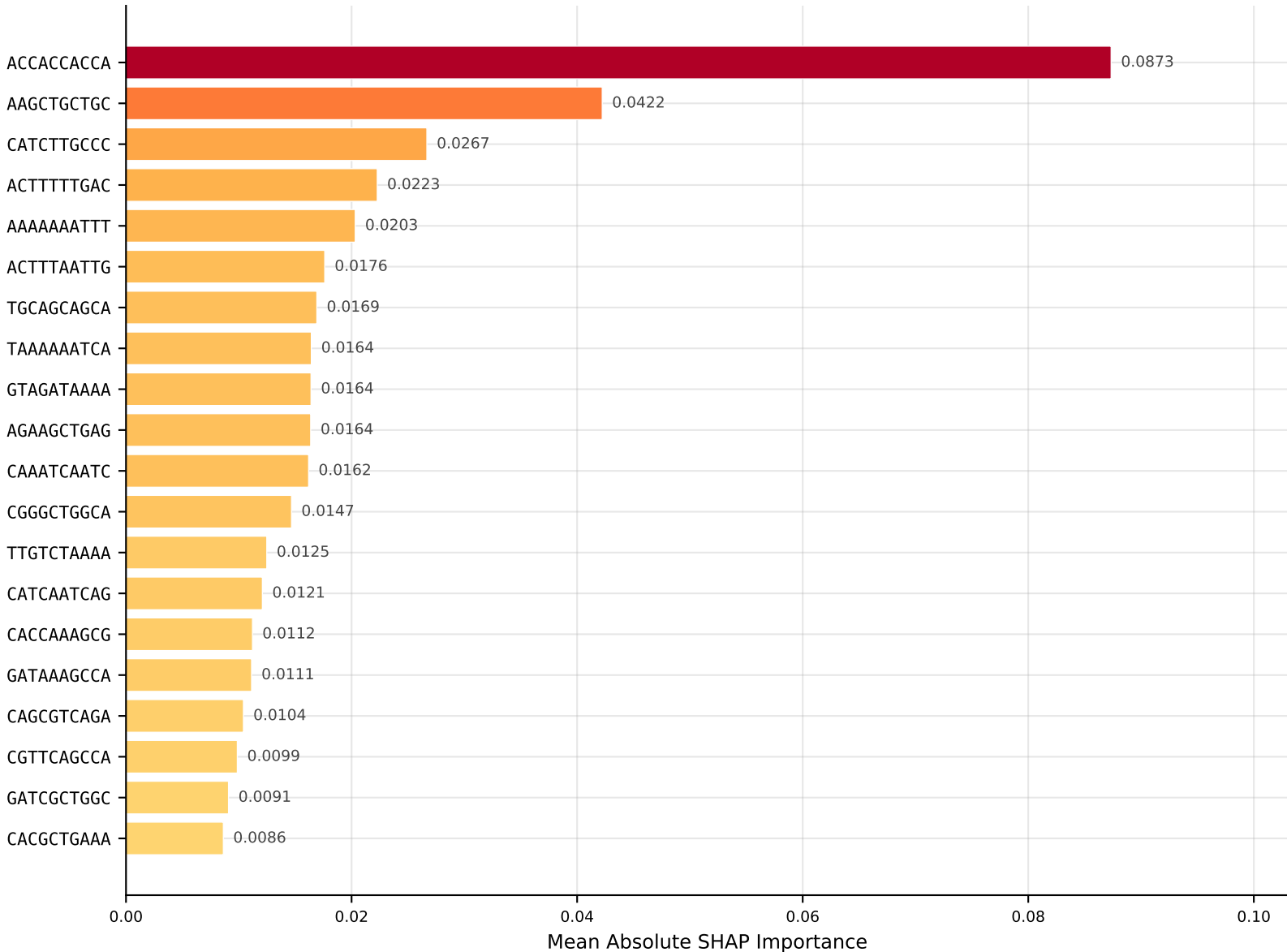
SHAP Feature Importance — Ceftriaxone/Tazobactam

Top 20 k-mer predictors

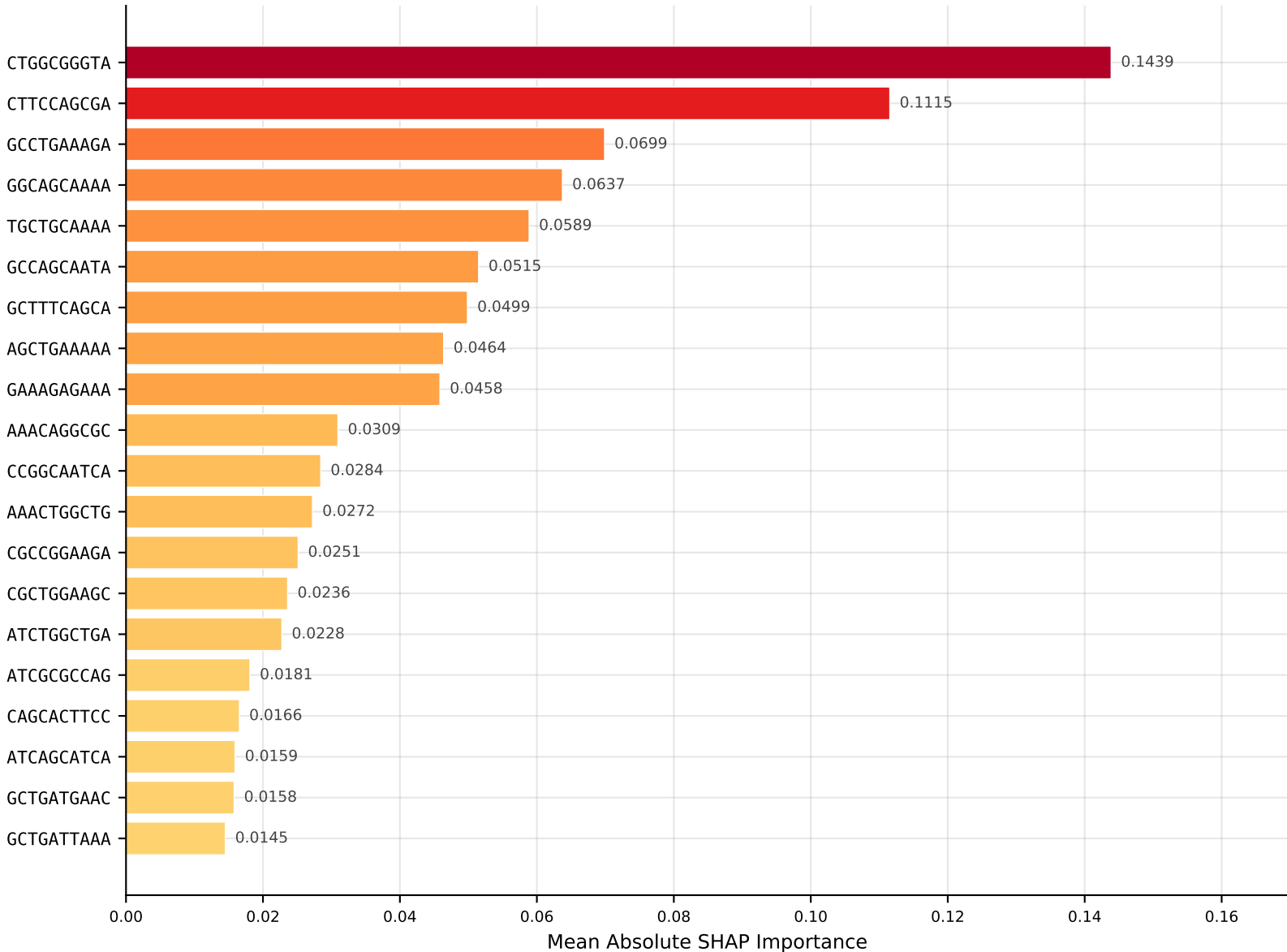


SHAP Feature Importance — Ceftriaxone

Top 20 k-mer predictors

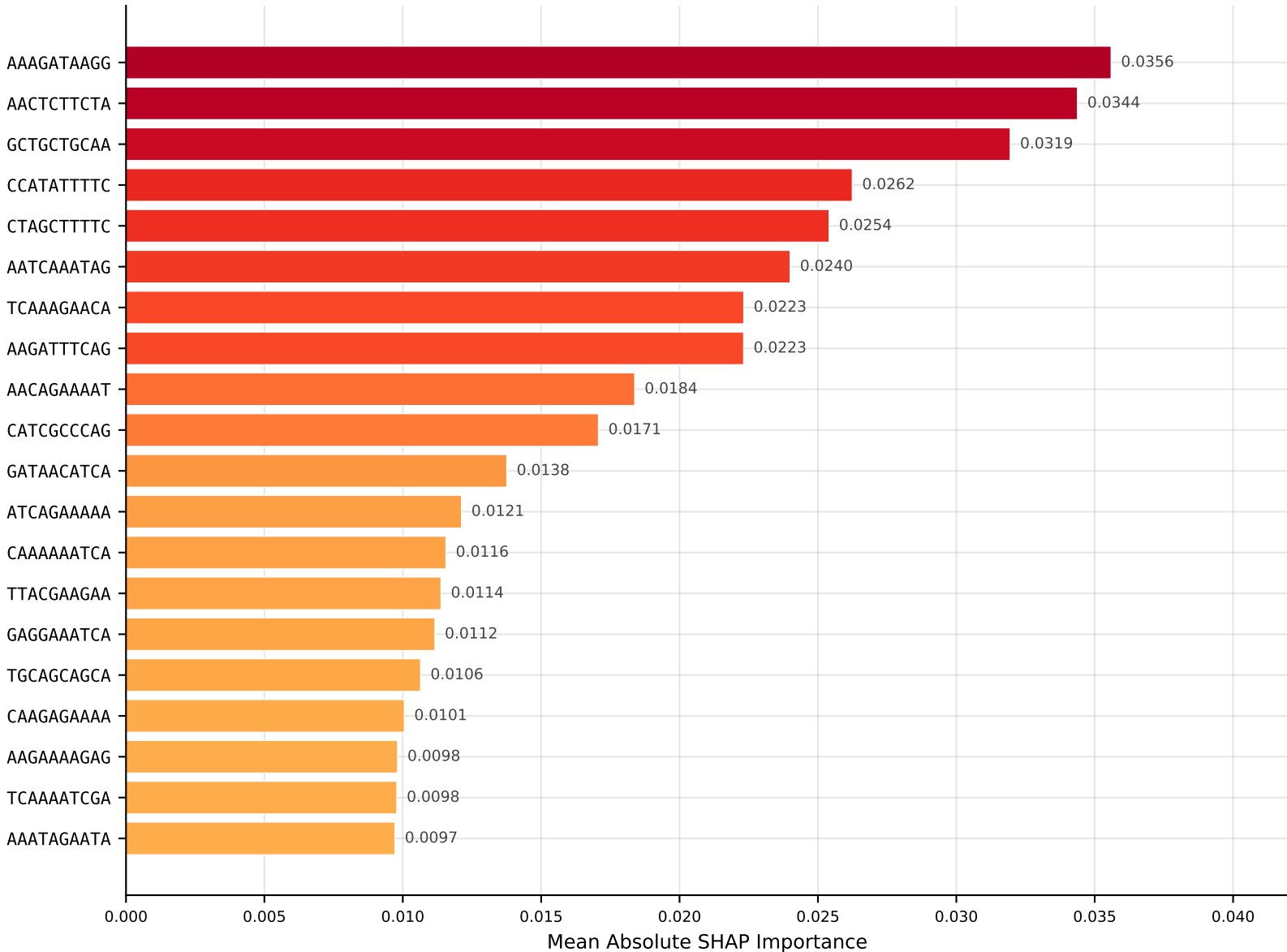


SHAP Feature Importance — Cephalothin Top 20 k-mer predictors



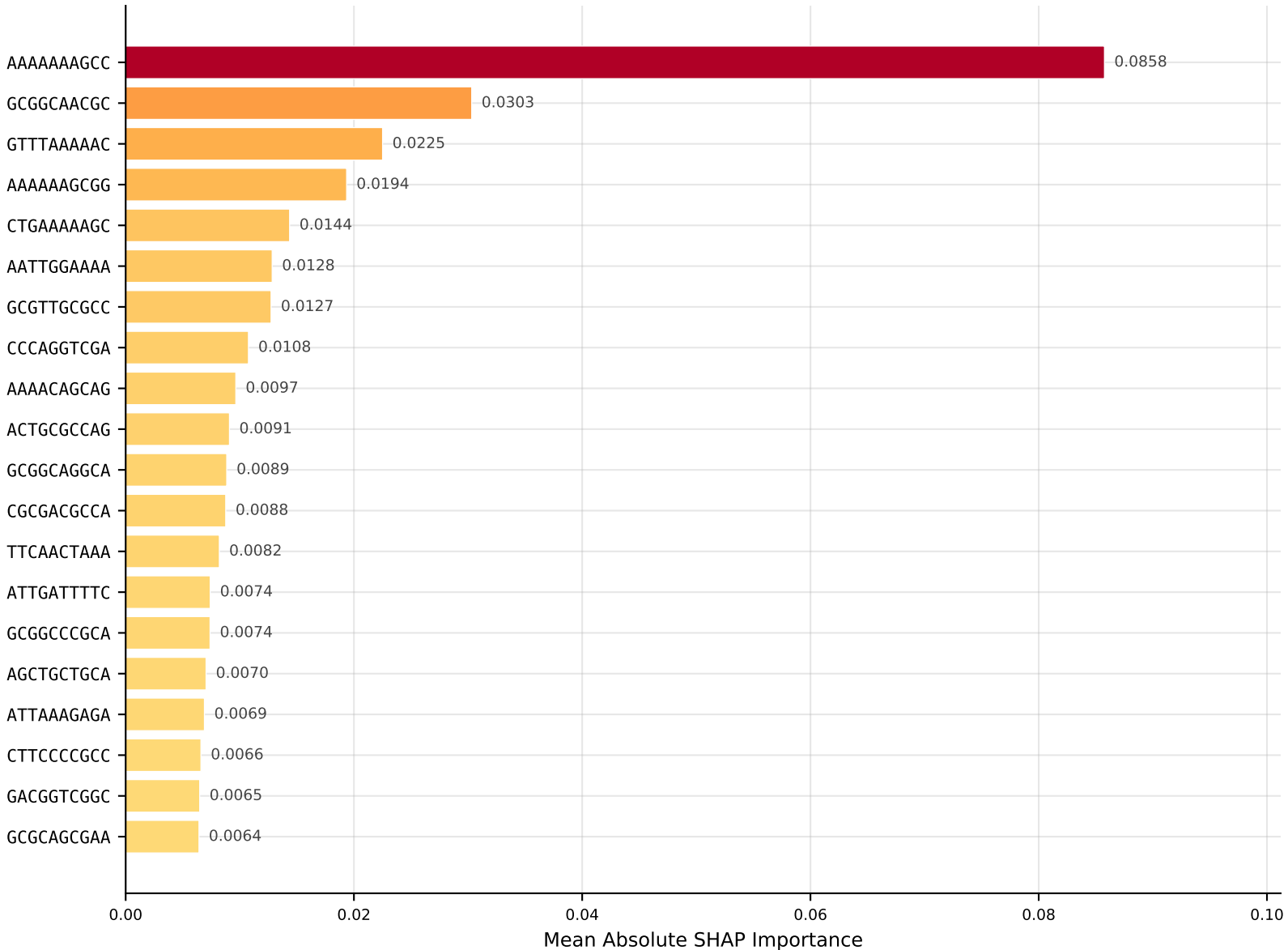
SHAP Feature Importance — Chloramphenicol

Top 20 k-mer predictors



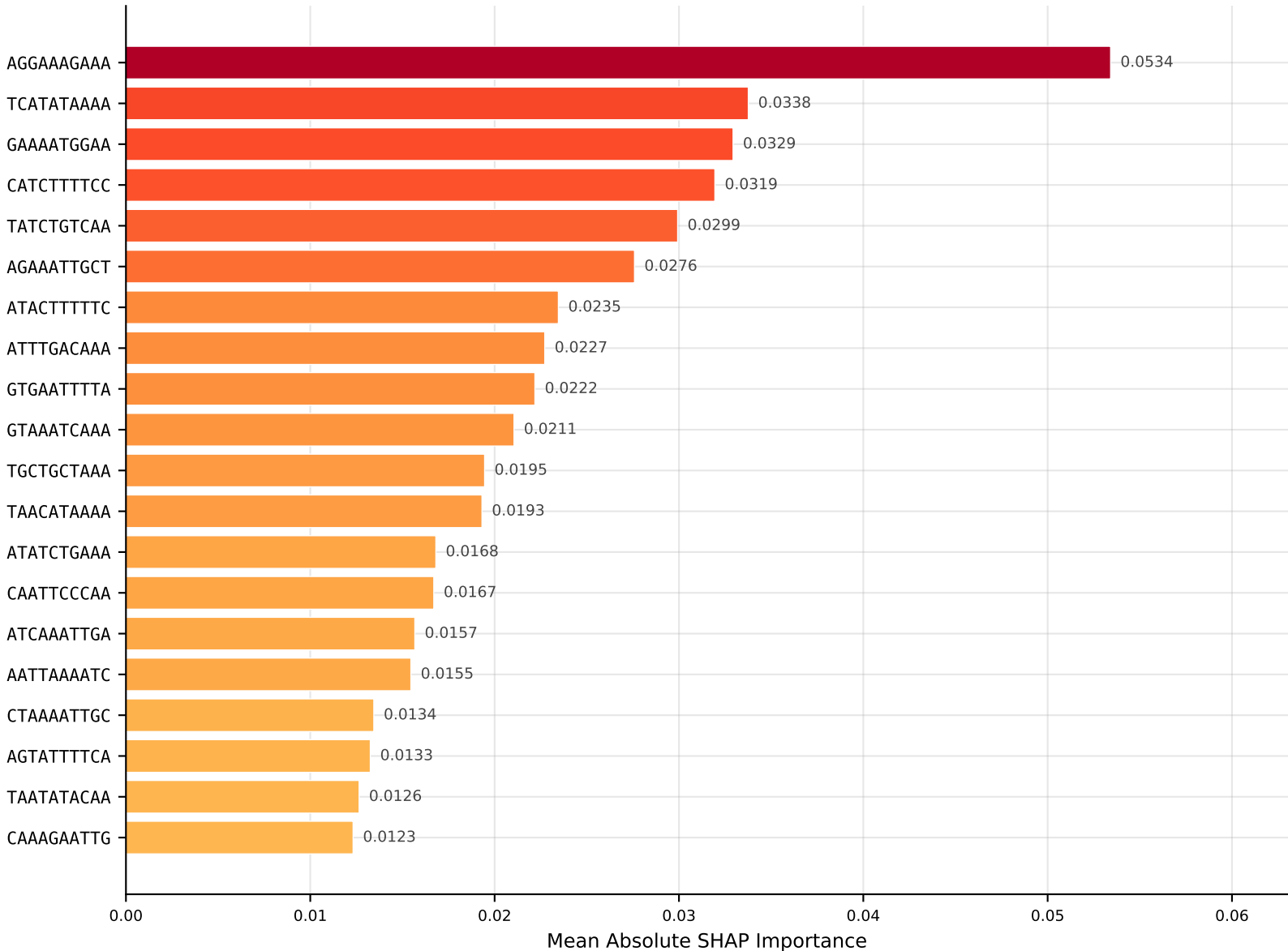
SHAP Feature Importance — Ciprofloxacin

Top 20 k-mer predictors



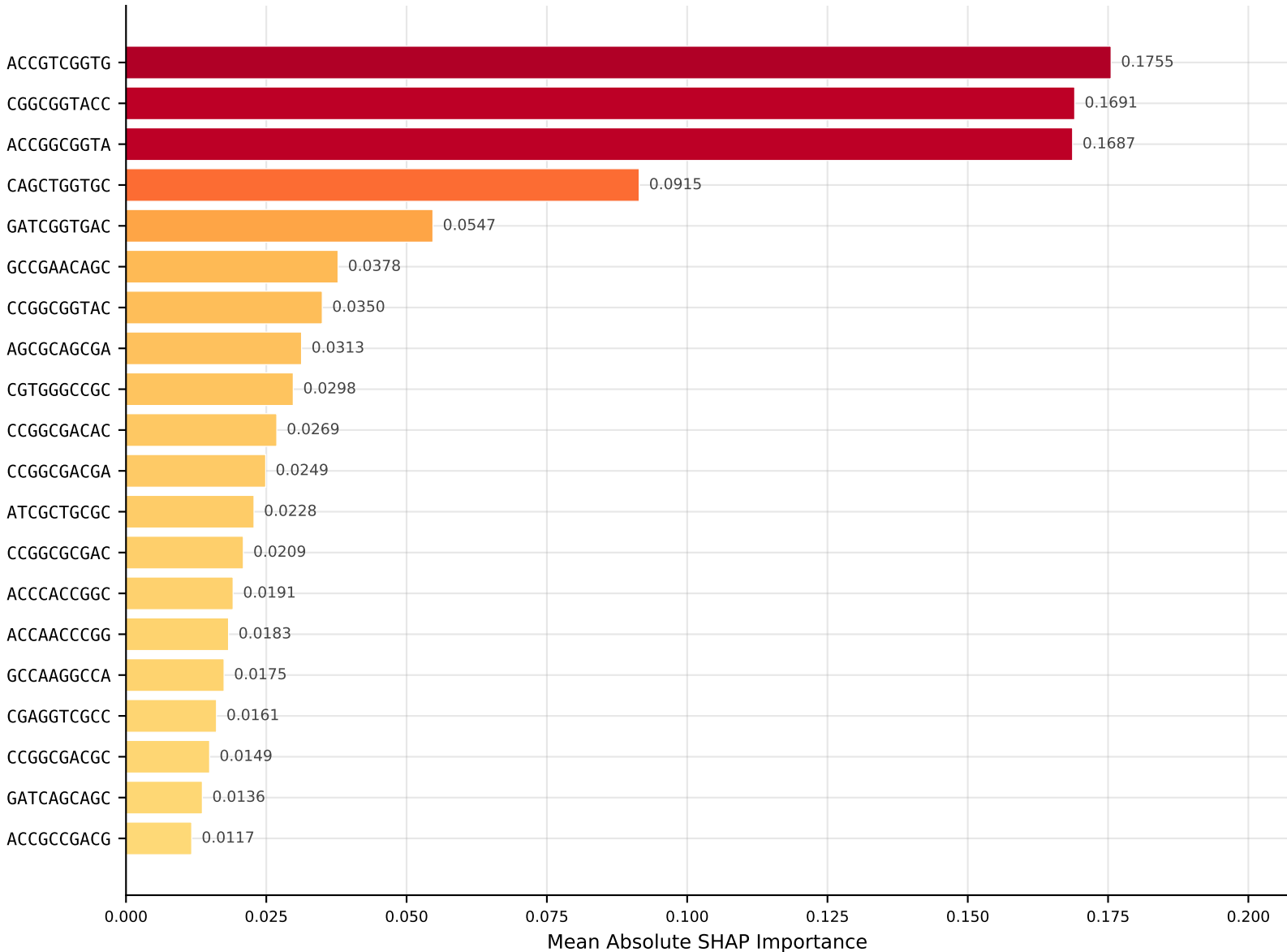
SHAP Feature Importance — Clindamycin

Top 20 k-mer predictors



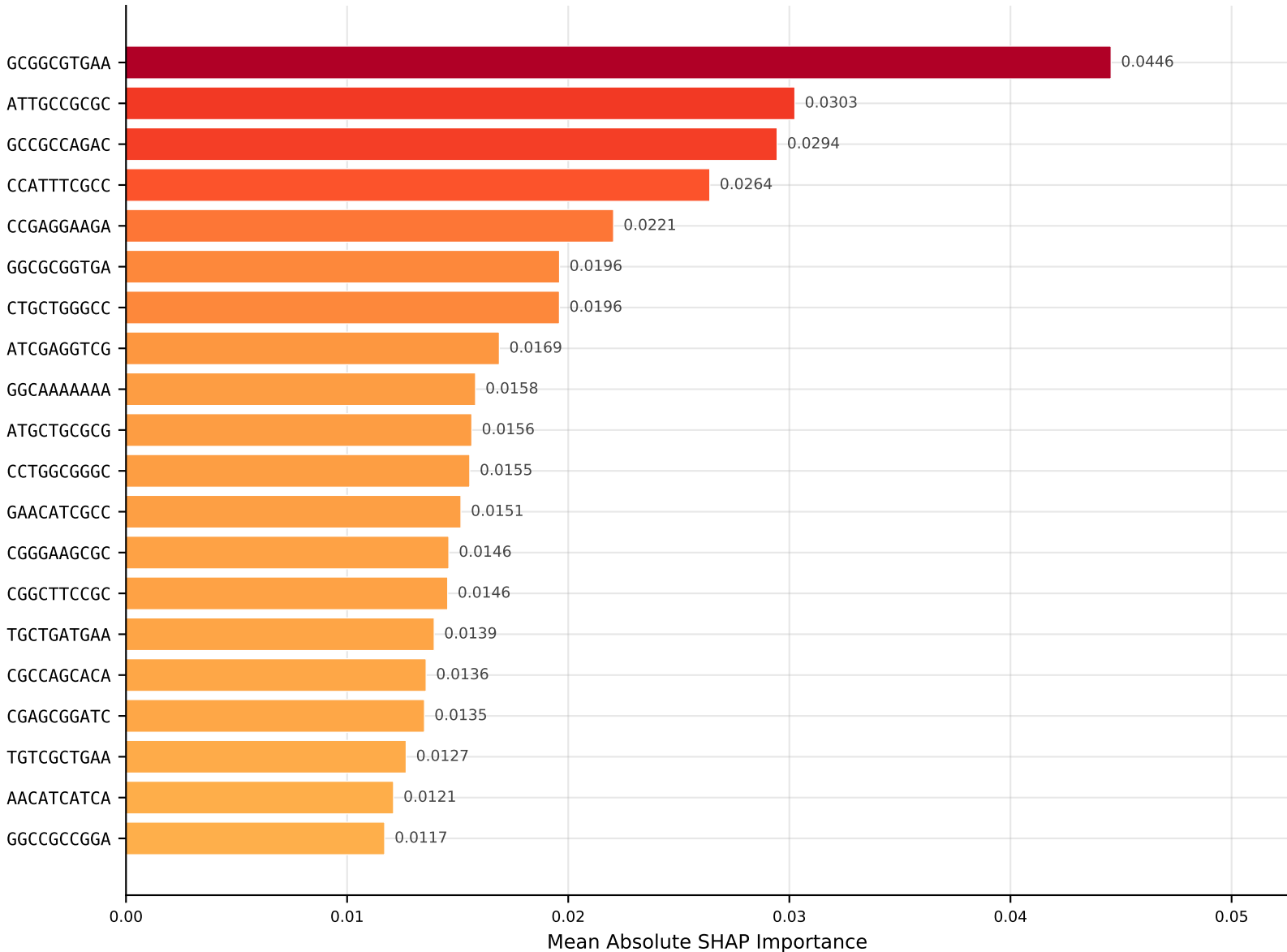
SHAP Feature Importance — Clofazimine

Top 20 k-mer predictors



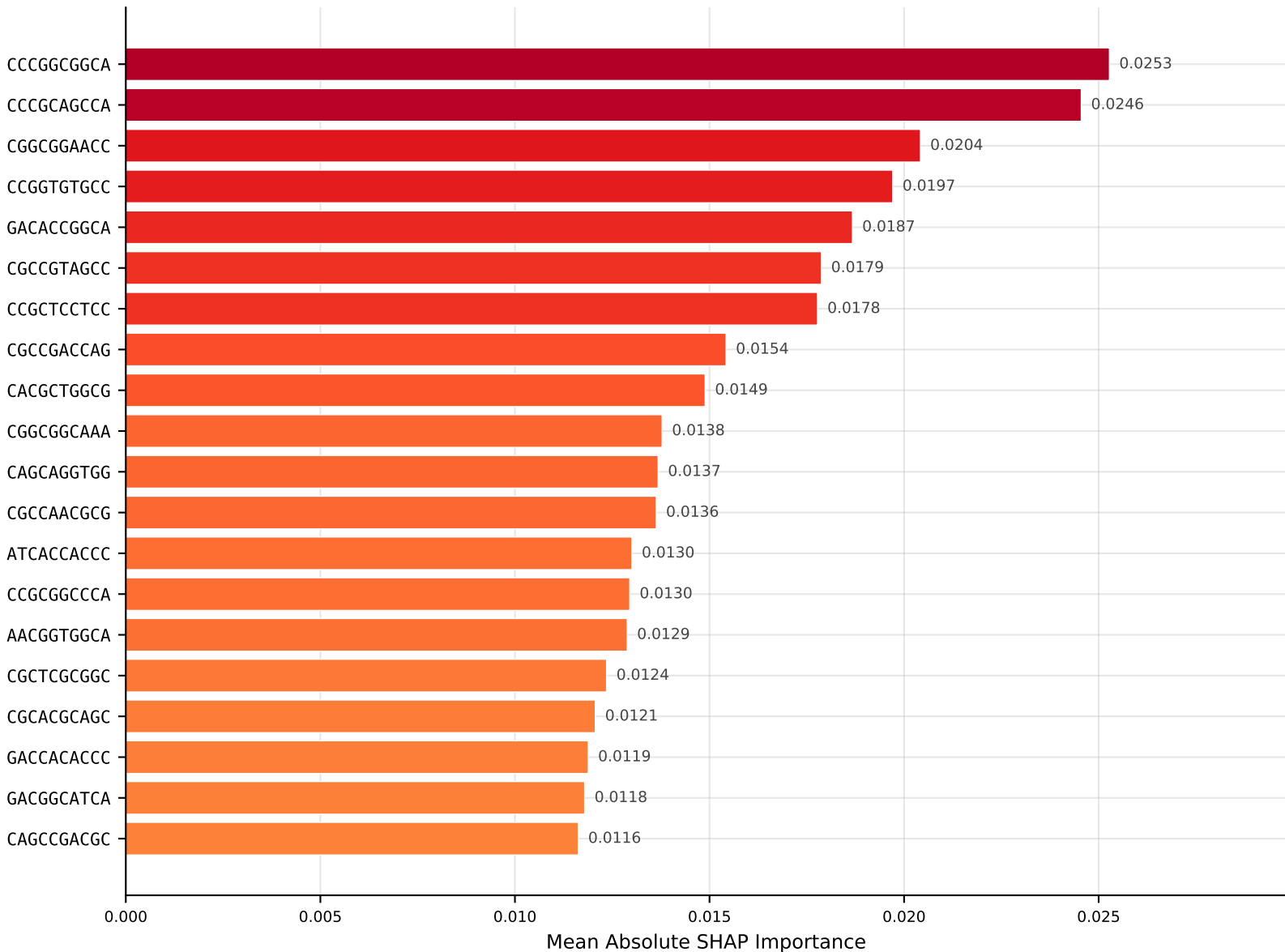
SHAP Feature Importance — Colistin

Top 20 k-mer predictors

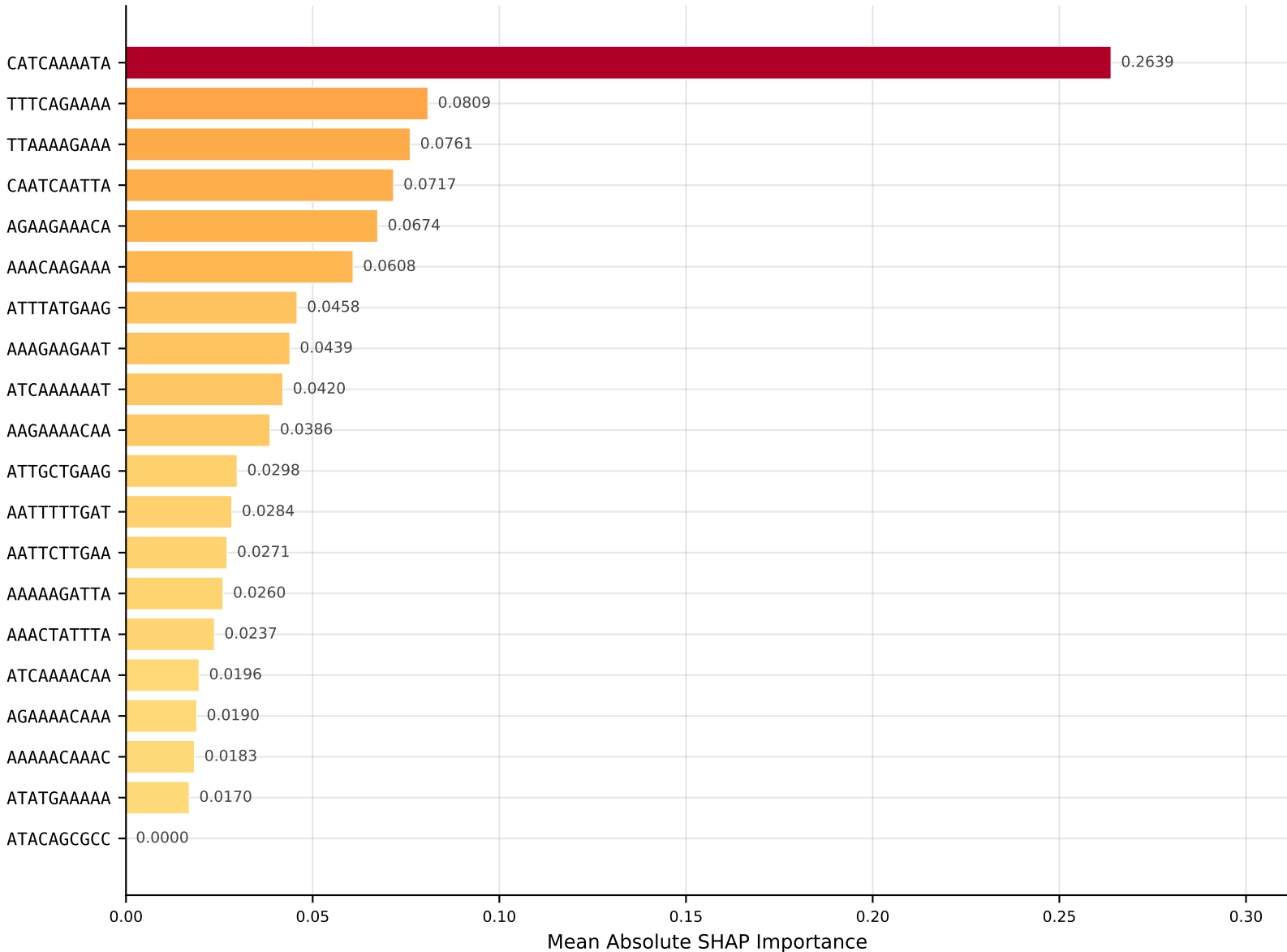


SHAP Feature Importance — Cycloserine

Top 20 k-mer predictors

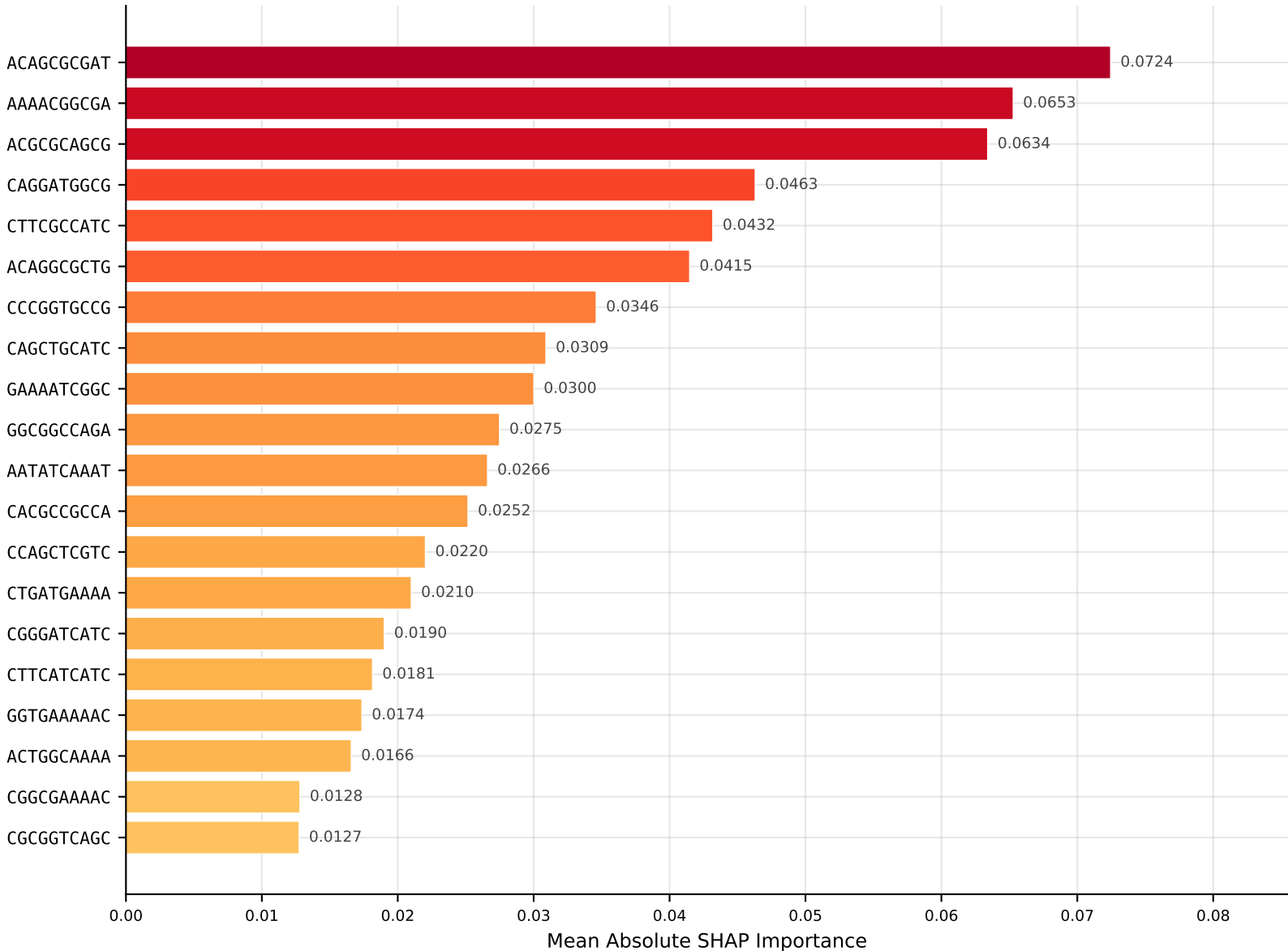


SHAP Feature Importance — Daptomycin Top 20 k-mer predictors

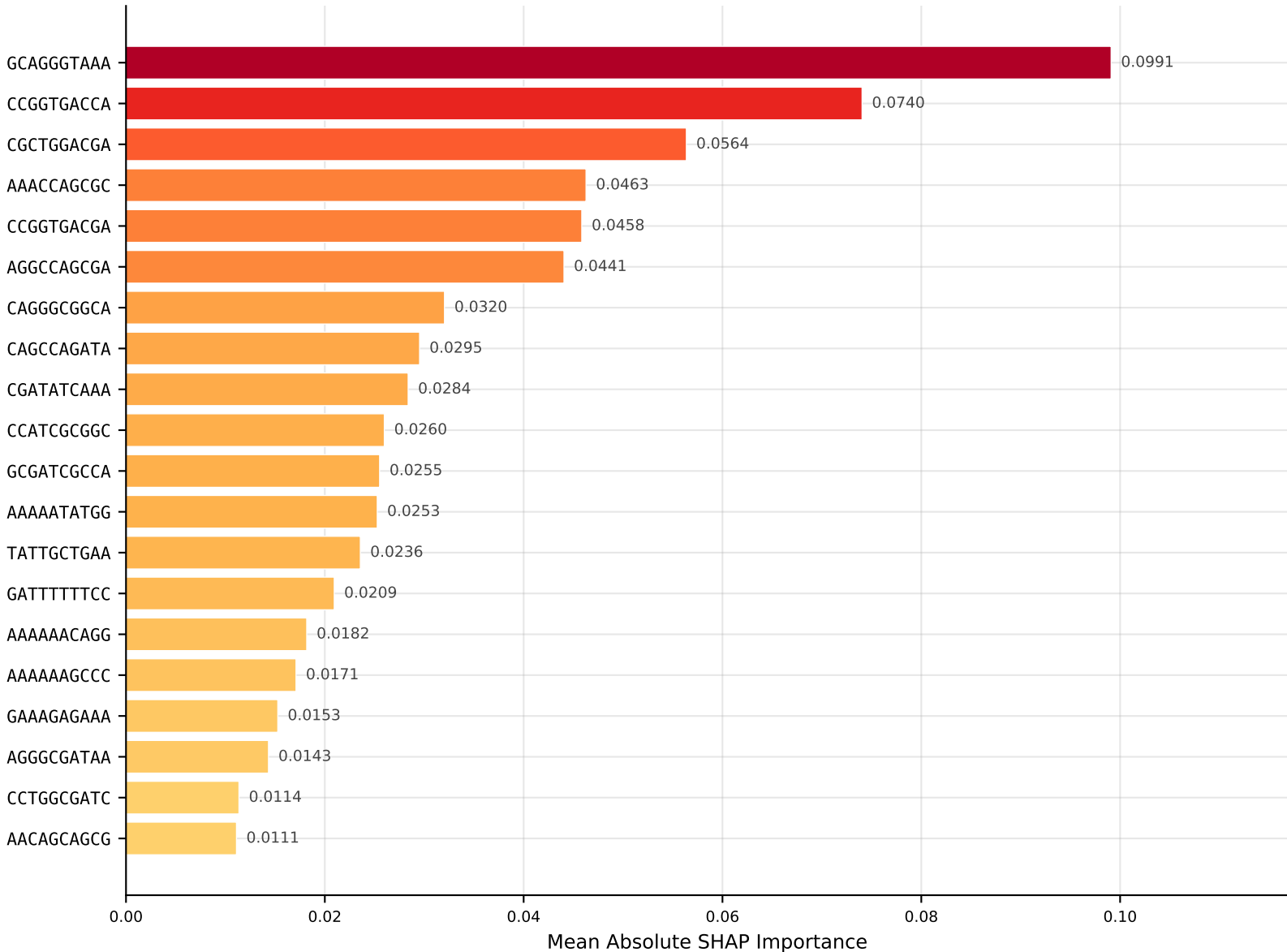


SHAP Feature Importance — Doripenem

Top 20 k-mer predictors

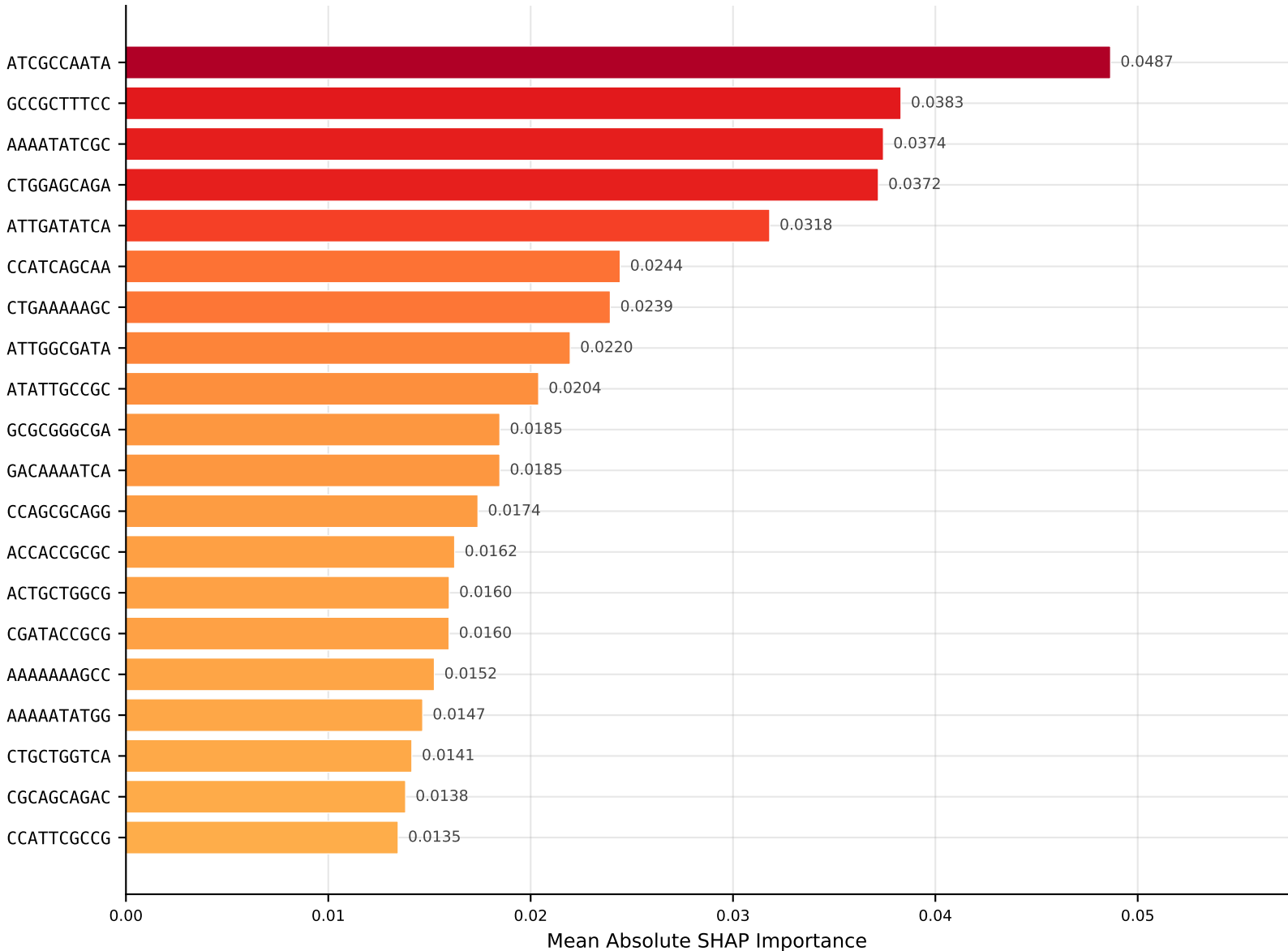


SHAP Feature Importance — Doxycycline Top 20 k-mer predictors

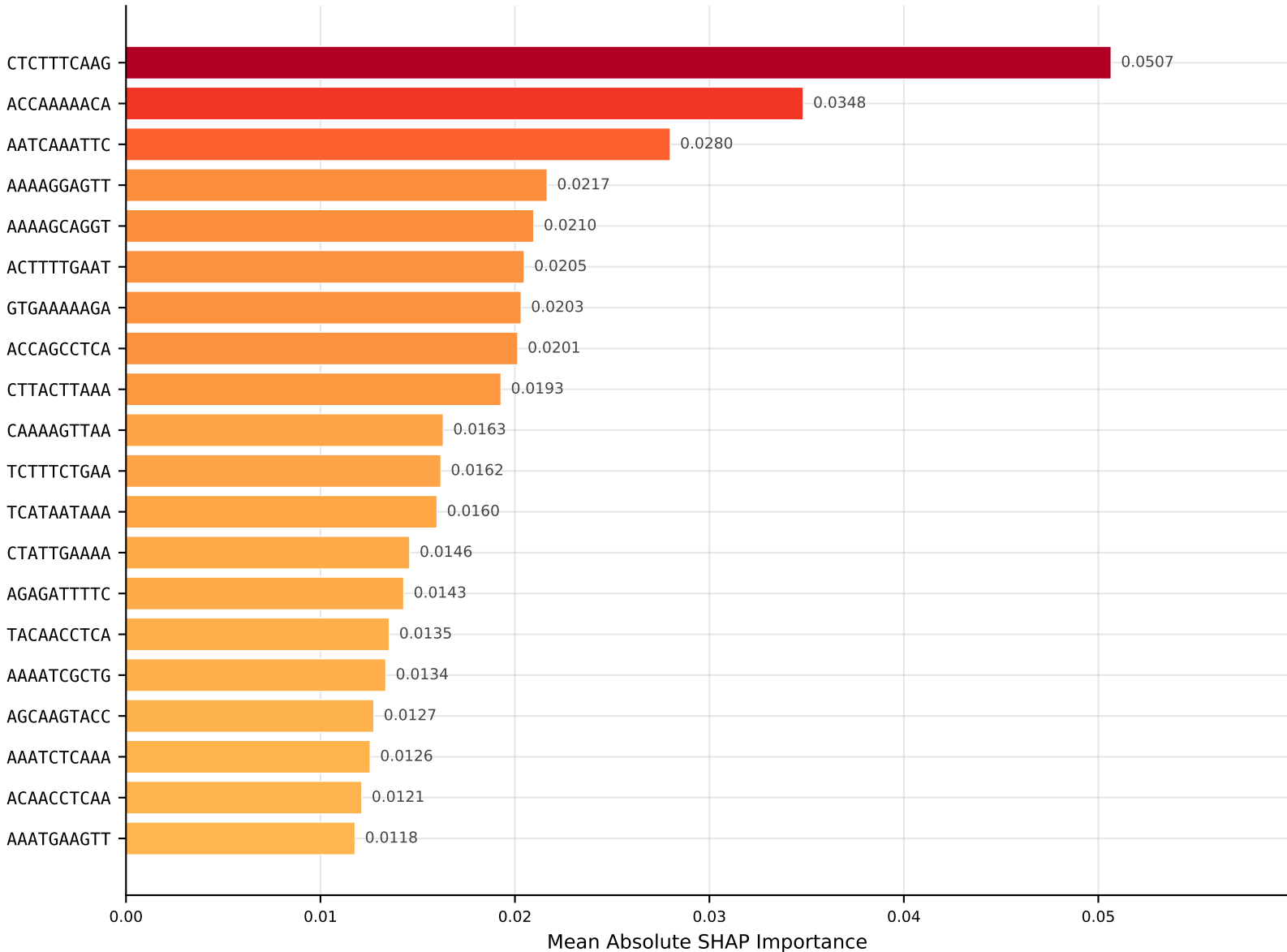


SHAP Feature Importance — Ertapenem

Top 20 k-mer predictors

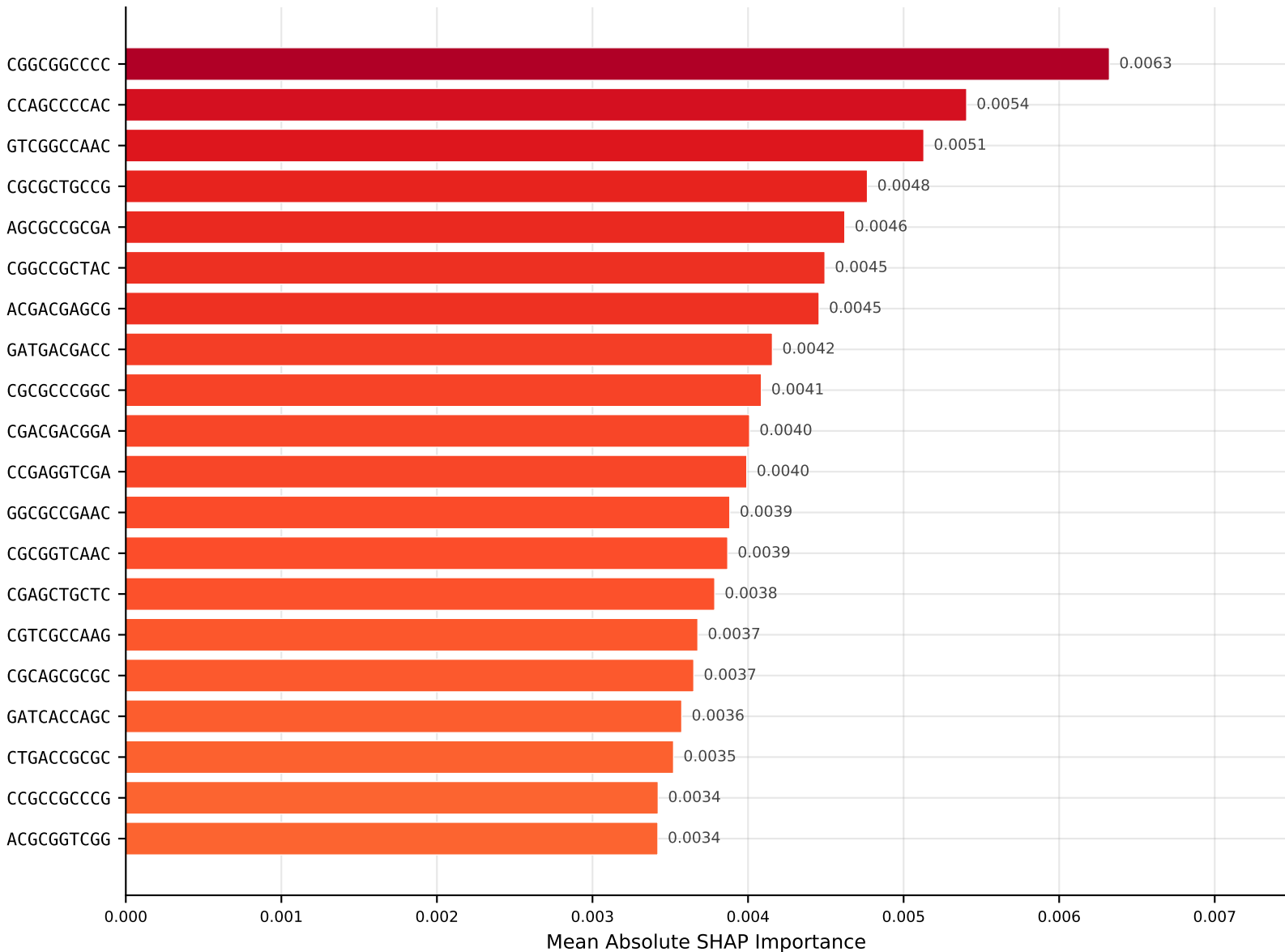


SHAP Feature Importance — Erythromycin Top 20 k-mer predictors



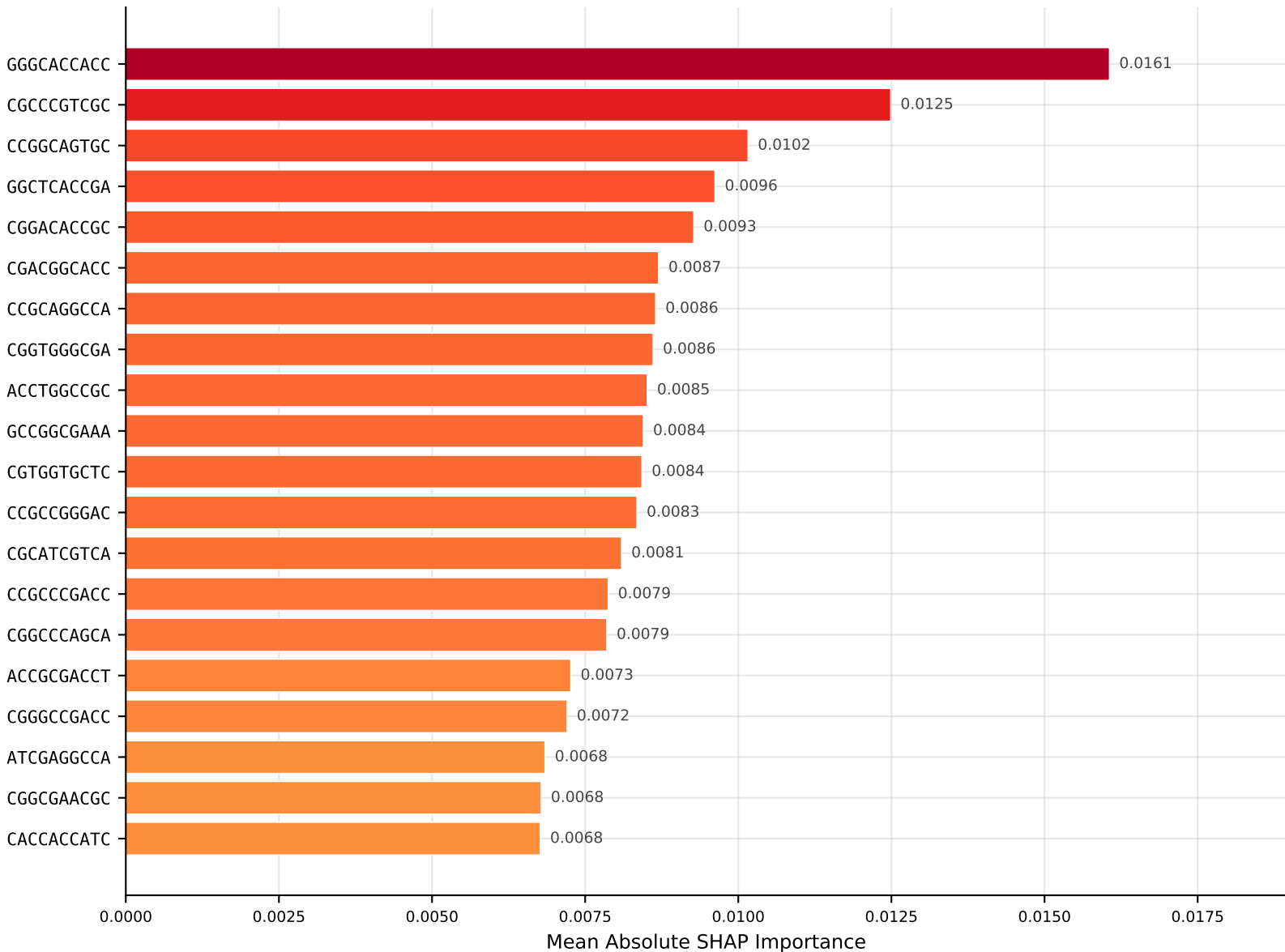
SHAP Feature Importance — Ethambutol

Top 20 k-mer predictors



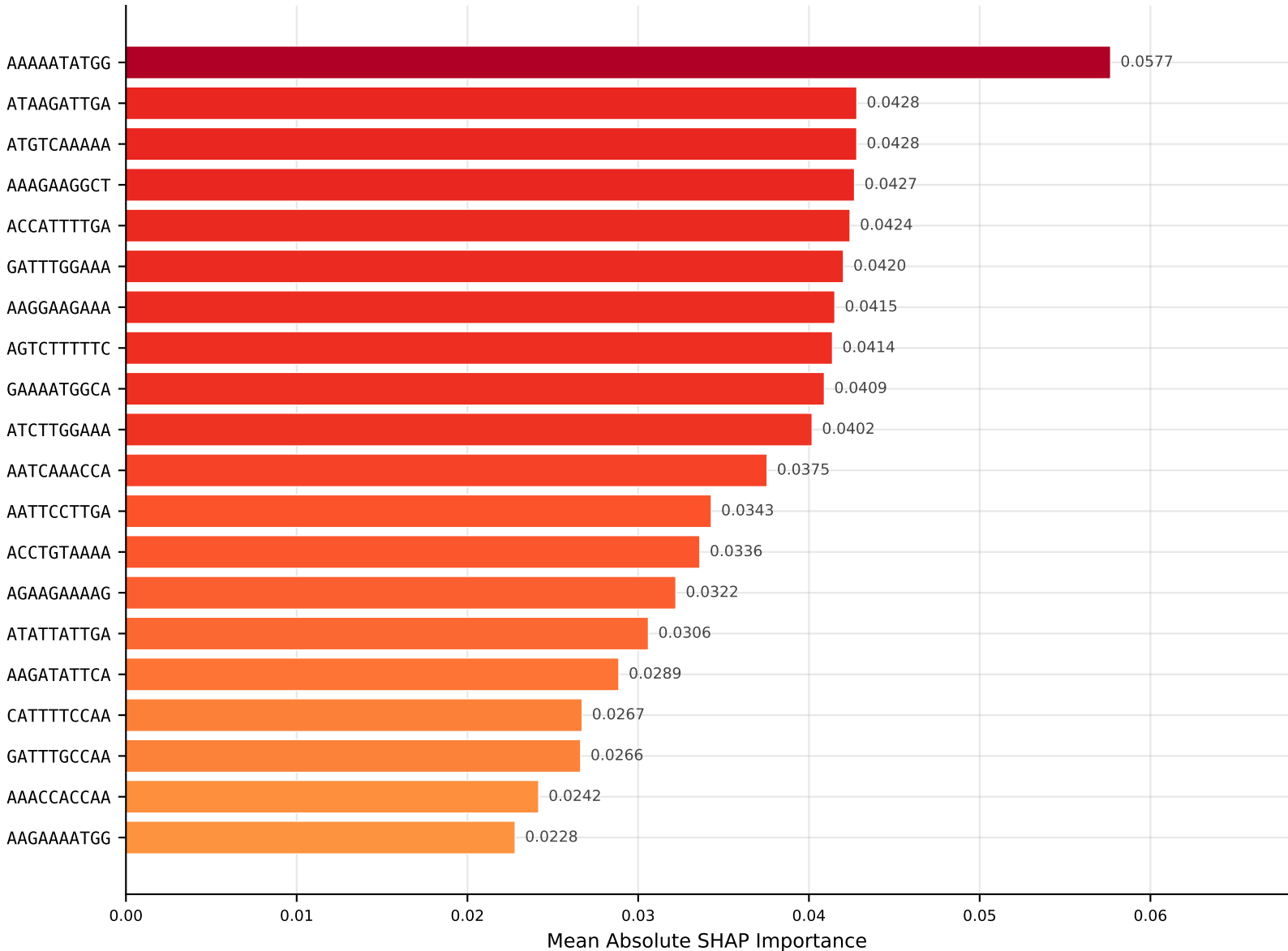
SHAP Feature Importance — Ethionamide

Top 20 k-mer predictors



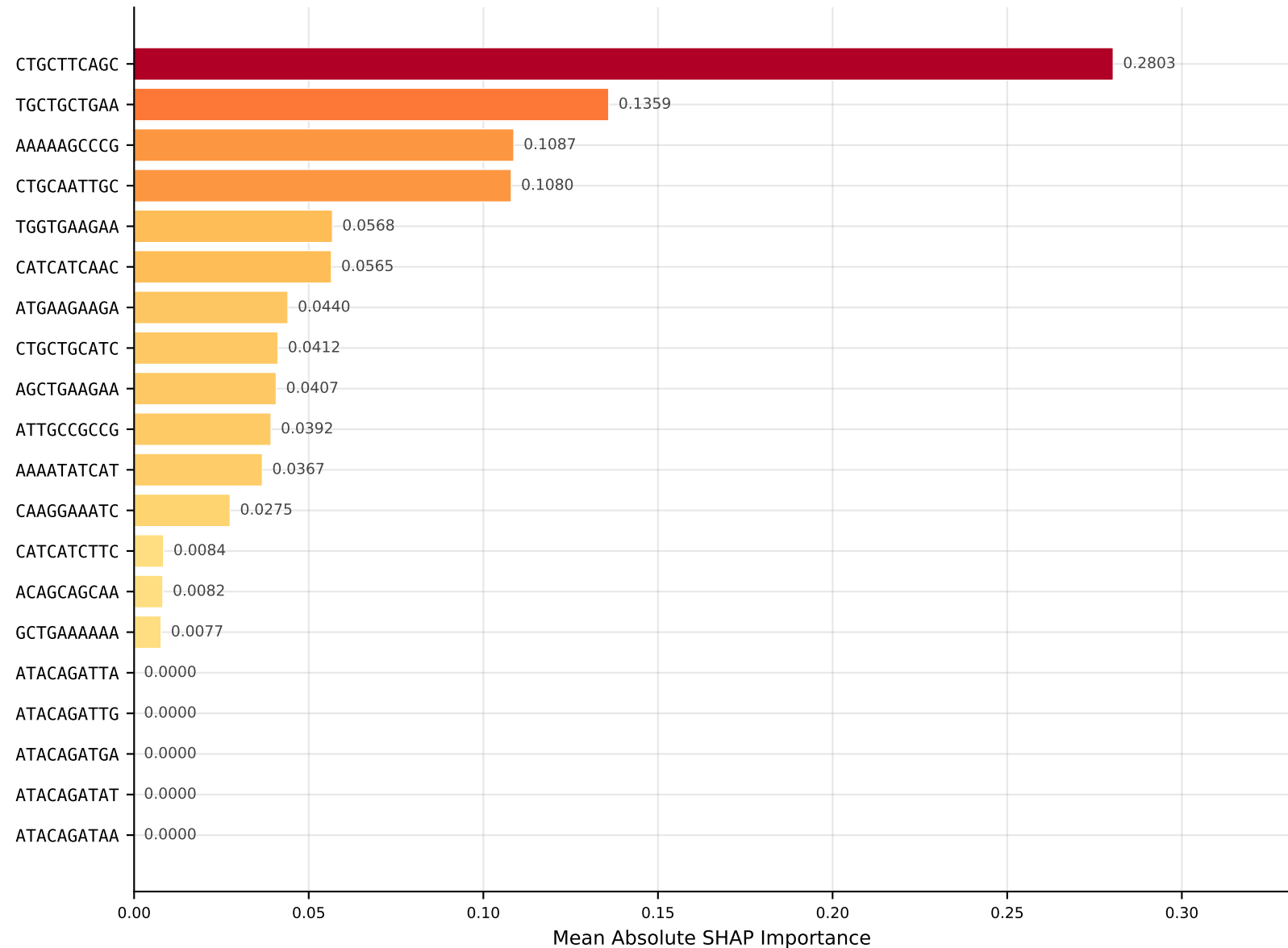
SHAP Feature Importance — Florfenicol

Top 20 k-mer predictors



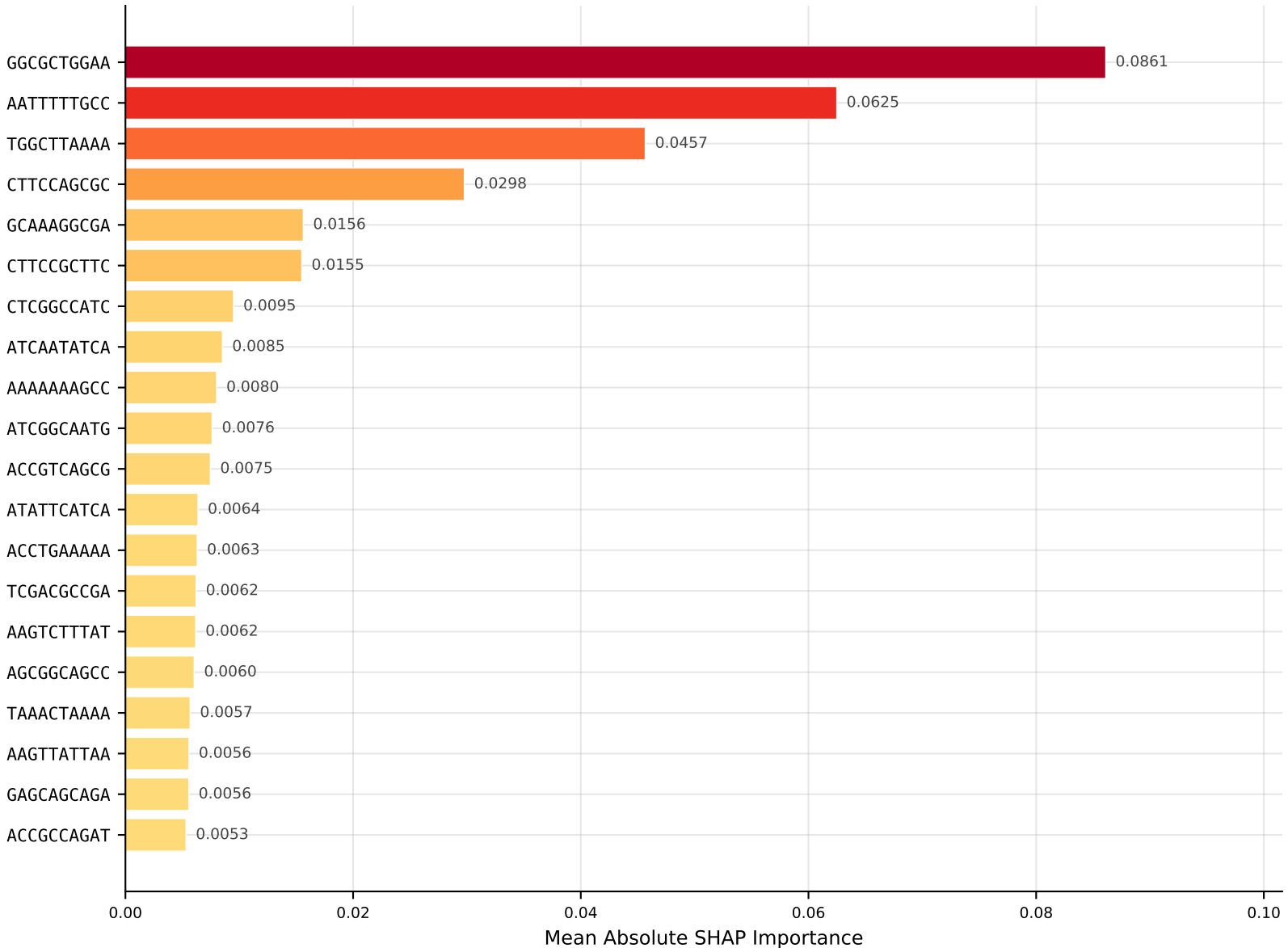
SHAP Feature Importance — Fosfomycin

Top 20 k-mer predictors

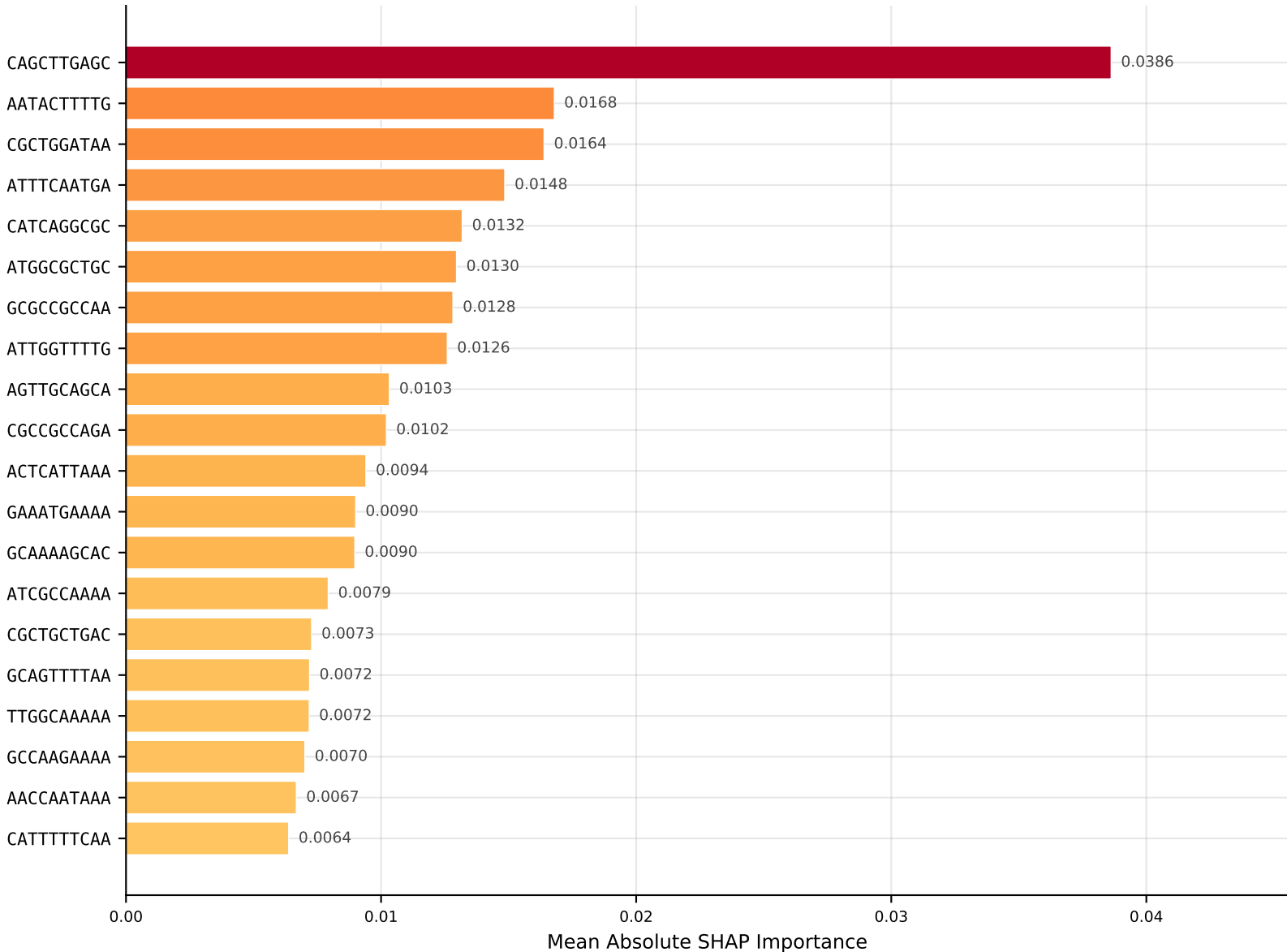


SHAP Feature Importance — Gentamicin

Top 20 k-mer predictors

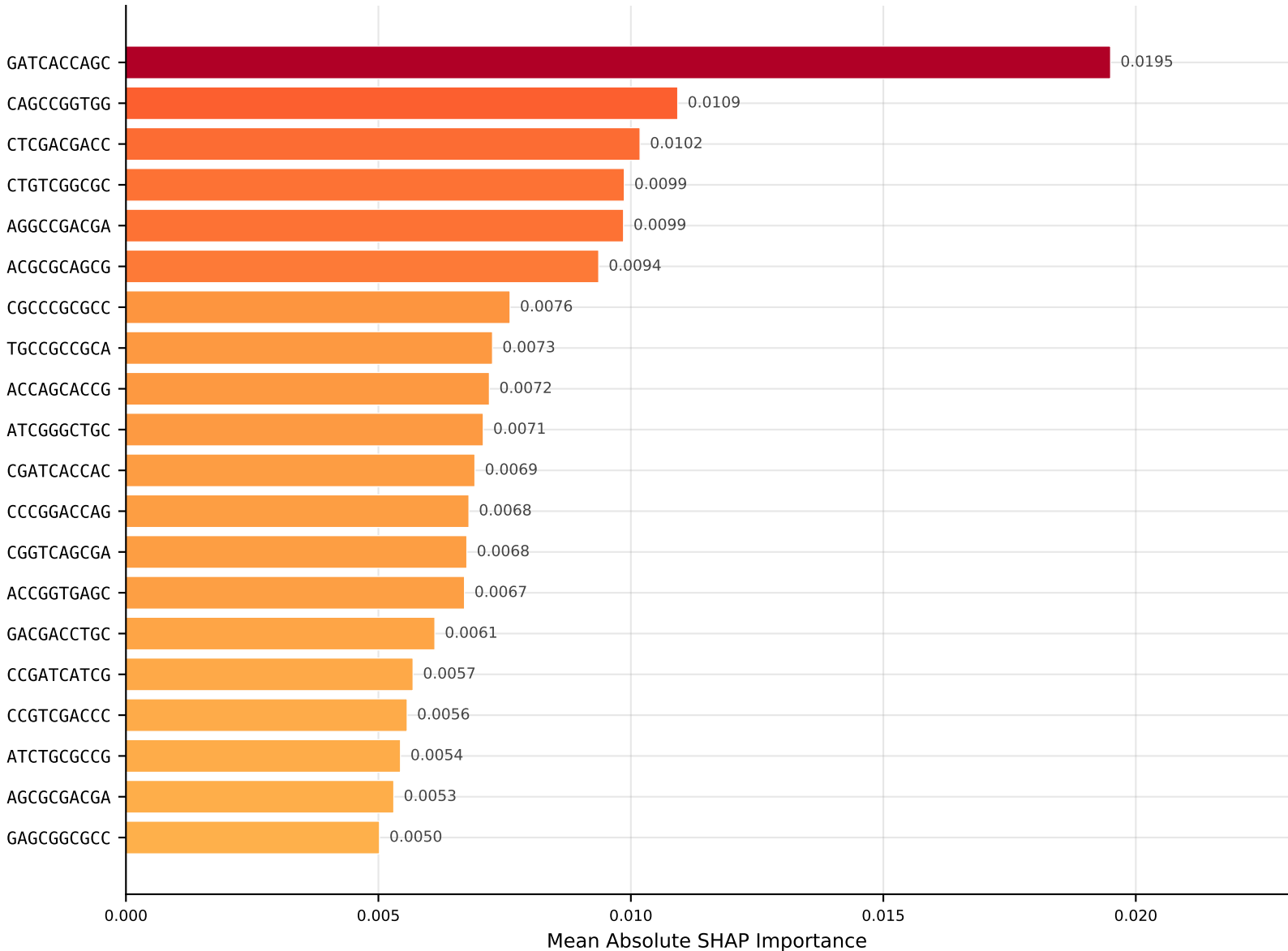


SHAP Feature Importance — Imipenem Top 20 k-mer predictors



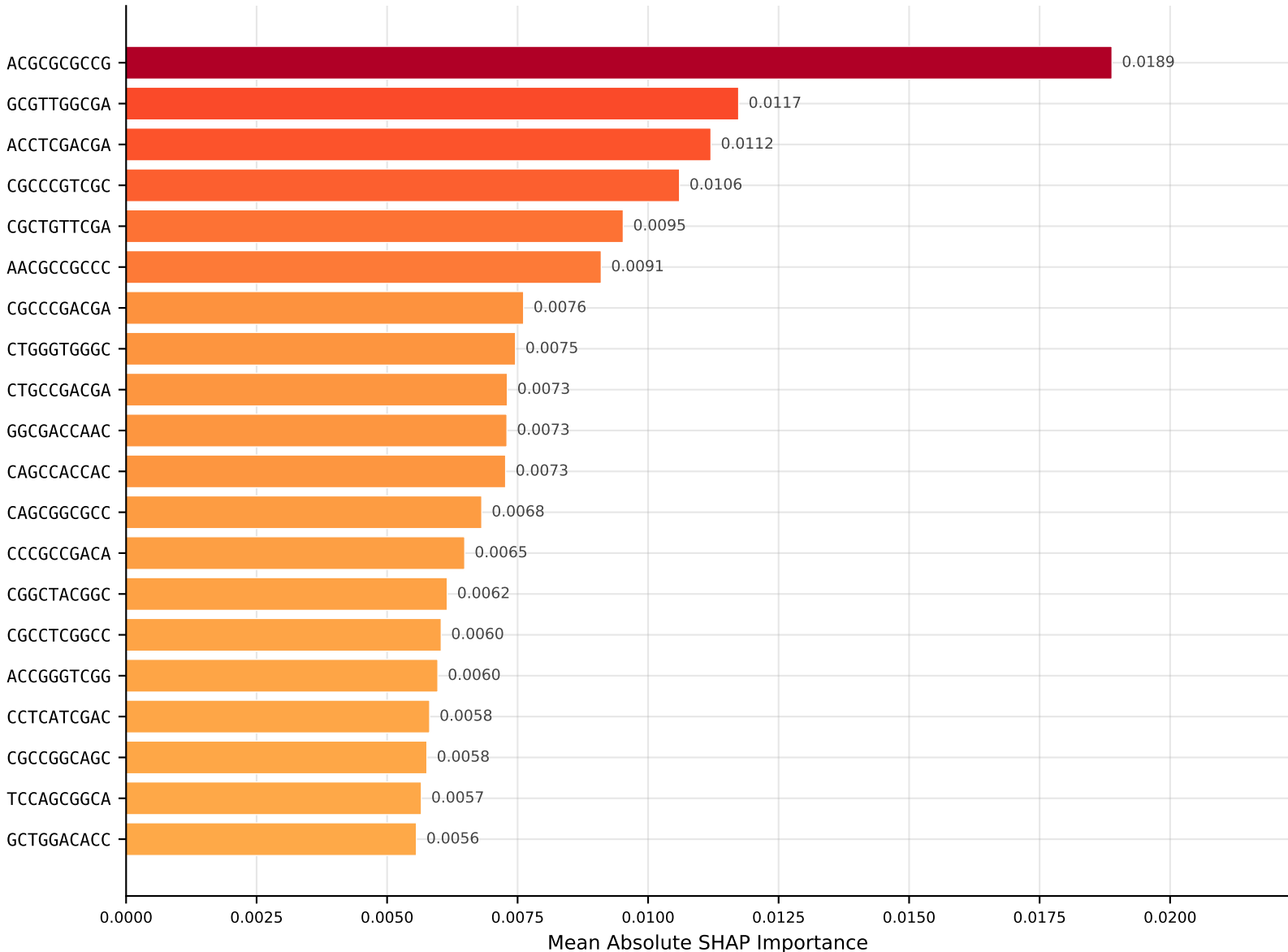
SHAP Feature Importance — Isoniazid

Top 20 k-mer predictors

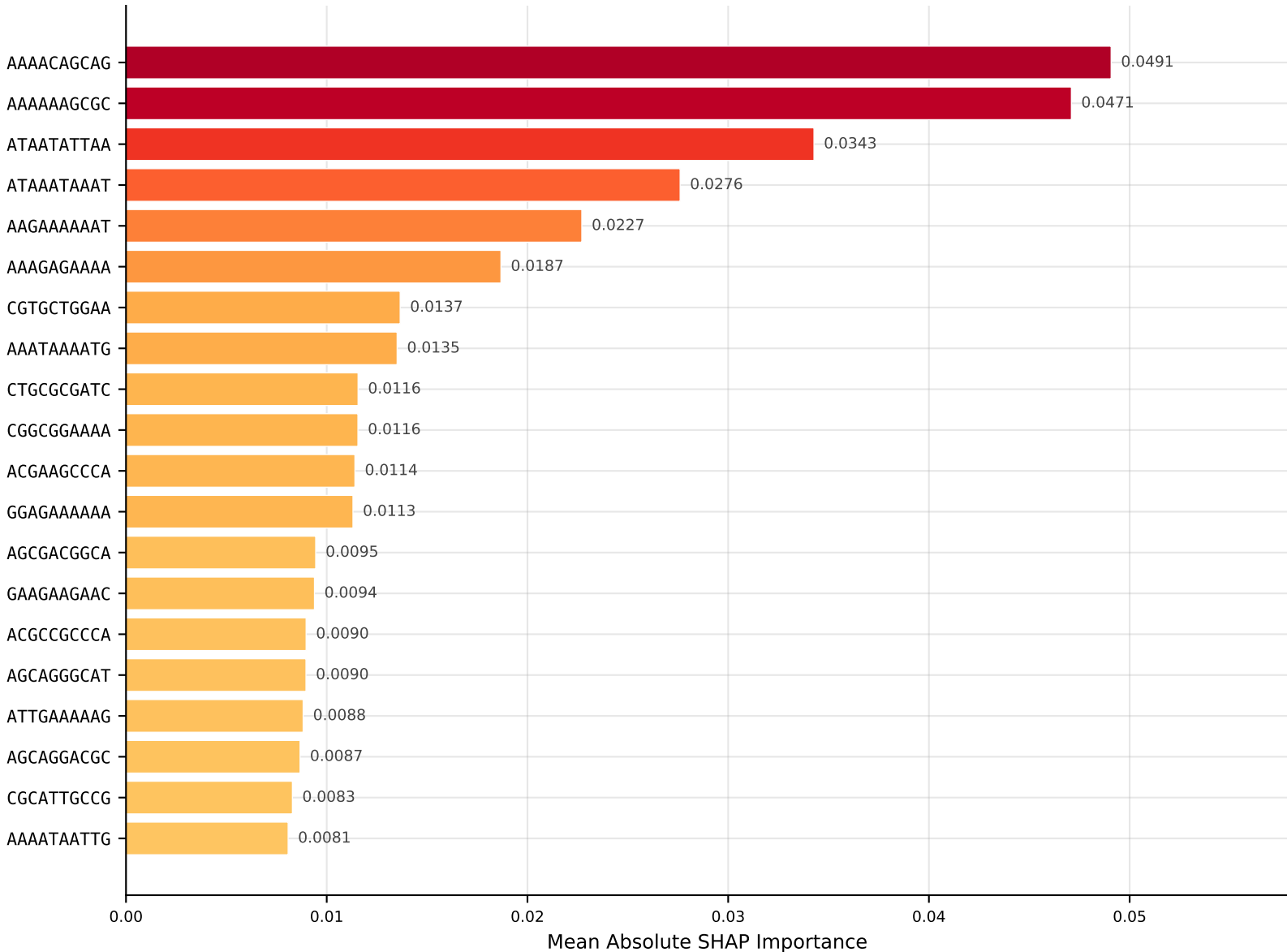


SHAP Feature Importance — Kanamycin

Top 20 k-mer predictors

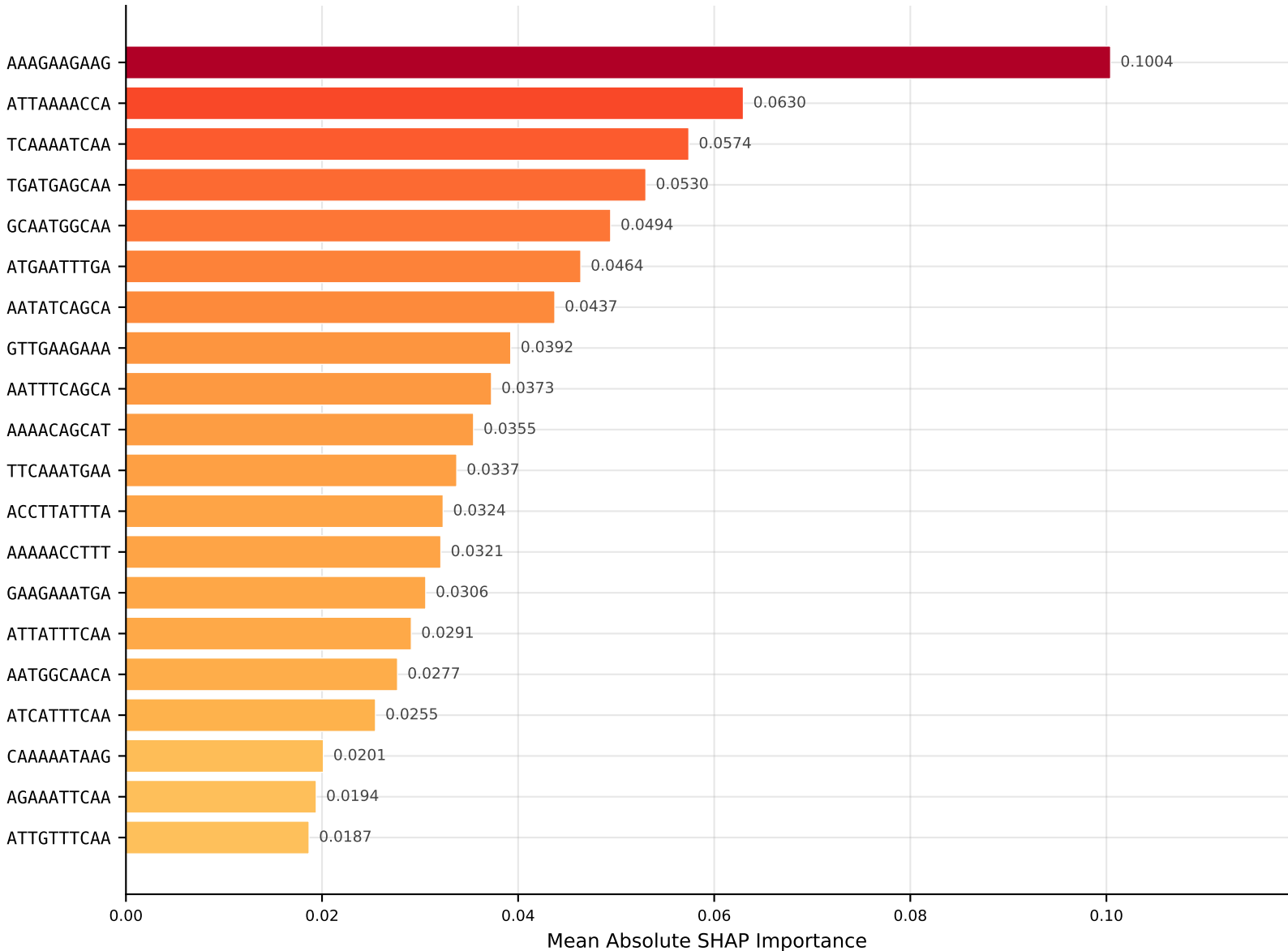


SHAP Feature Importance — Levofloxacin Top 20 k-mer predictors



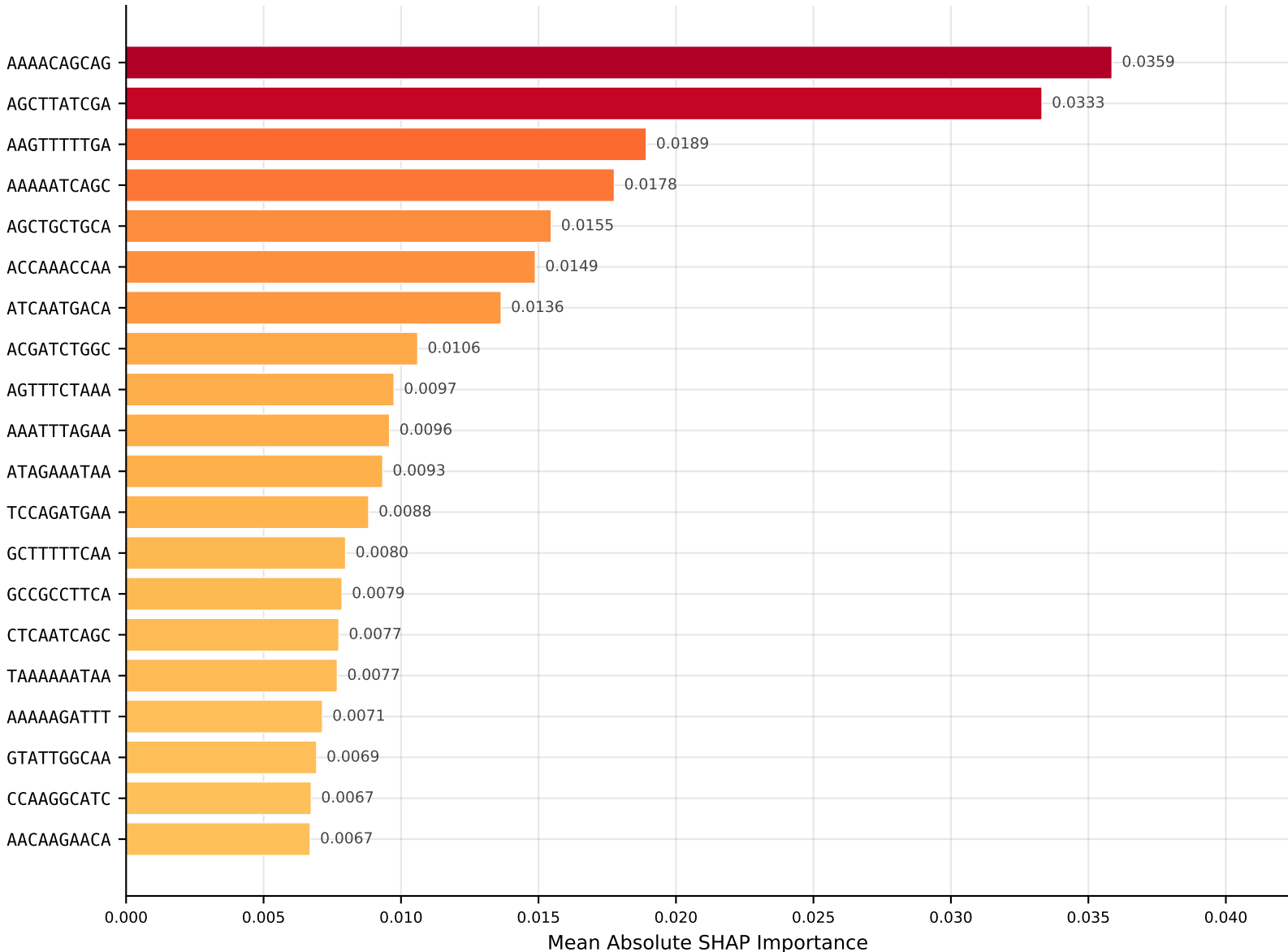
SHAP Feature Importance — Linezolid

Top 20 k-mer predictors

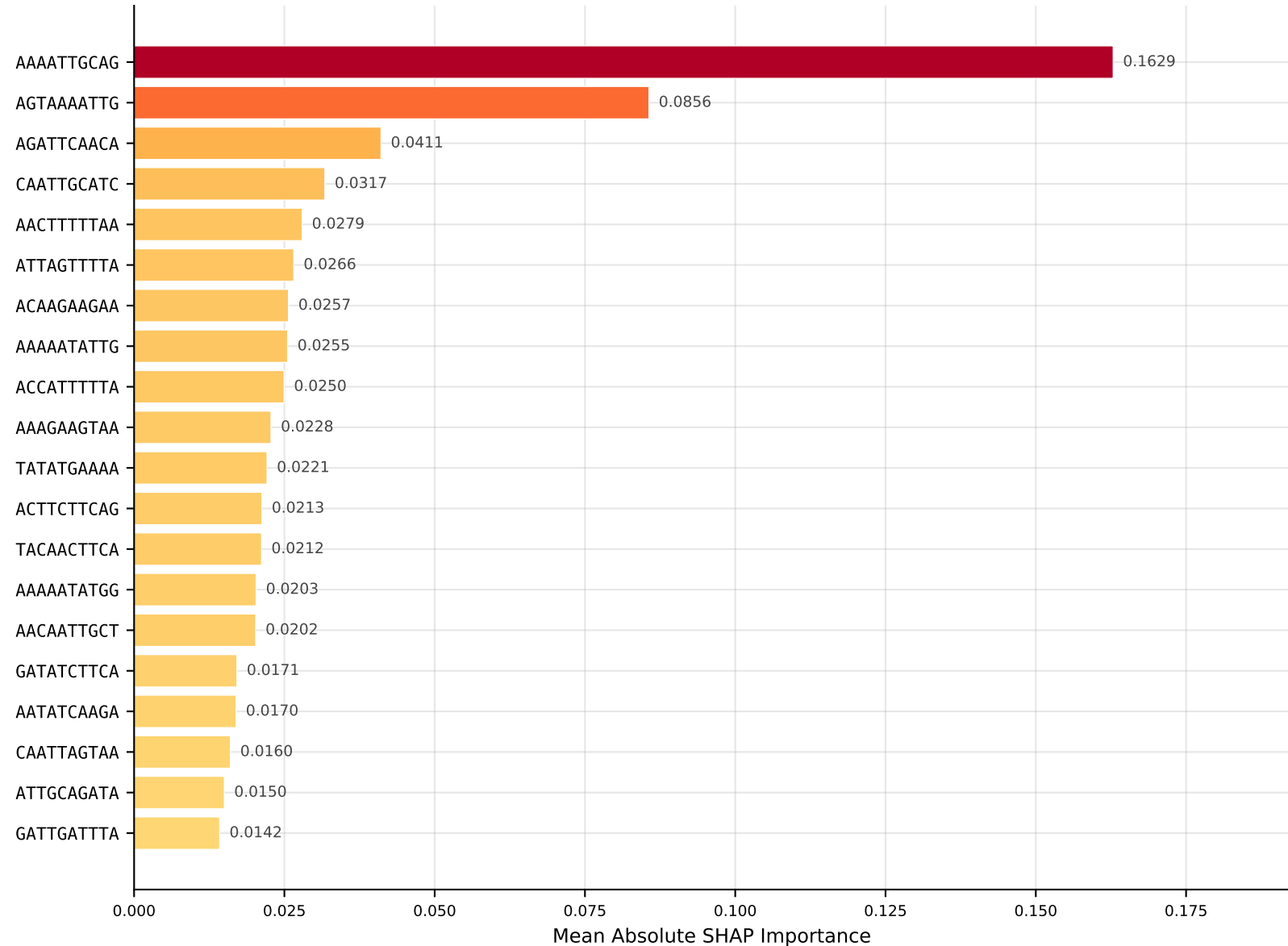


SHAP Feature Importance — Meropenem

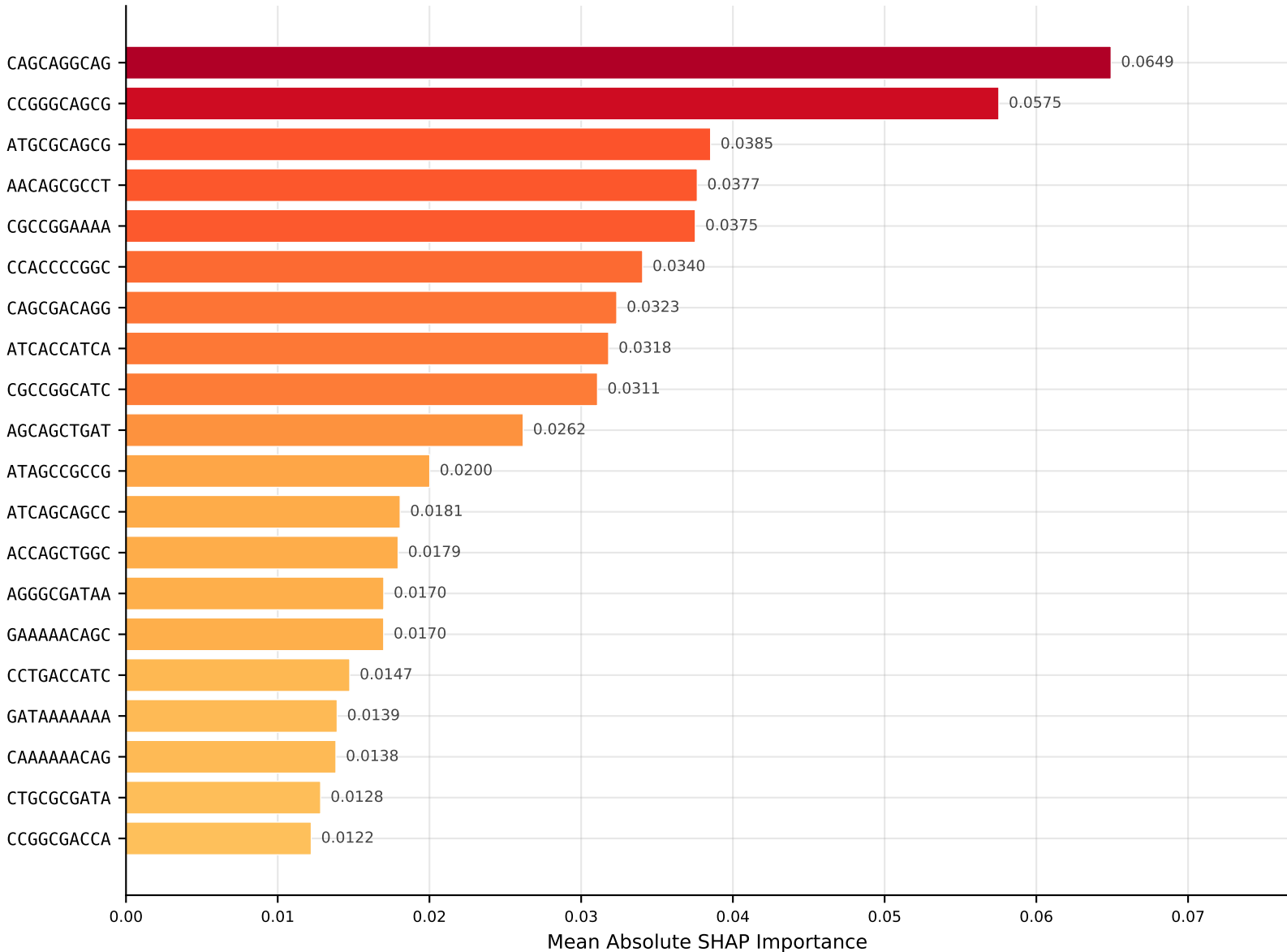
Top 20 k-mer predictors



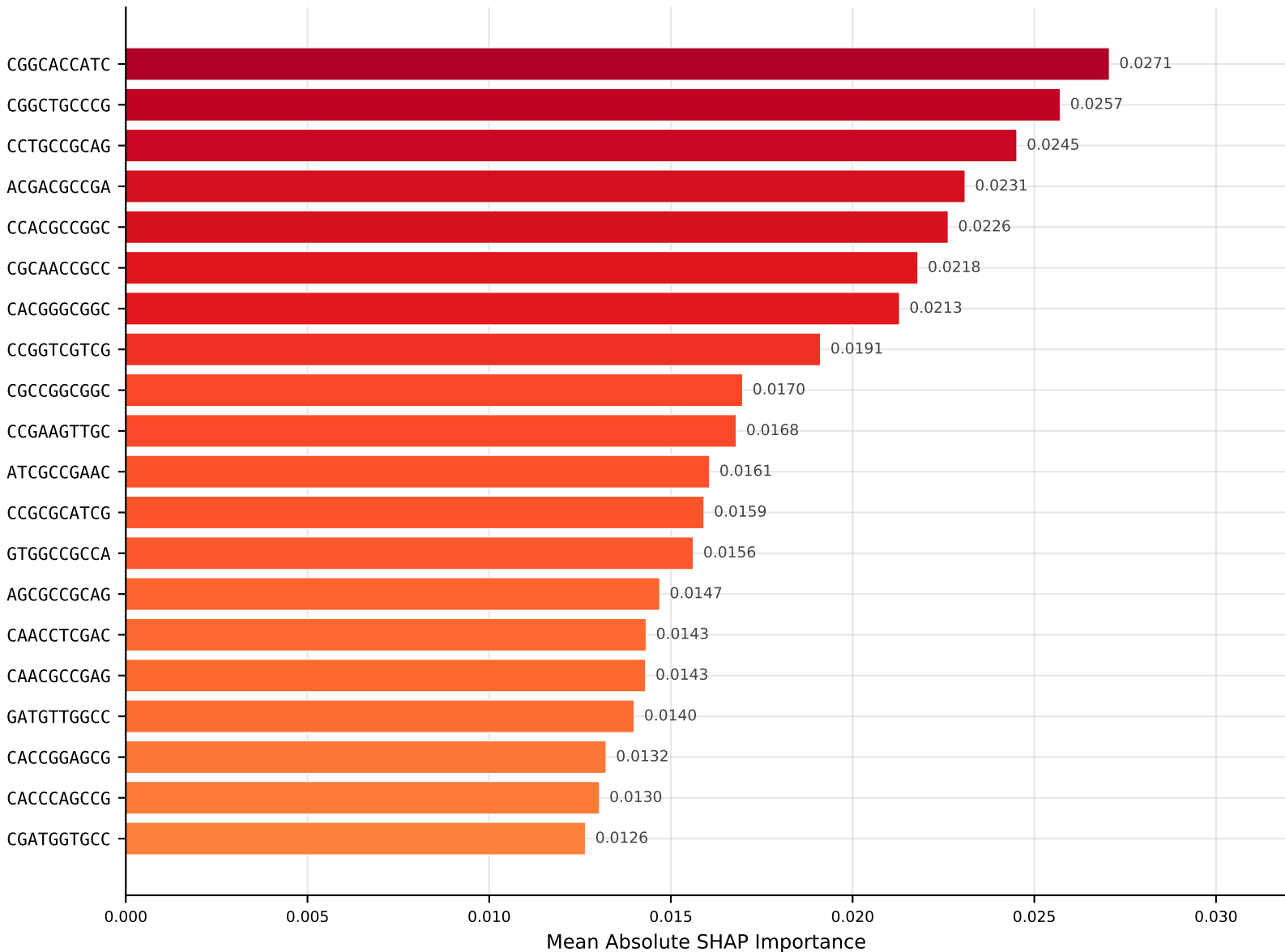
SHAP Feature Importance — Methicillin Top 20 k-mer predictors



SHAP Feature Importance — Minocycline Top 20 k-mer predictors

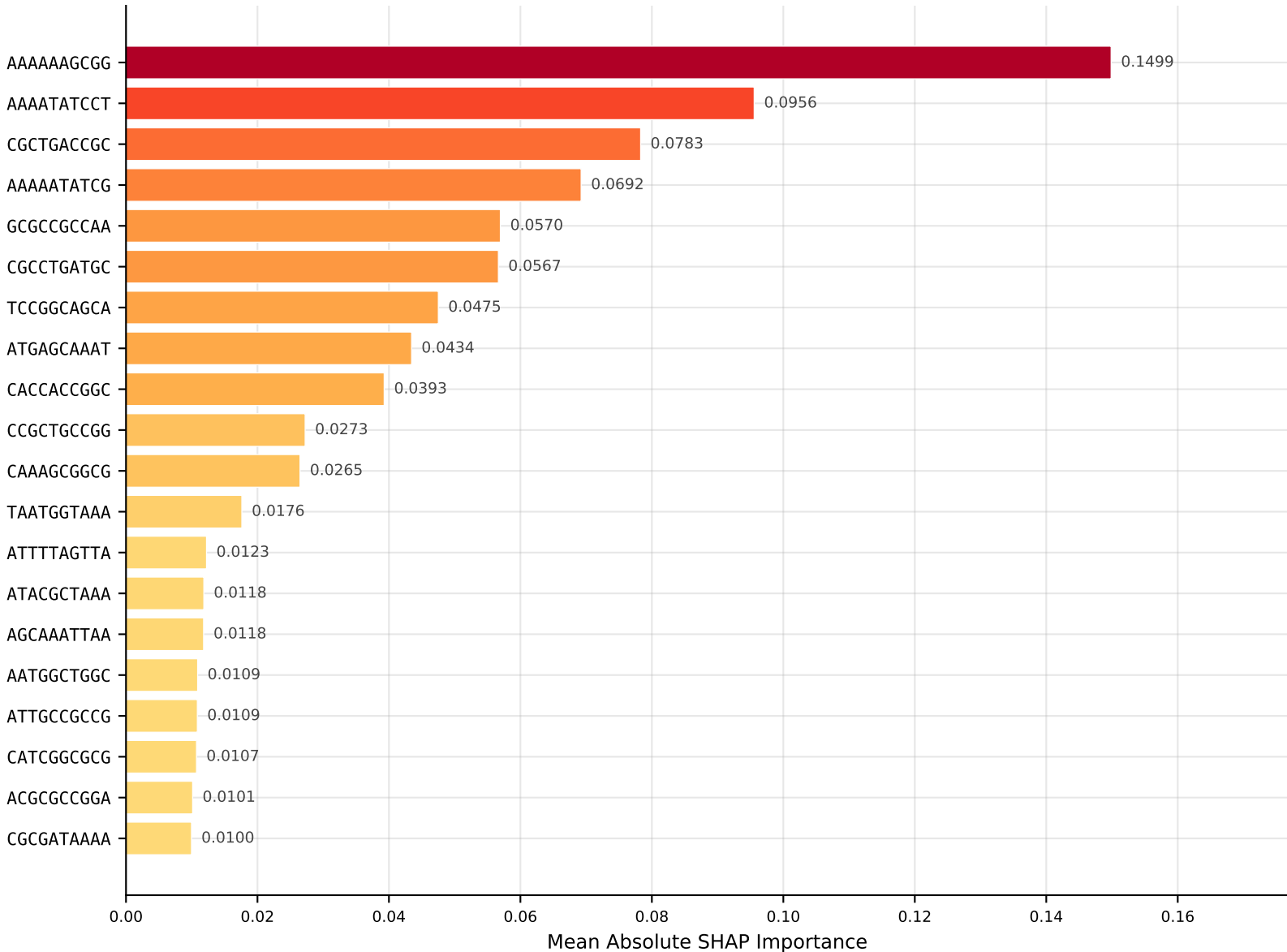


SHAP Feature Importance — Moxifloxacin Top 20 k-mer predictors



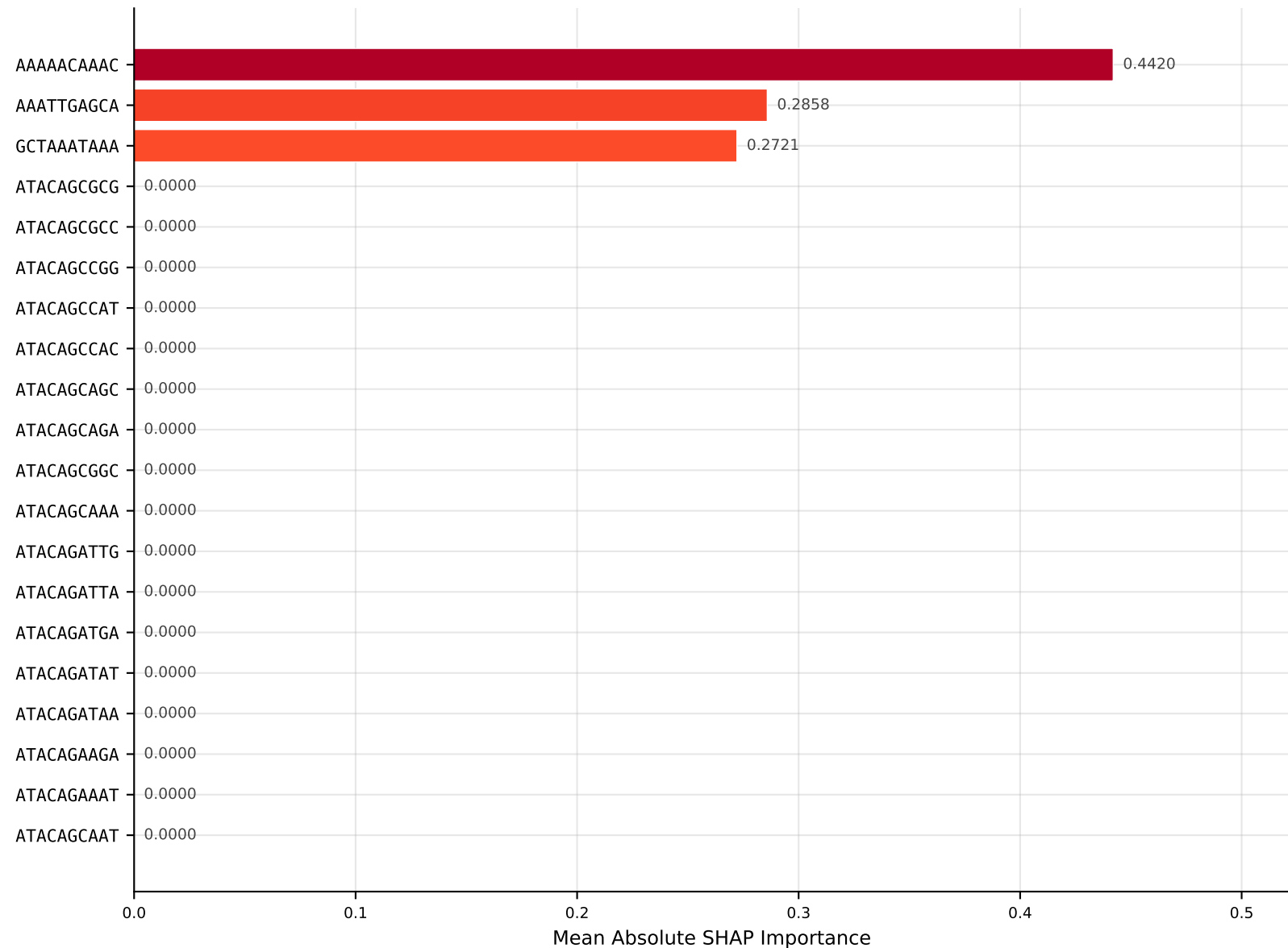
SHAP Feature Importance — Nalidixic Acid

Top 20 k-mer predictors



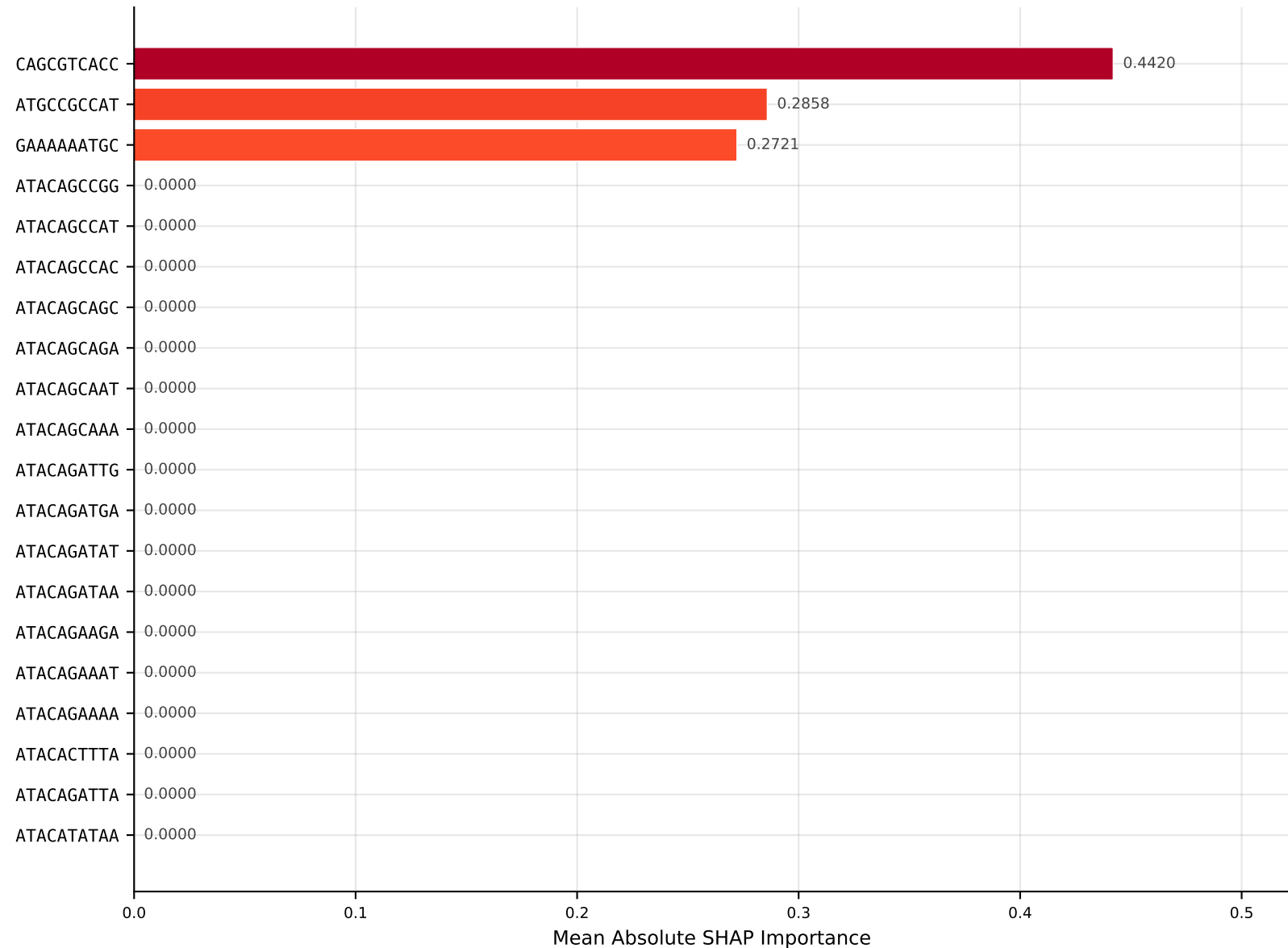
SHAP Feature Importance — Neomycin

Top 20 k-mer predictors

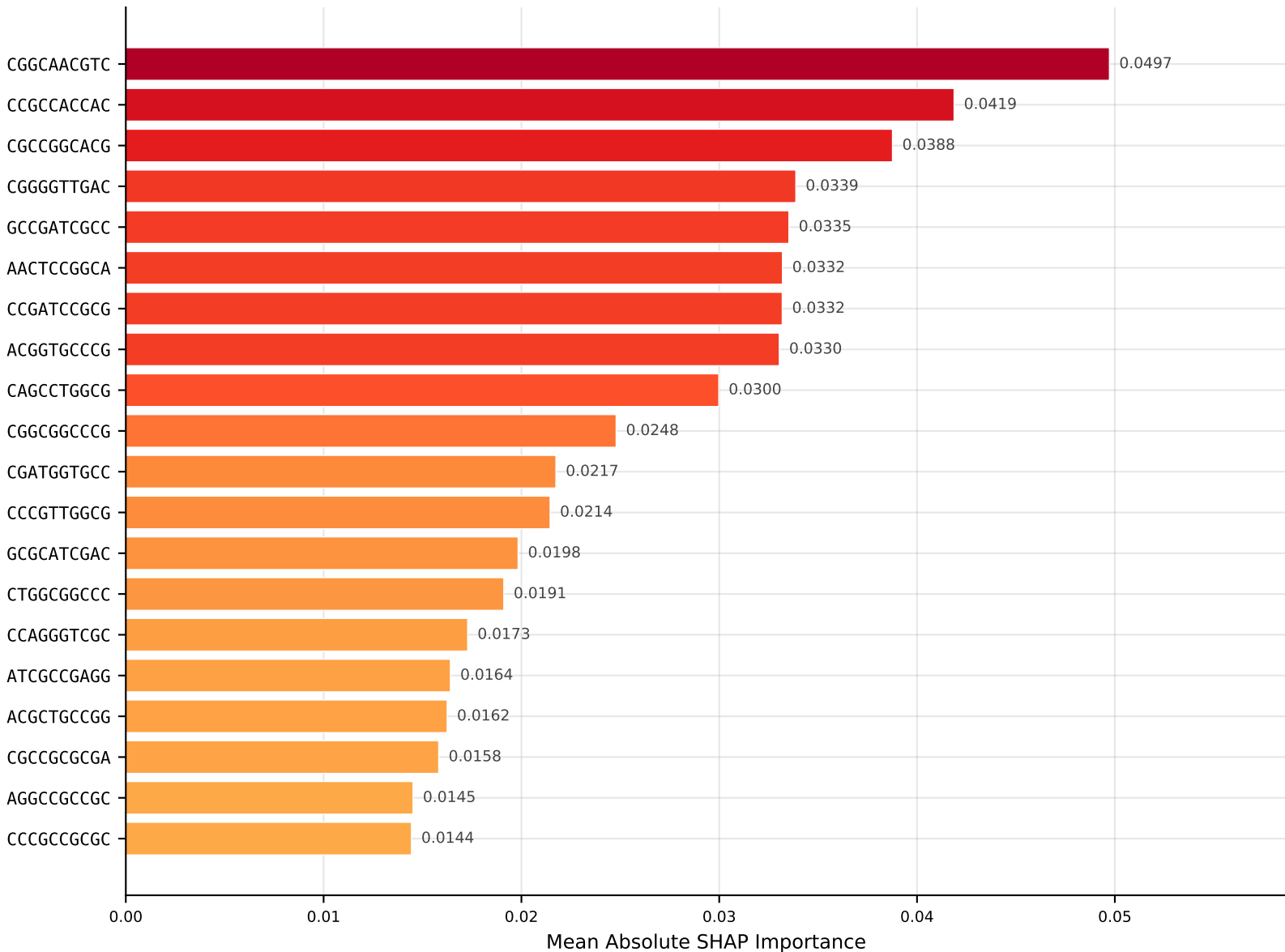


SHAP Feature Importance — Netilmicin

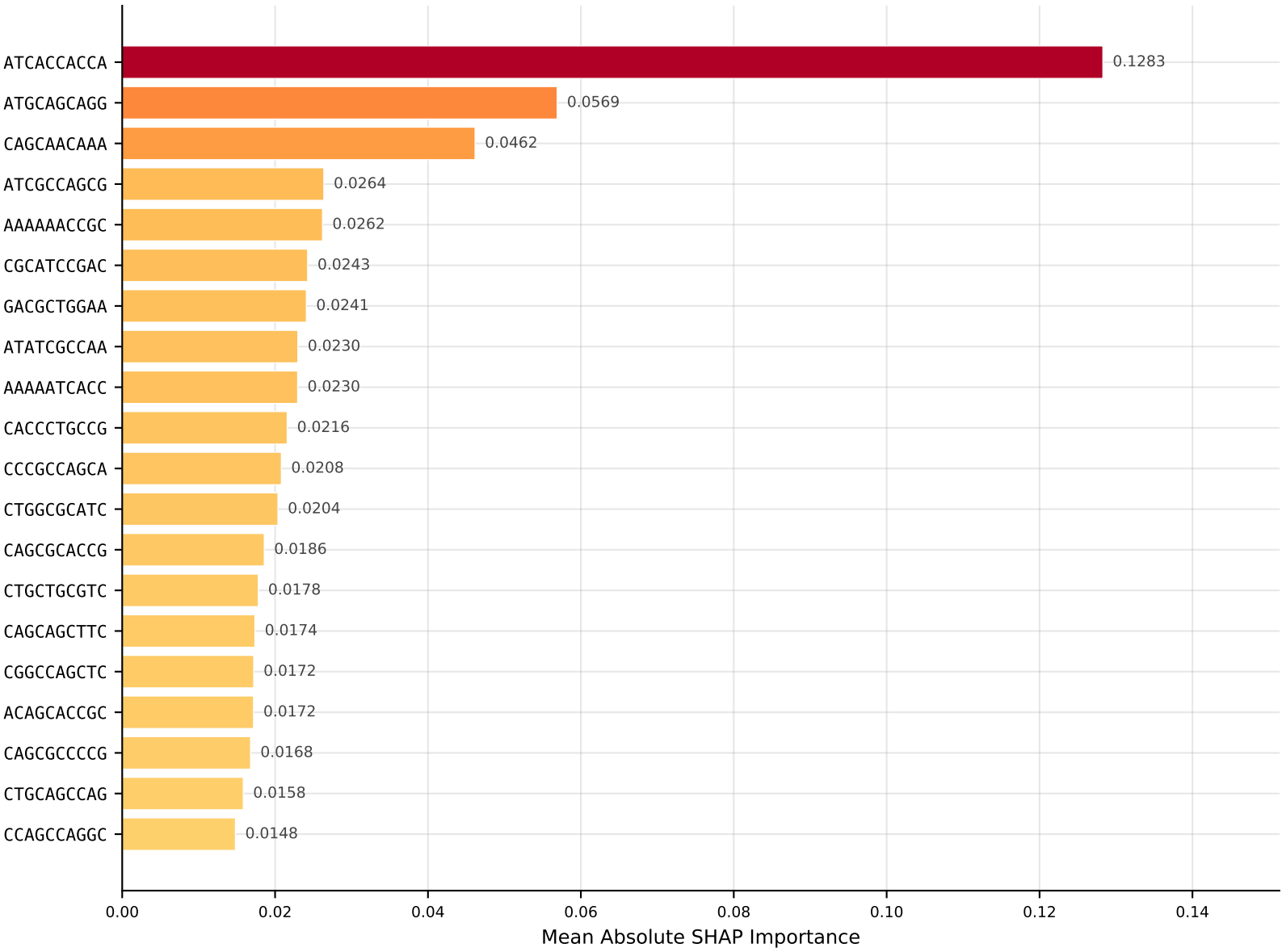
Top 20 k-mer predictors



SHAP Feature Importance — Nicotinamide Top 20 k-mer predictors

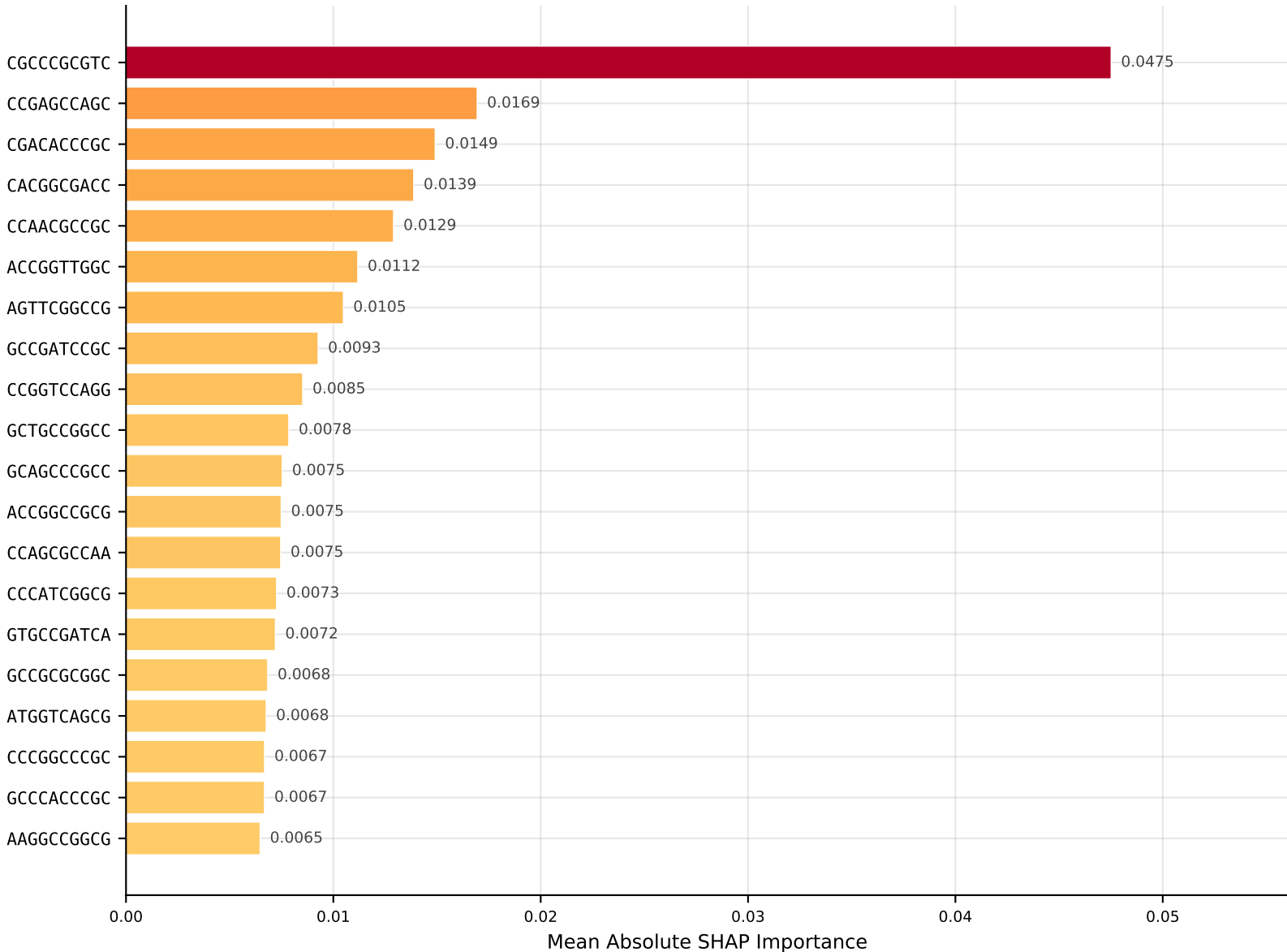


SHAP Feature Importance — Nitrofurantoin
Top 20 k-mer predictors

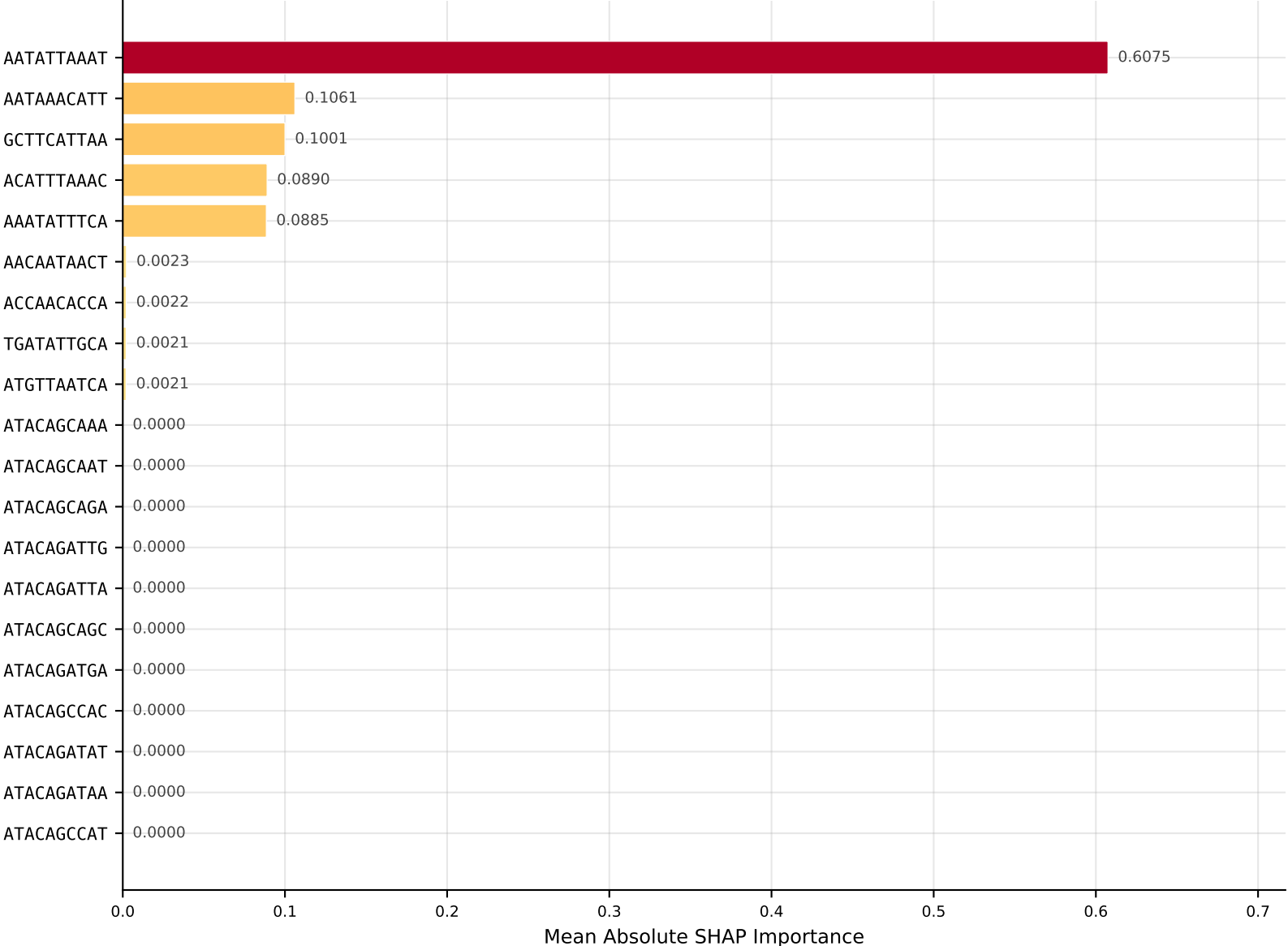


SHAP Feature Importance — Ofloxacin

Top 20 k-mer predictors

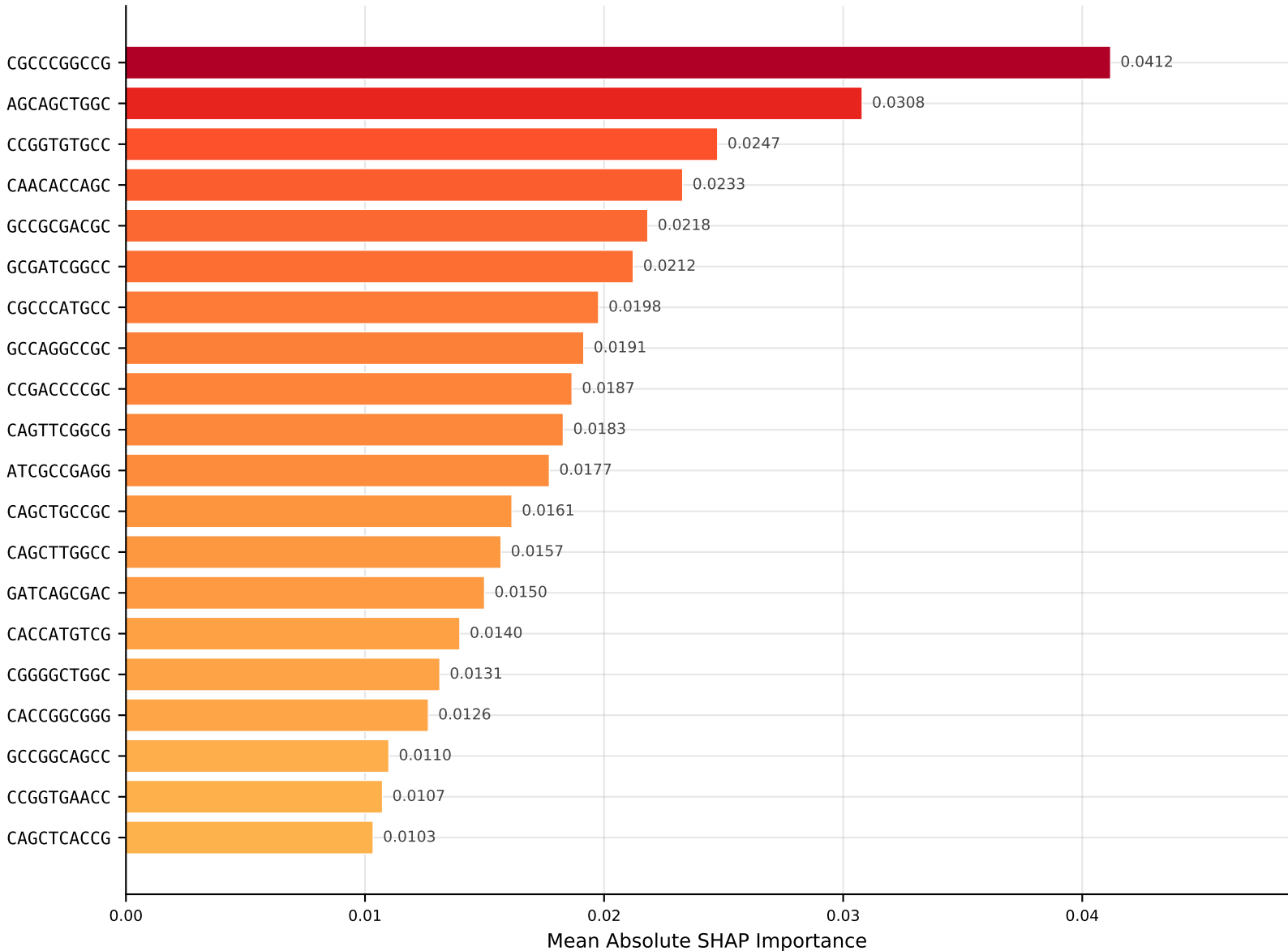


SHAP Feature Importance — Oxacillin
Top 20 k-mer predictors

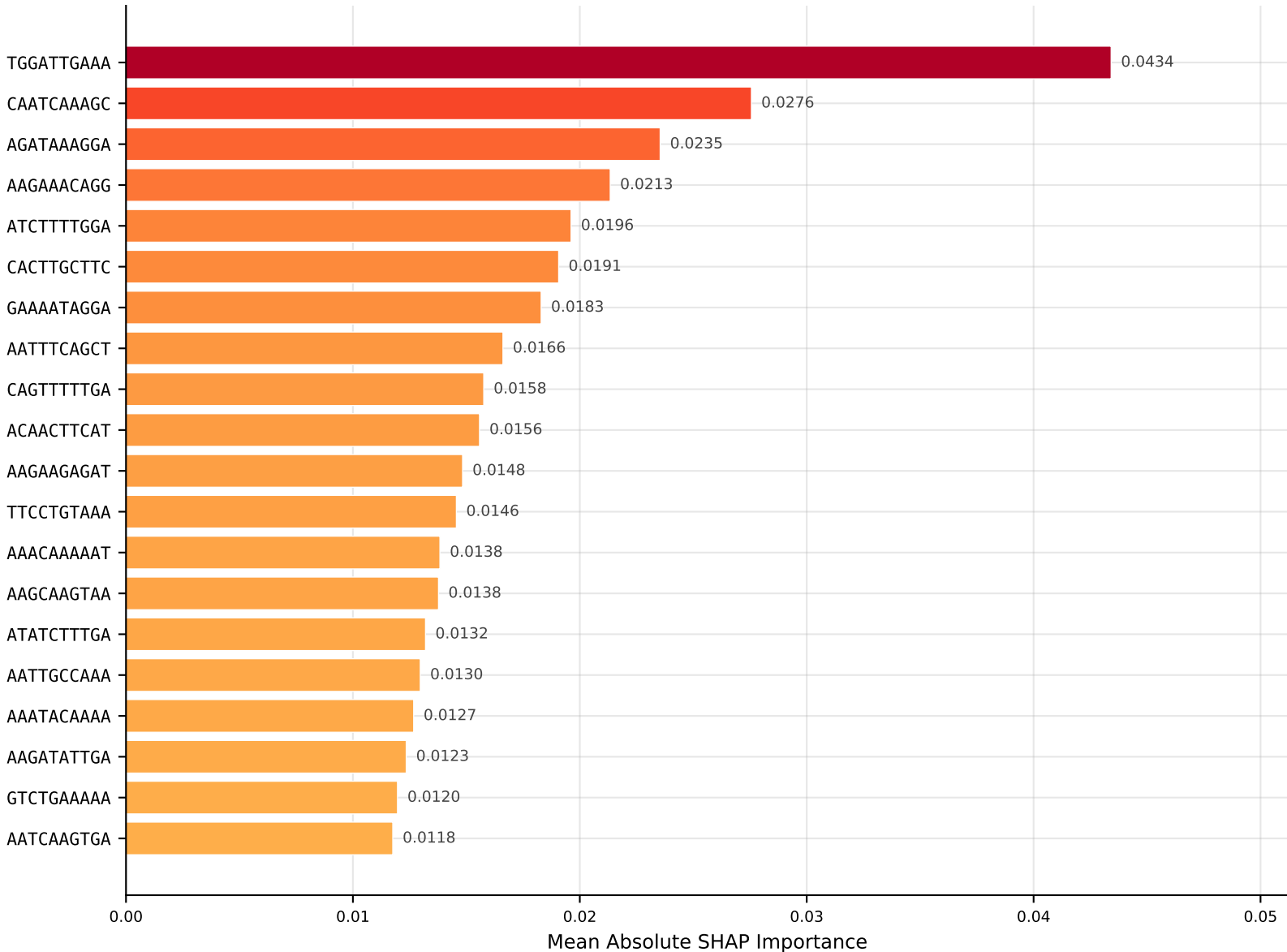


SHAP Feature Importance — Para-Aminosalicylic Acid

Top 20 k-mer predictors

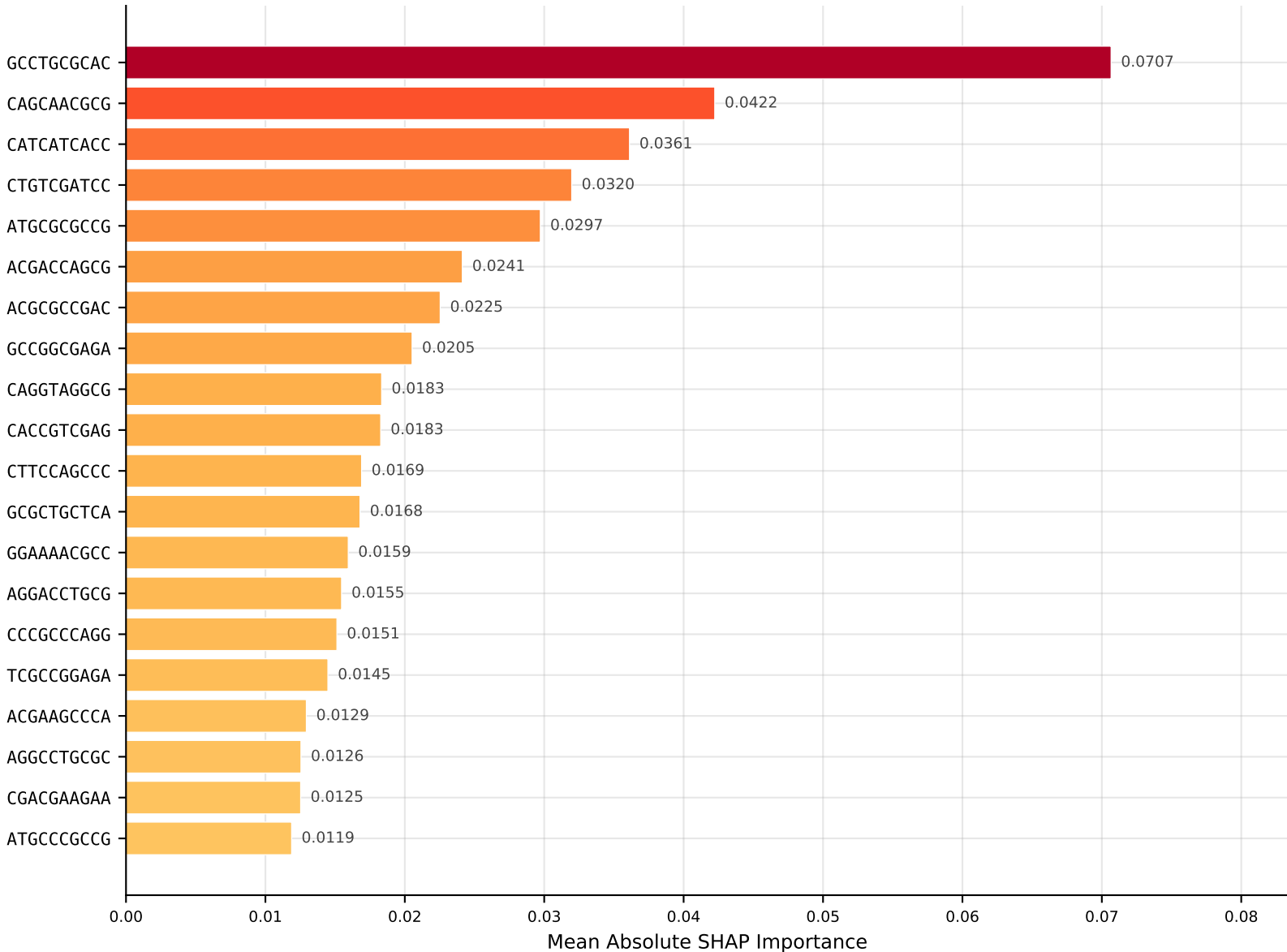


SHAP Feature Importance — Penicillin Top 20 k-mer predictors

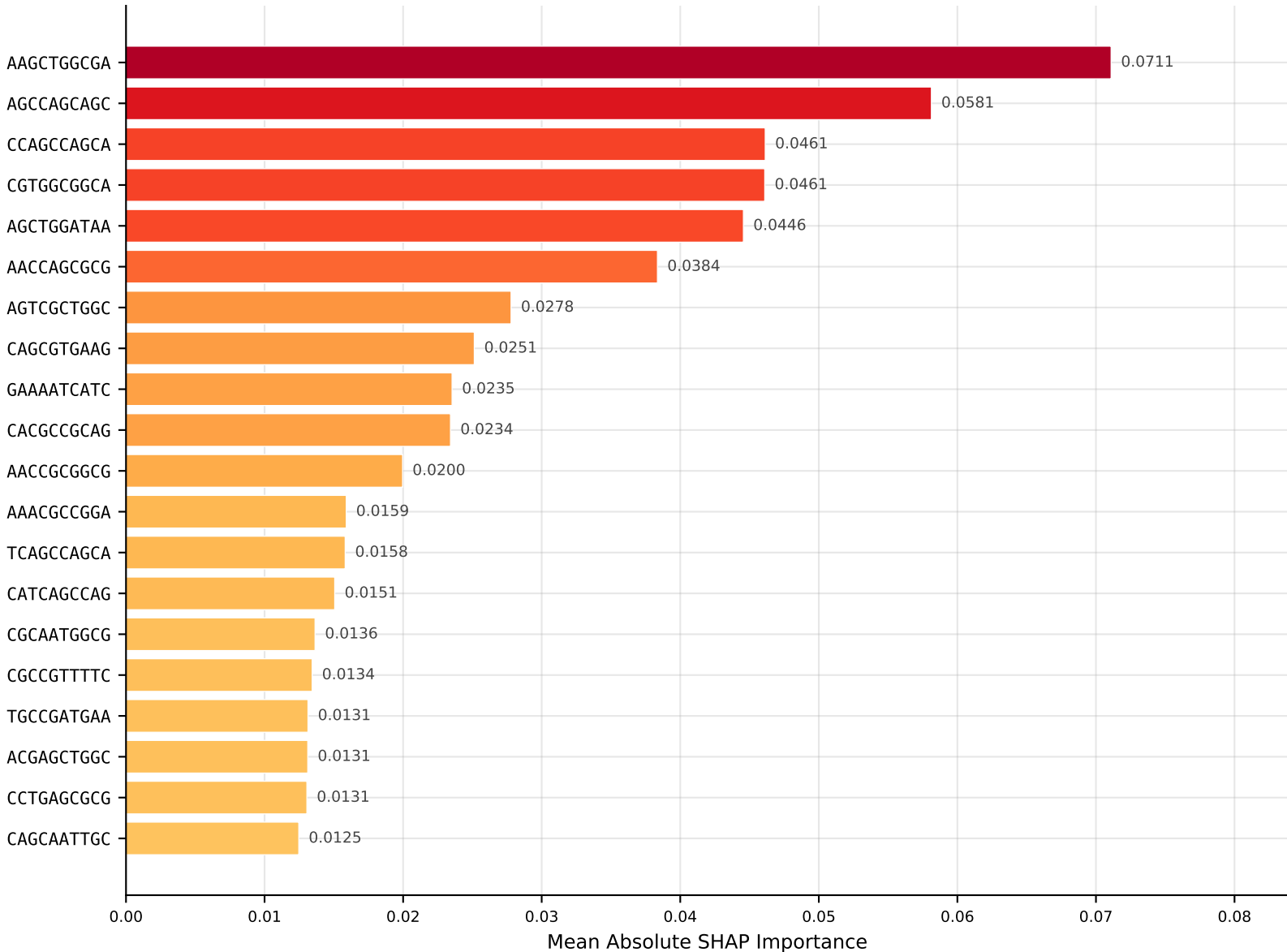


SHAP Feature Importance — Piperacillin

Top 20 k-mer predictors

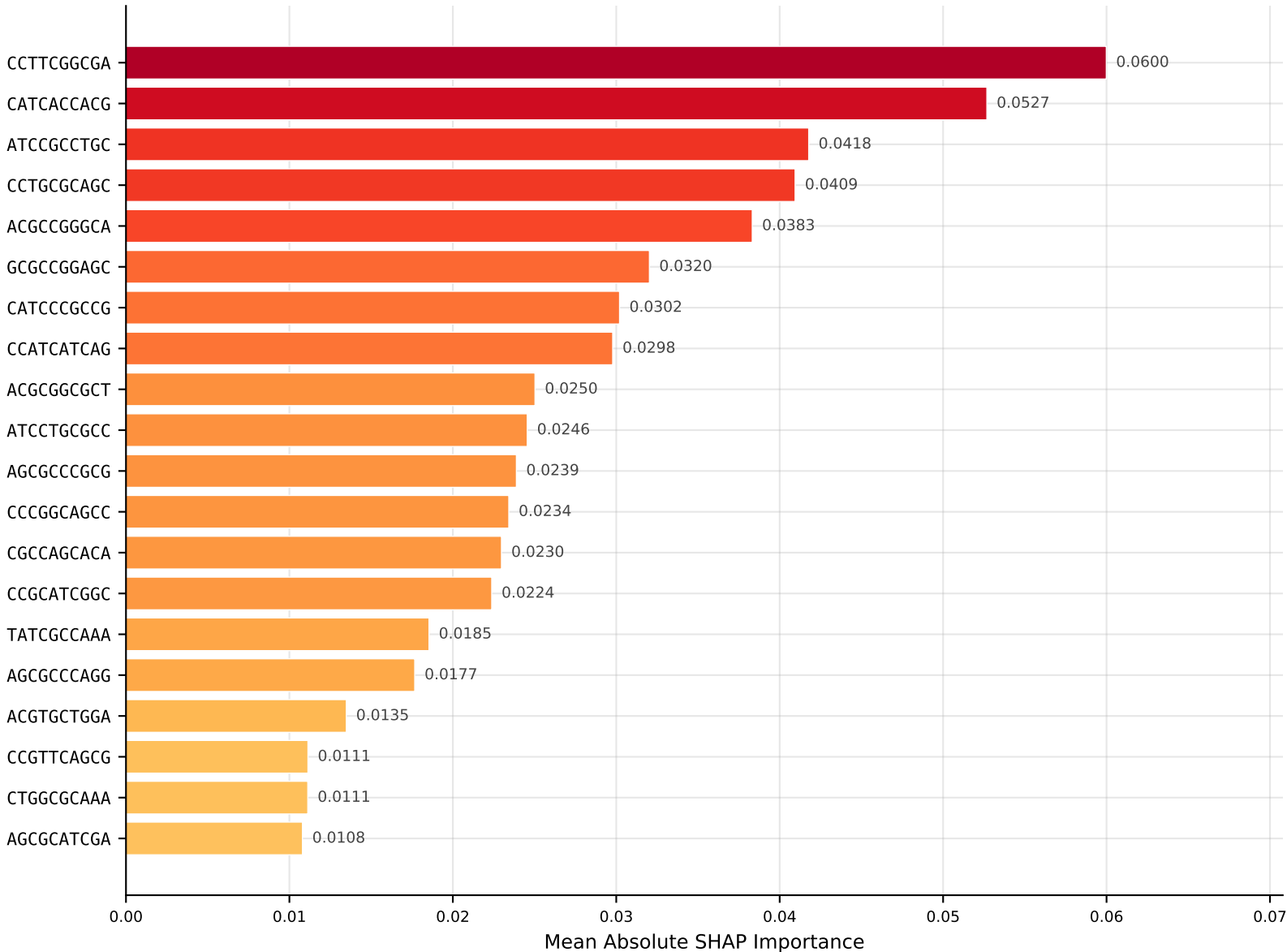


SHAP Feature Importance — Piperacillin/Tazobactam Top 20 k-mer predictors



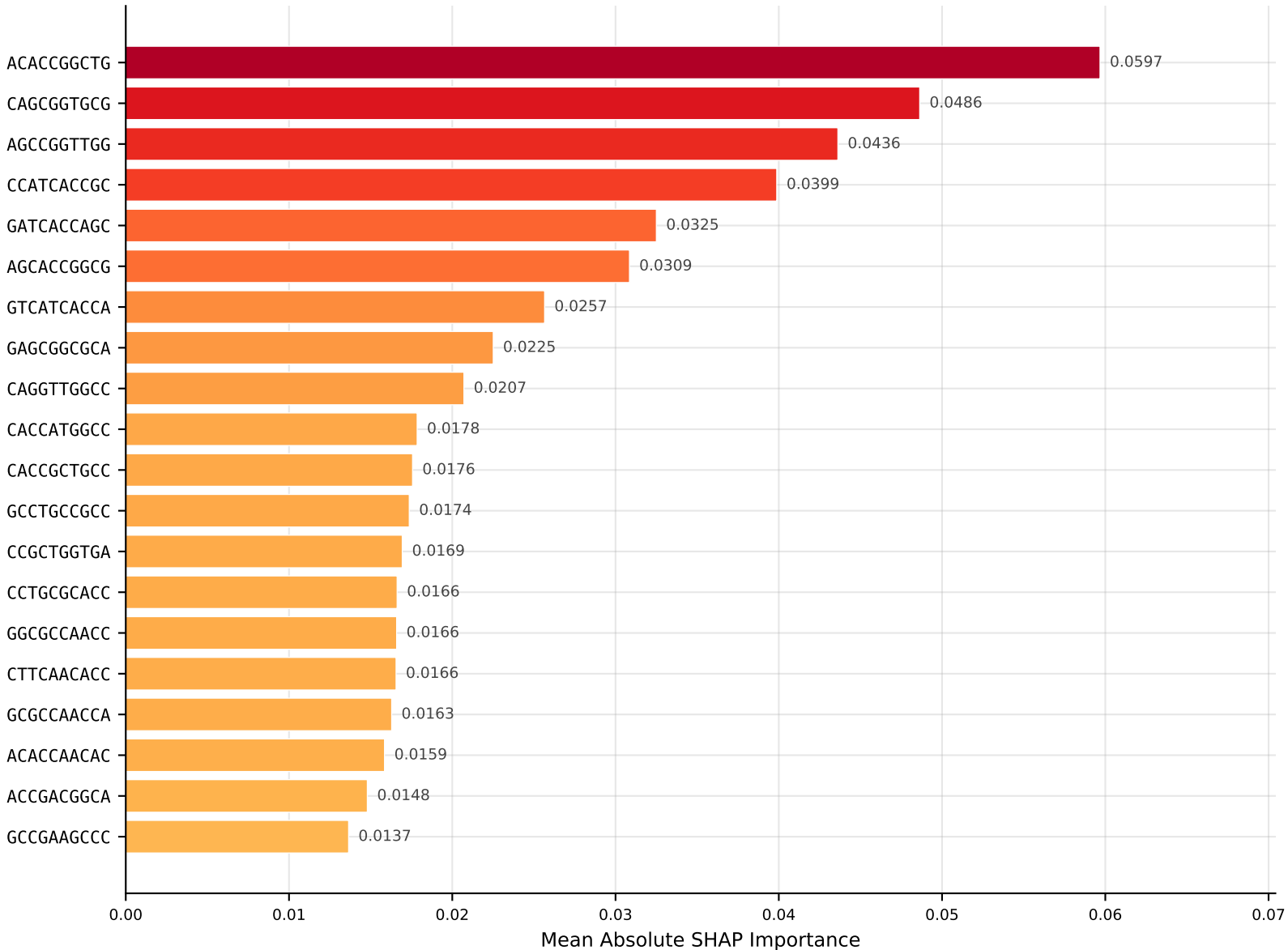
SHAP Feature Importance — Polymyxin B

Top 20 k-mer predictors



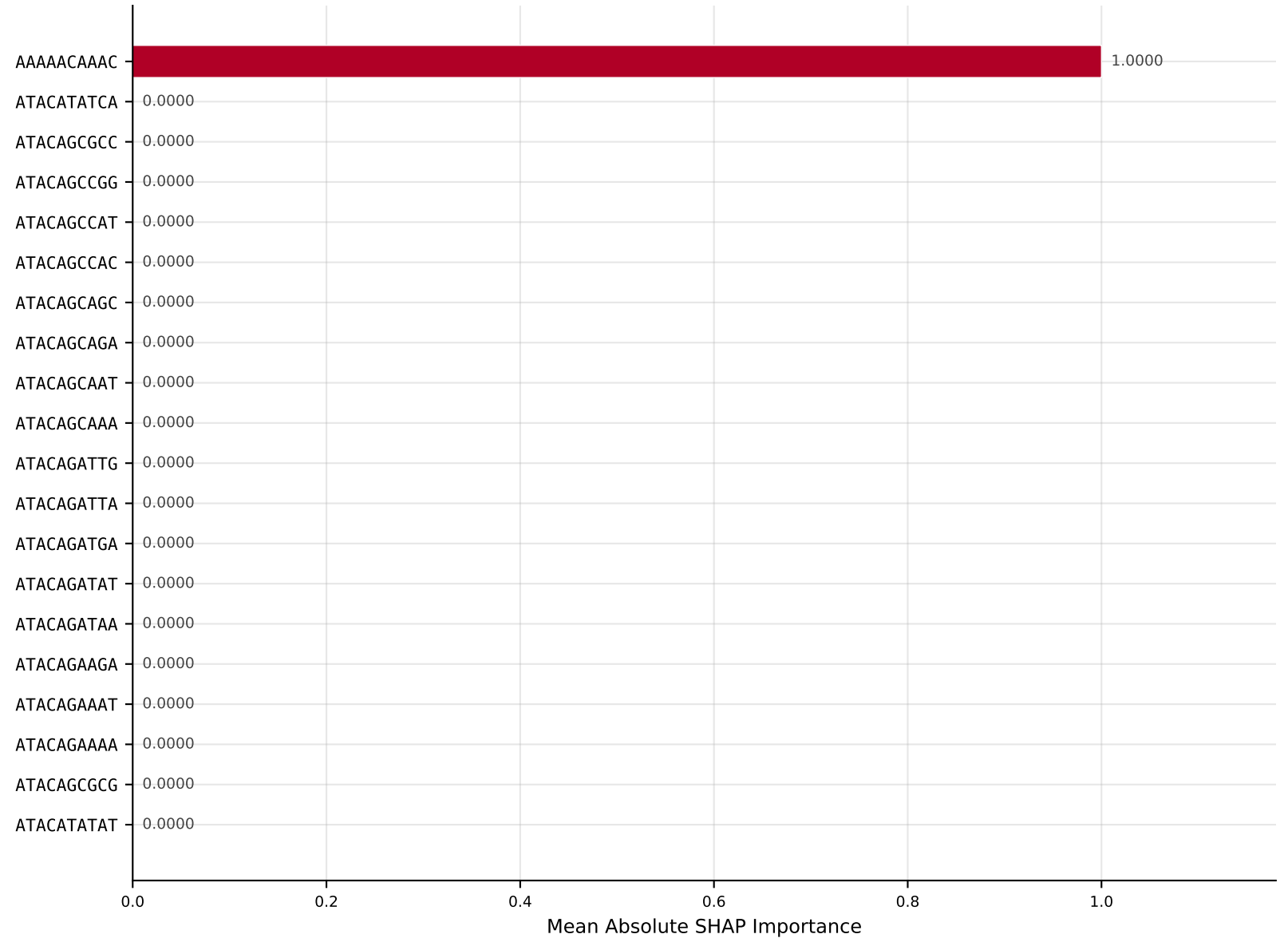
SHAP Feature Importance — Pyrazinamide

Top 20 k-mer predictors



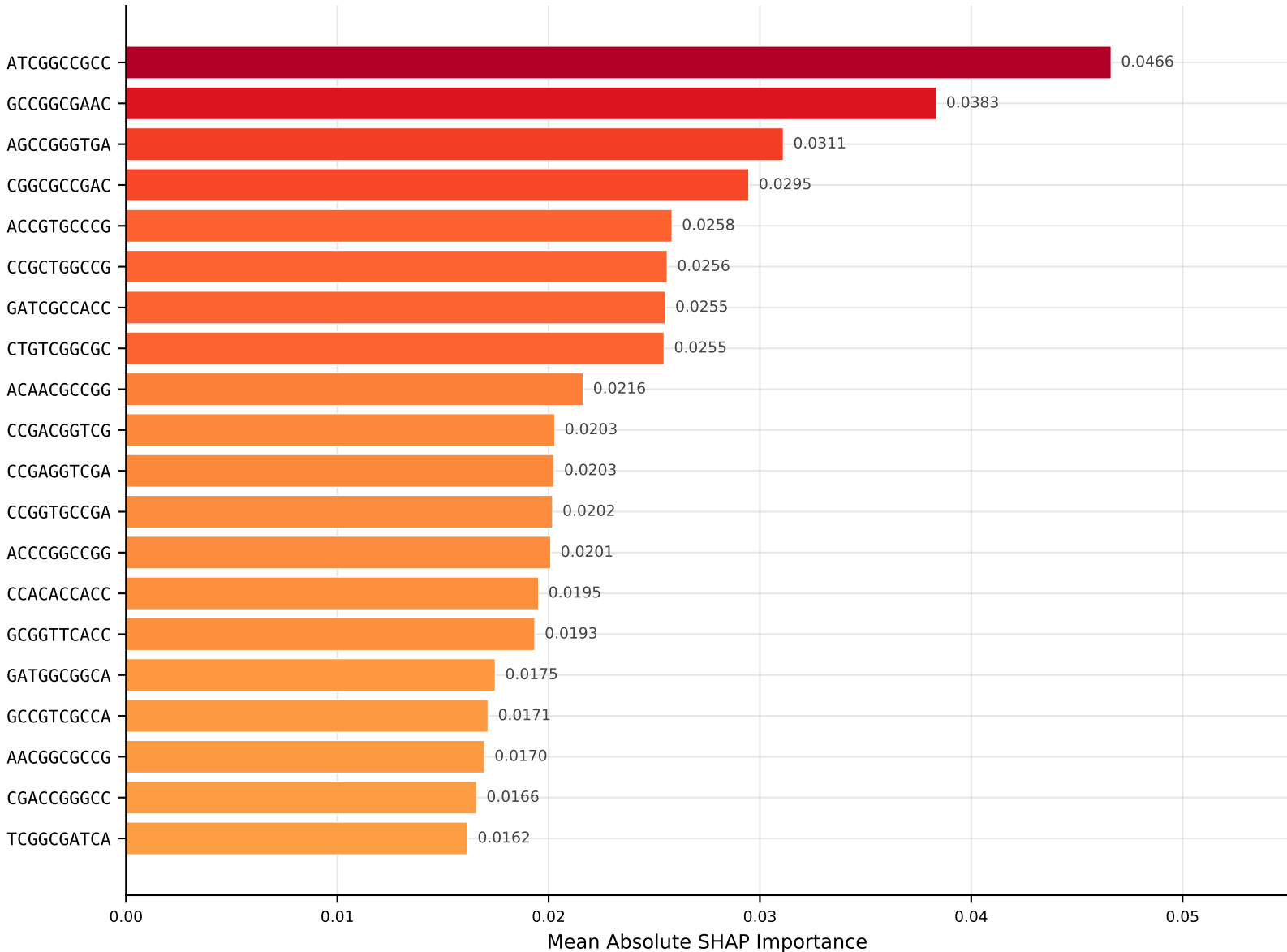
SHAP Feature Importance — Quinupristin/Dalfopristin

Top 20 k-mer predictors

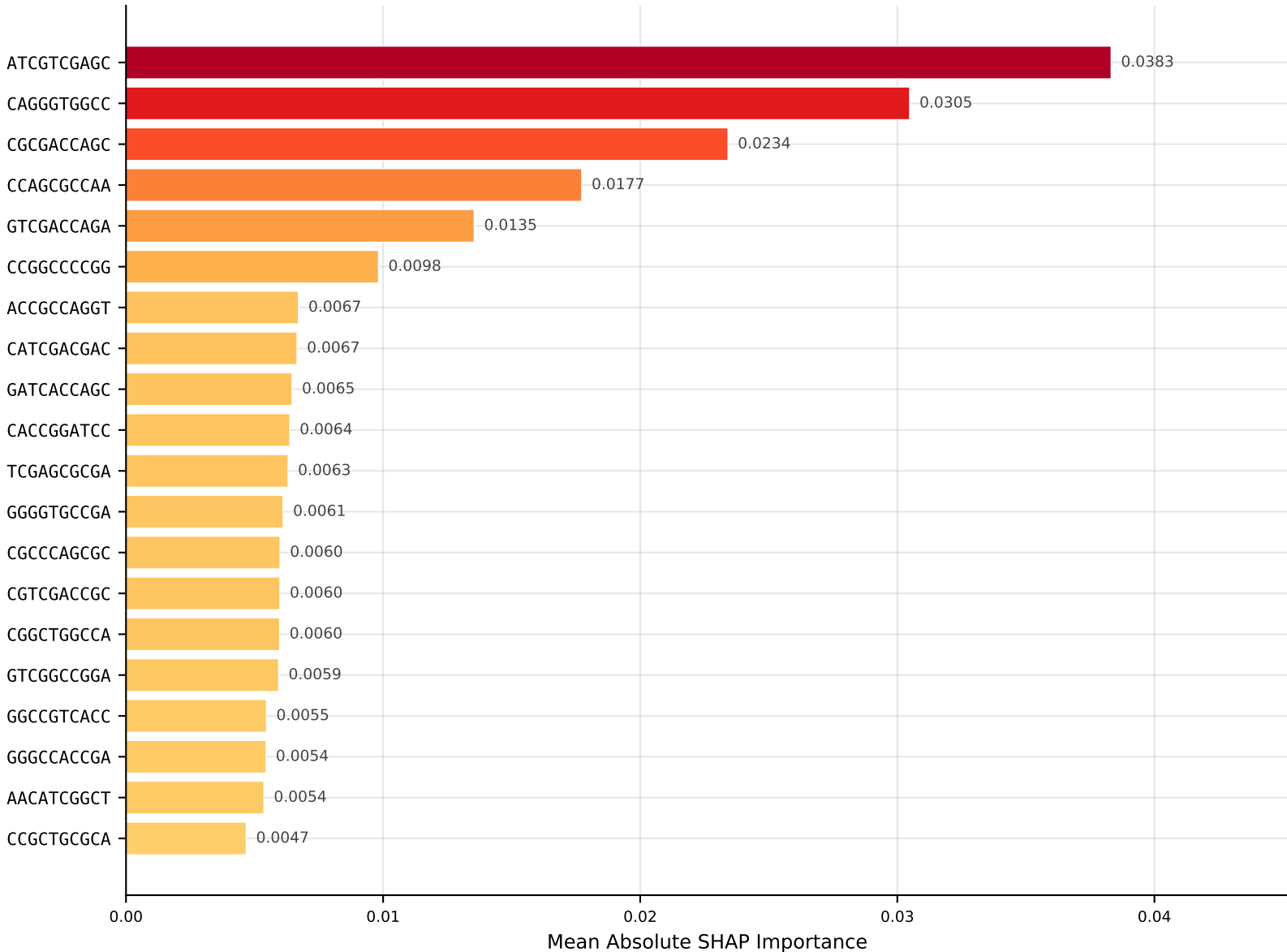


SHAP Feature Importance — Rifabutin

Top 20 k-mer predictors

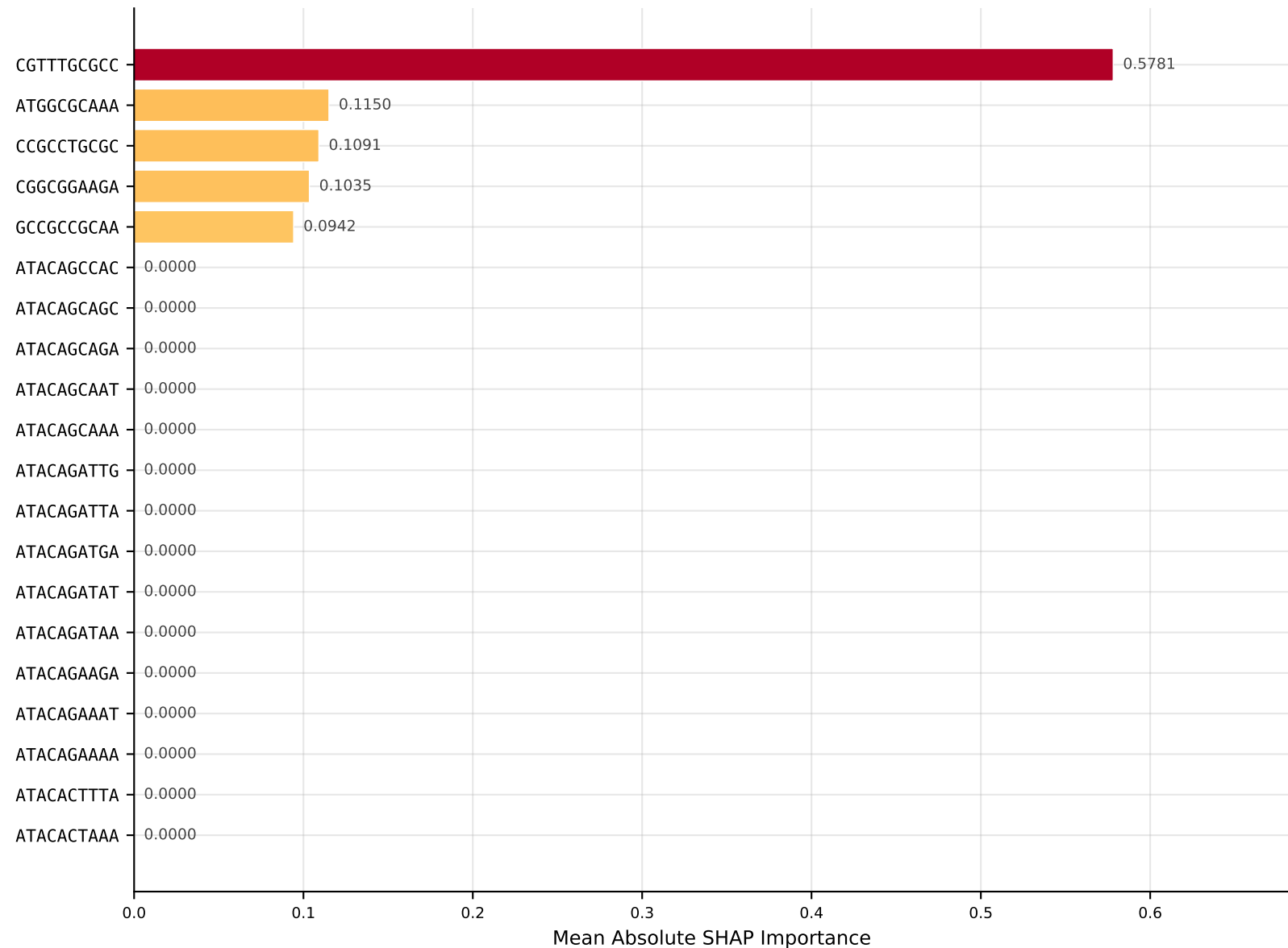


SHAP Feature Importance — Rifampin Top 20 k-mer predictors

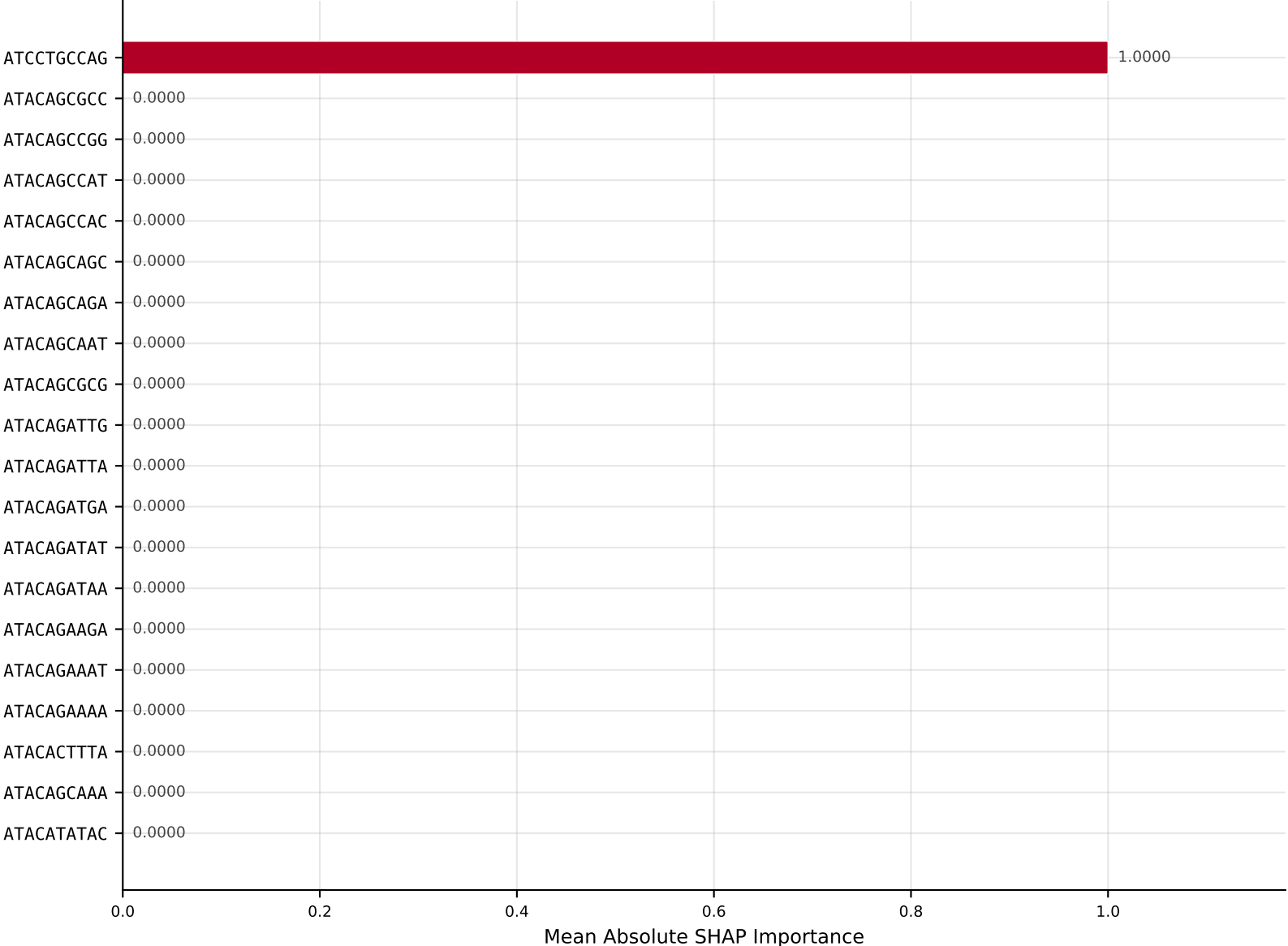


SHAP Feature Importance — Spectinomycin

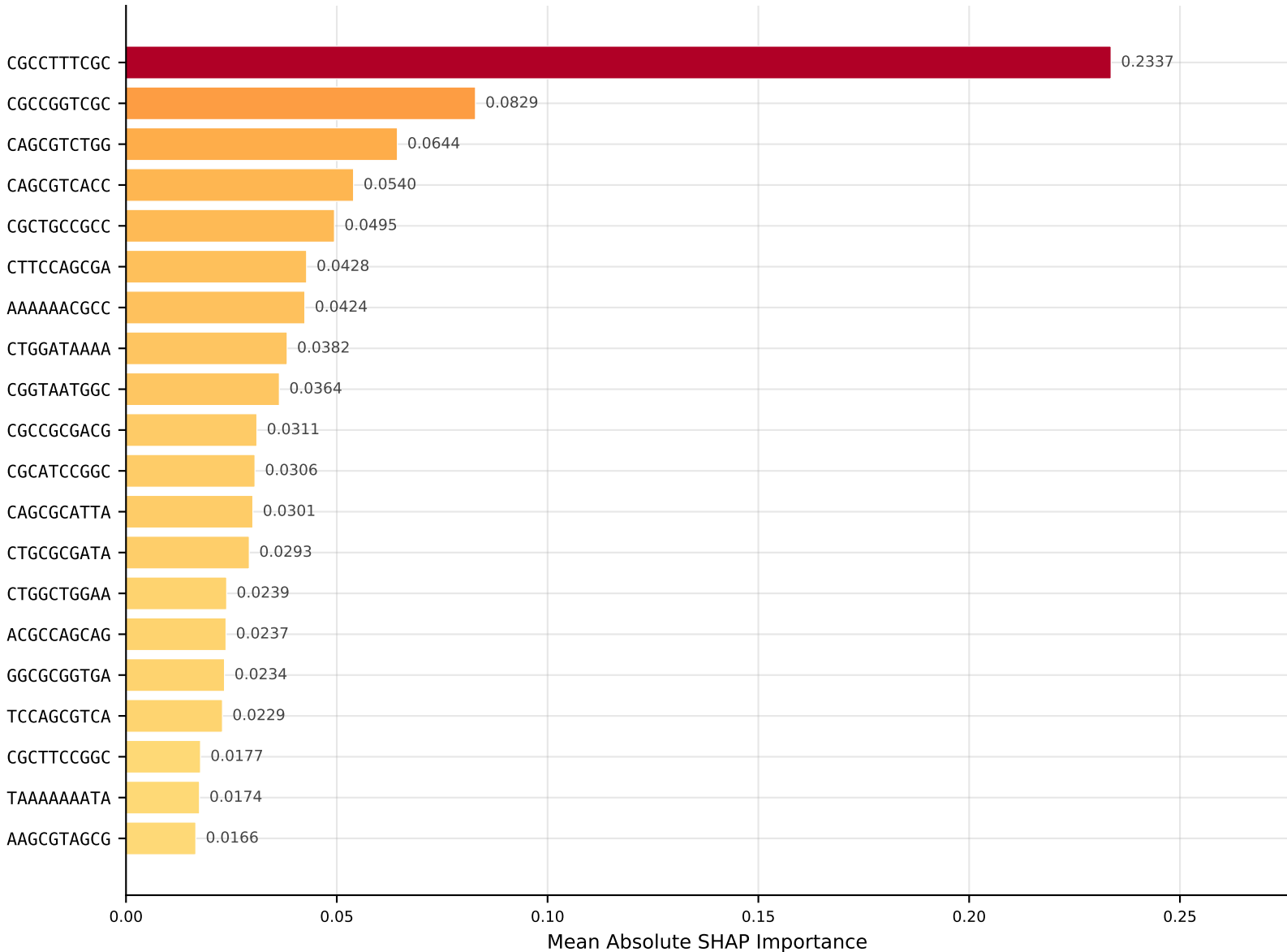
Top 20 k-mer predictors



SHAP Feature Importance — Sulfamethoxazole
Top 20 k-mer predictors

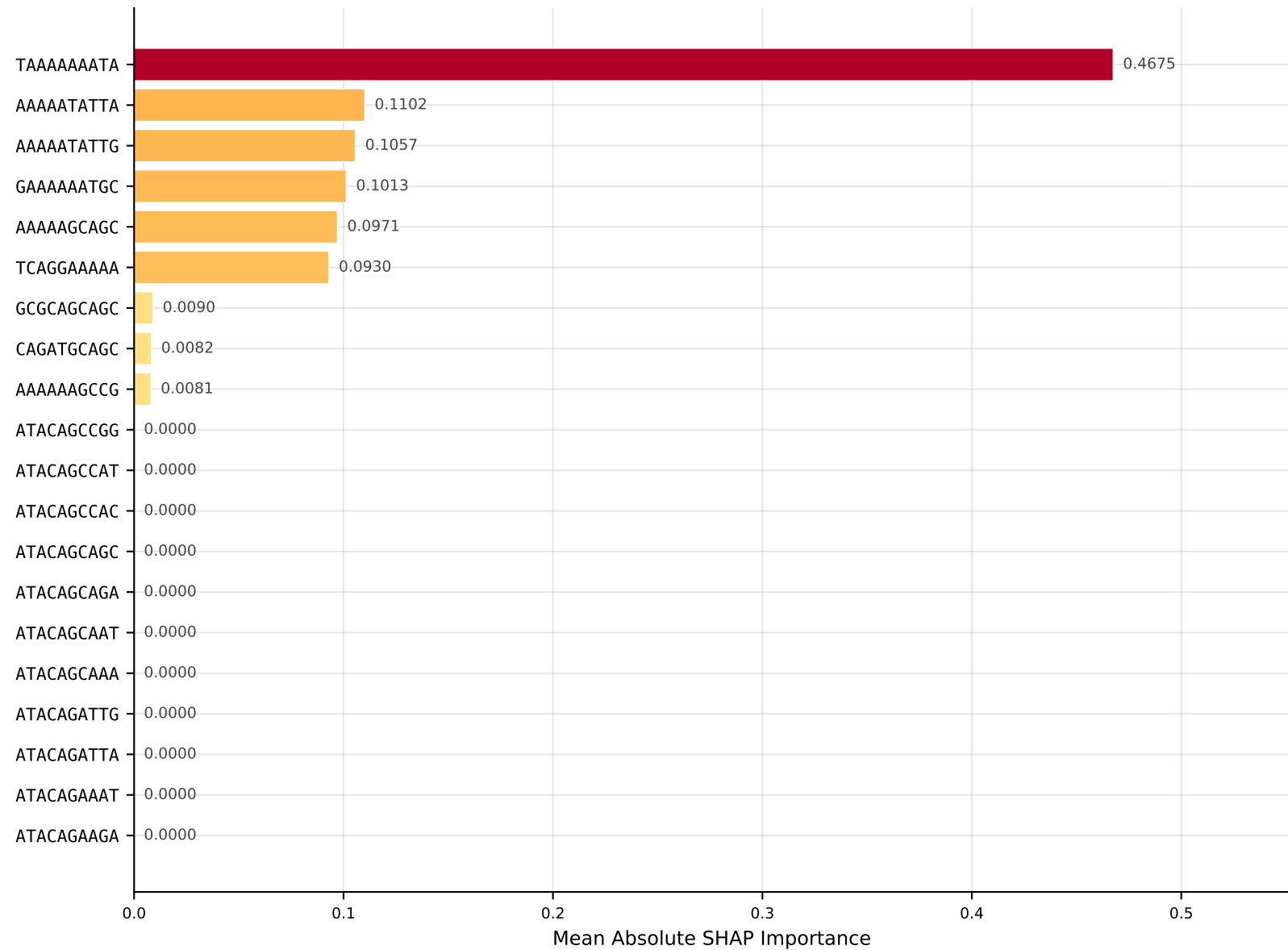


SHAP Feature Importance — Sulfisoxazole Top 20 k-mer predictors



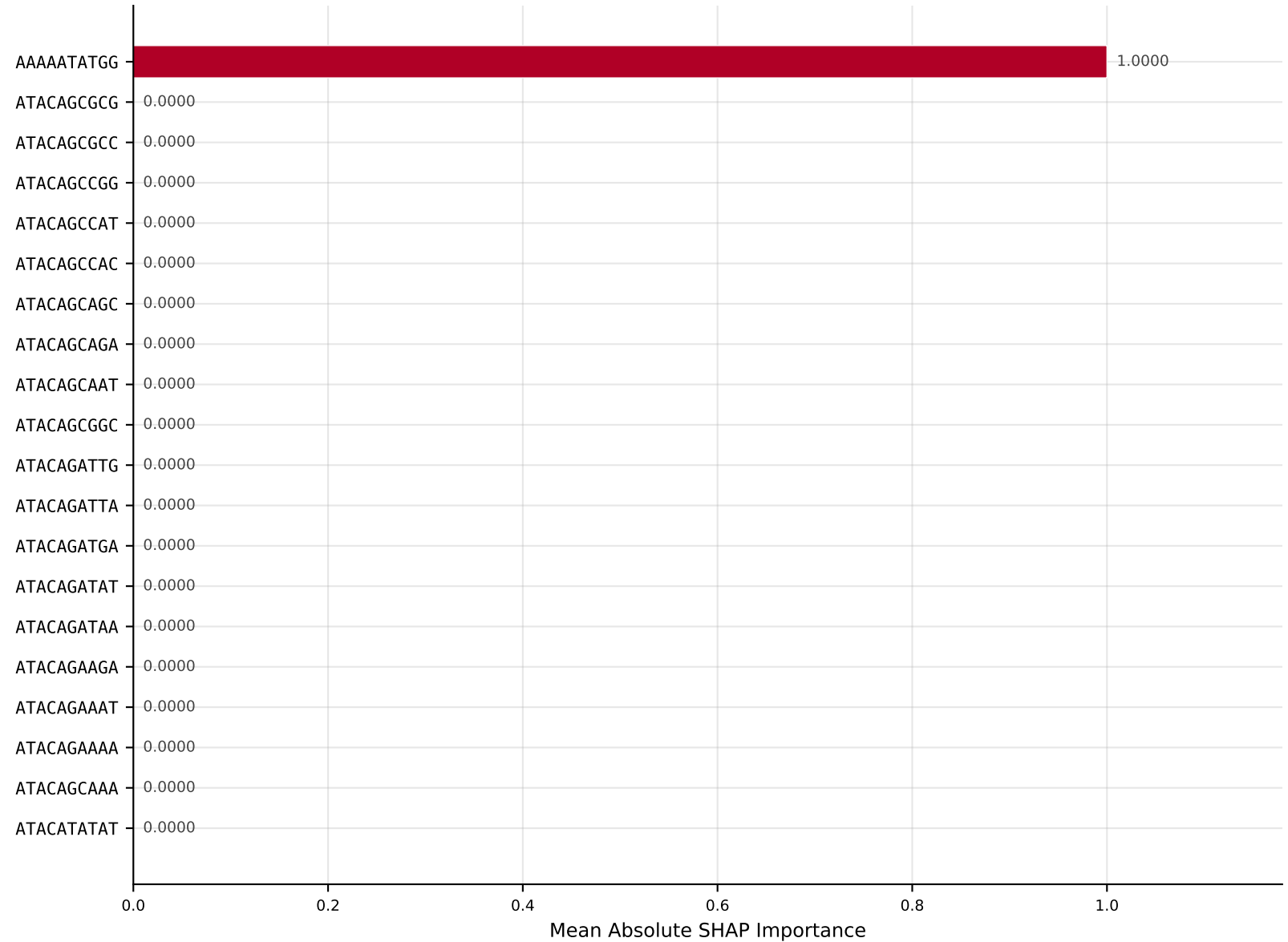
SHAP Feature Importance — Sulfonamides

Top 20 k-mer predictors



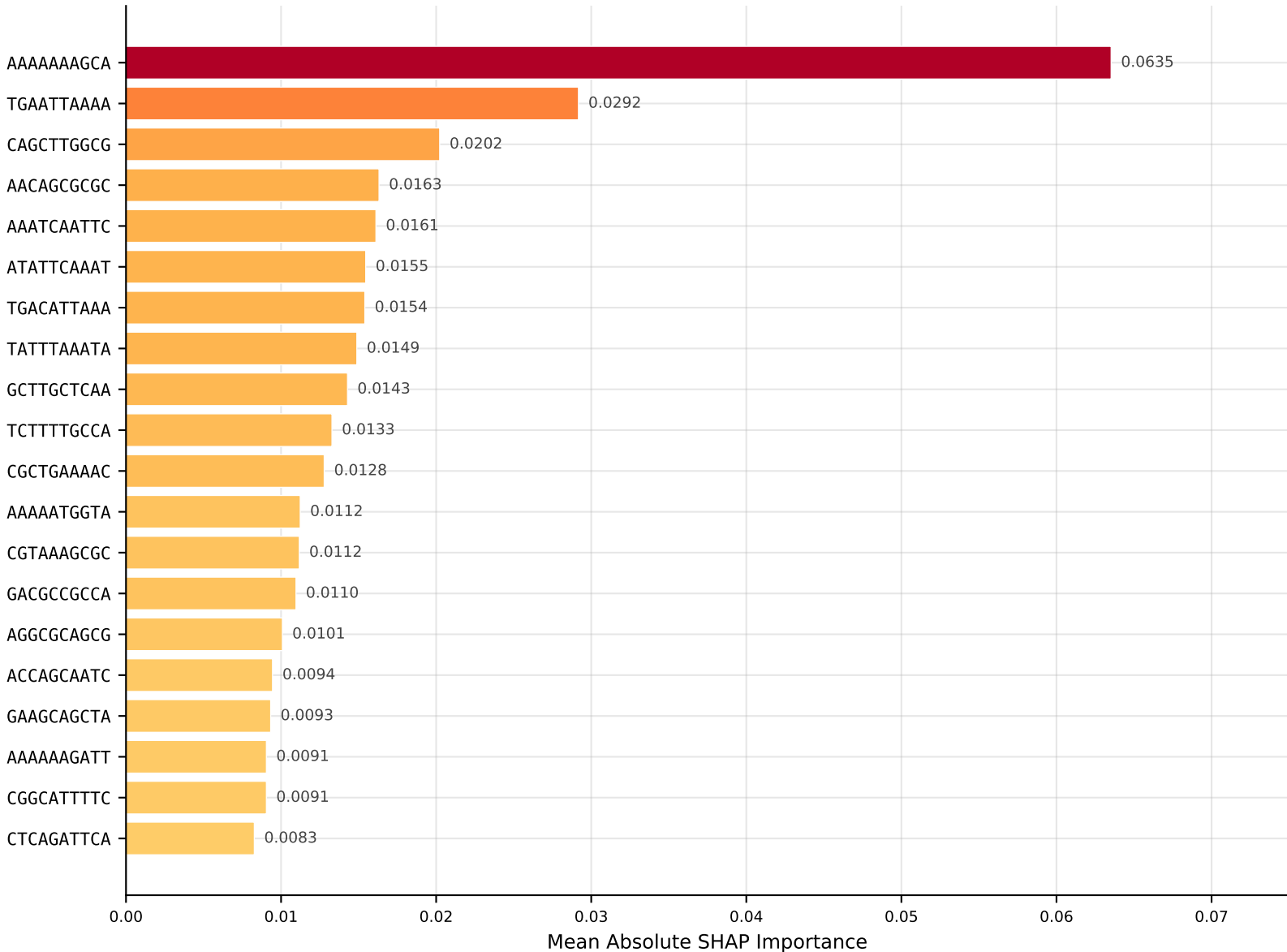
SHAP Feature Importance — Teicoplanin

Top 20 k-mer predictors

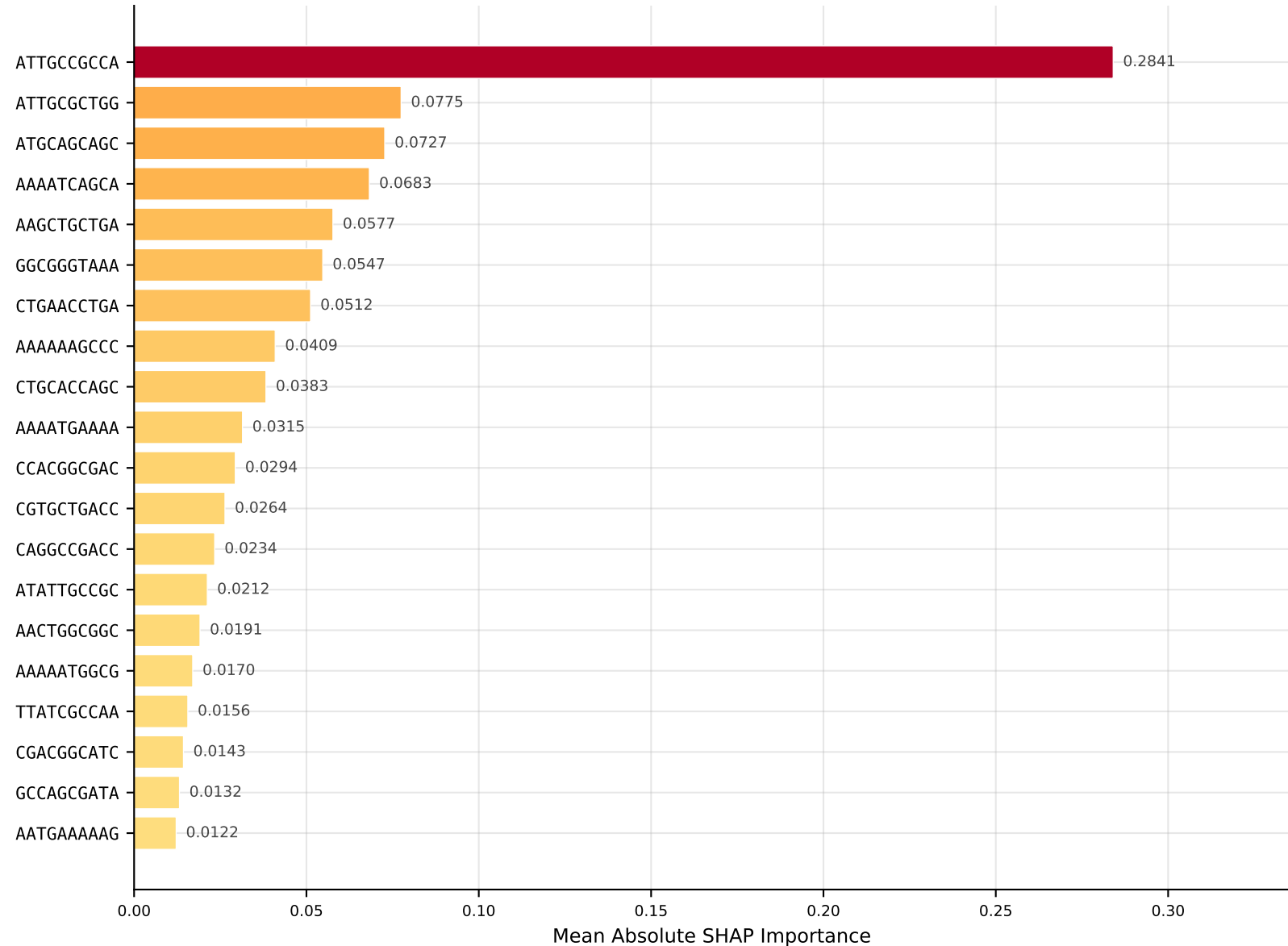


SHAP Feature Importance — Tetracycline

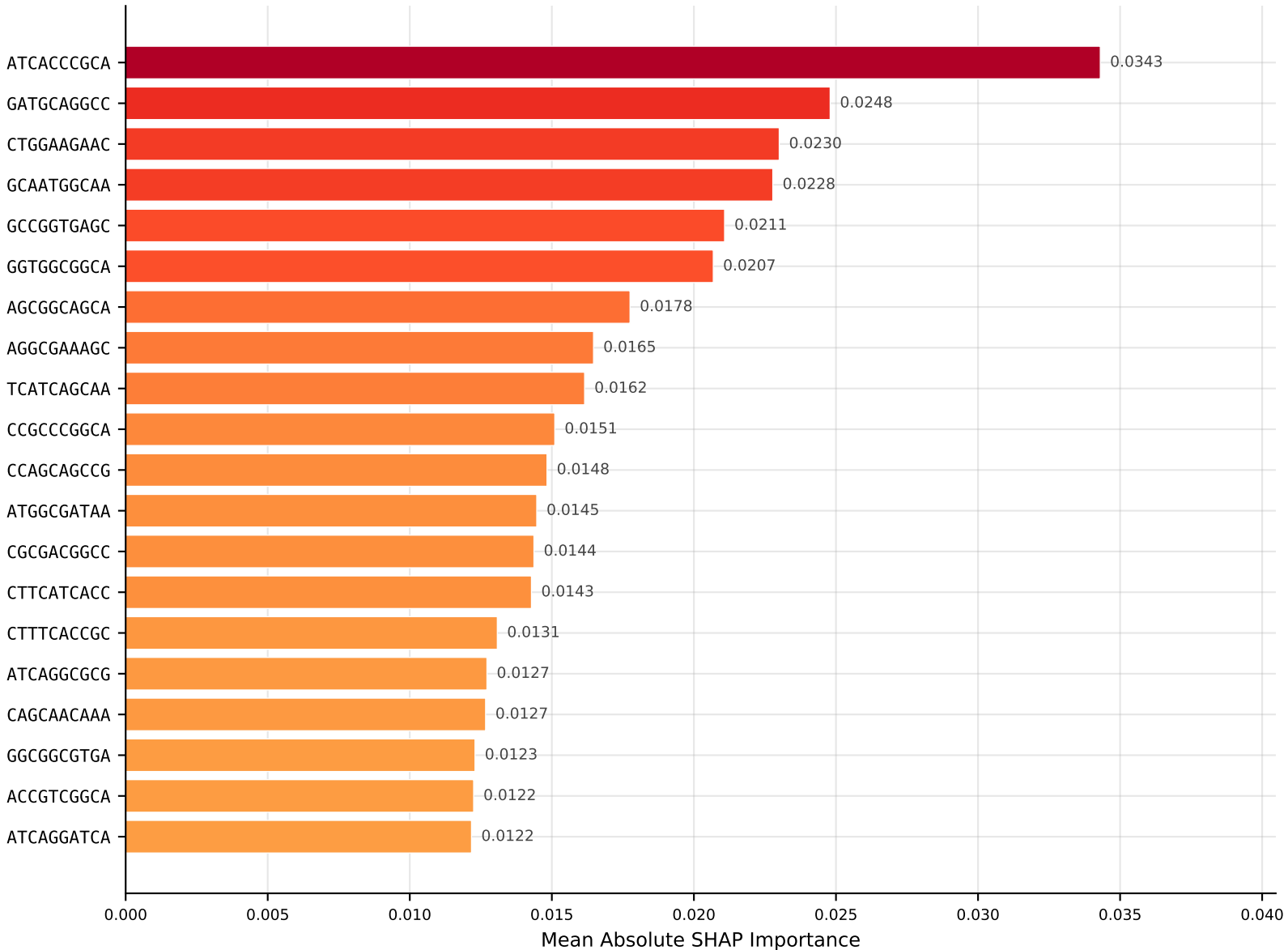
Top 20 k-mer predictors



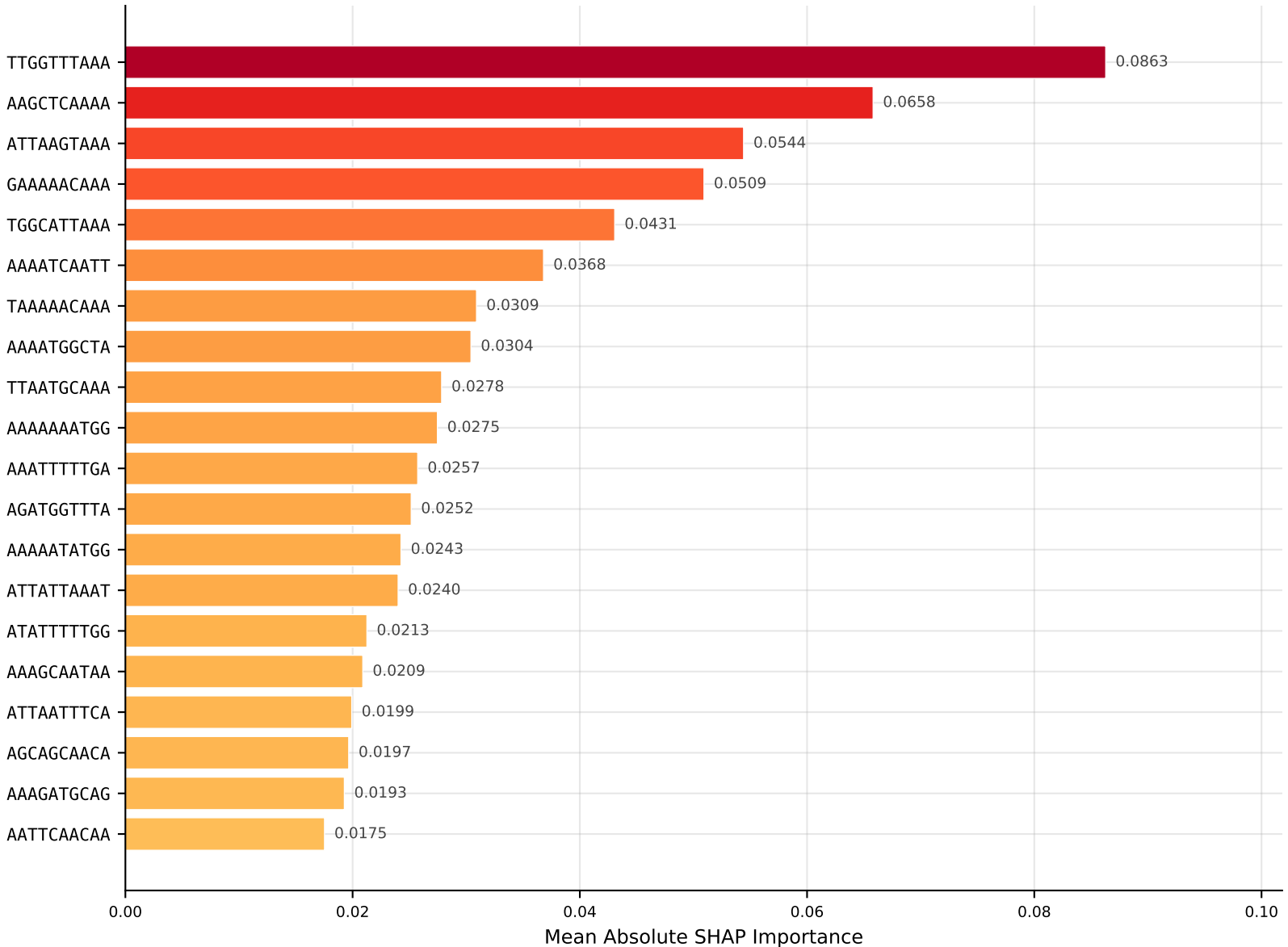
SHAP Feature Importance — Ticarcillin/Clavulanic Acid Top 20 k-mer predictors



SHAP Feature Importance — Tigecycline Top 20 k-mer predictors

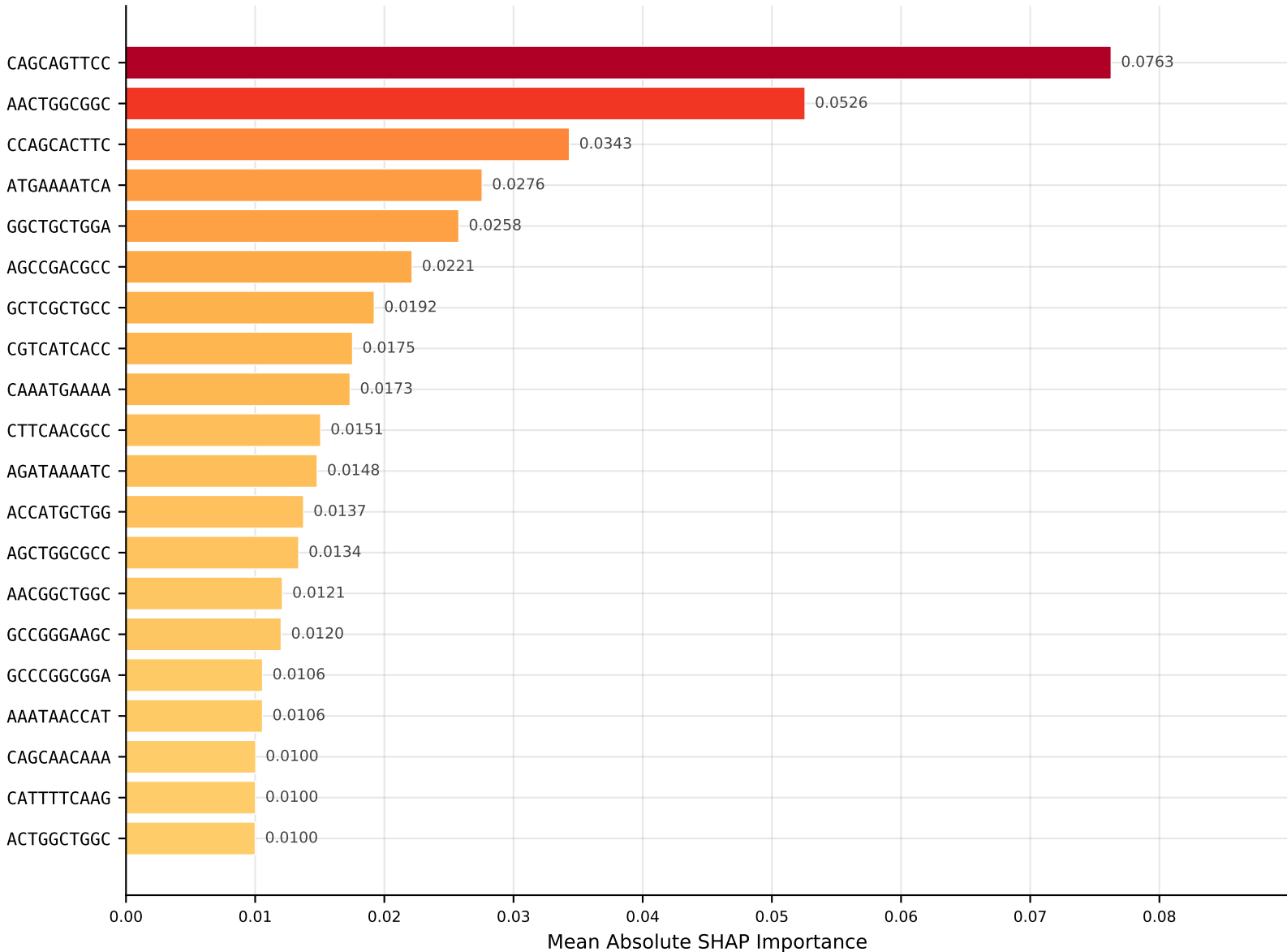


SHAP Feature Importance — Timentin Top 20 k-mer predictors

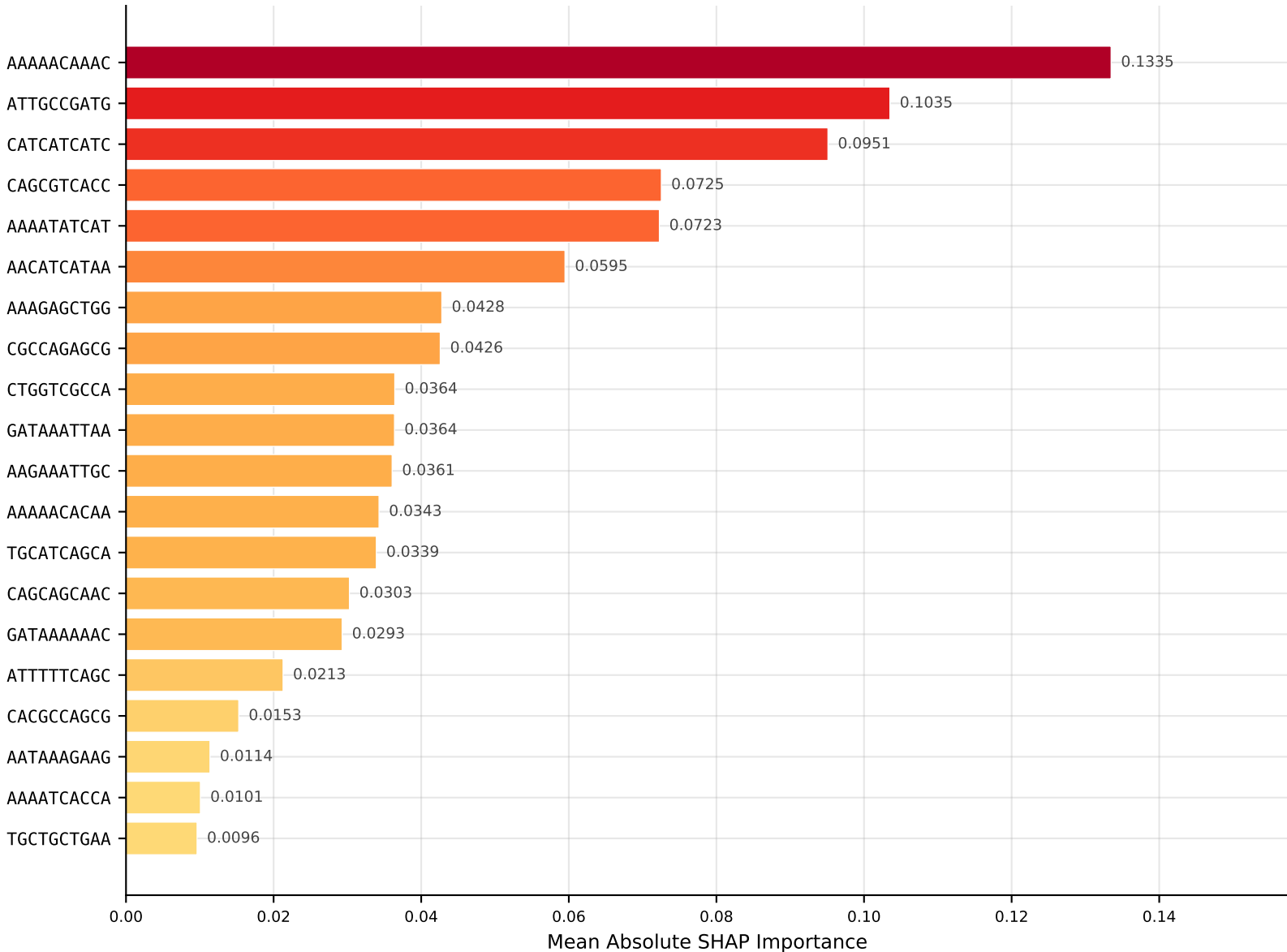


SHAP Feature Importance — Tobramycin

Top 20 k-mer predictors

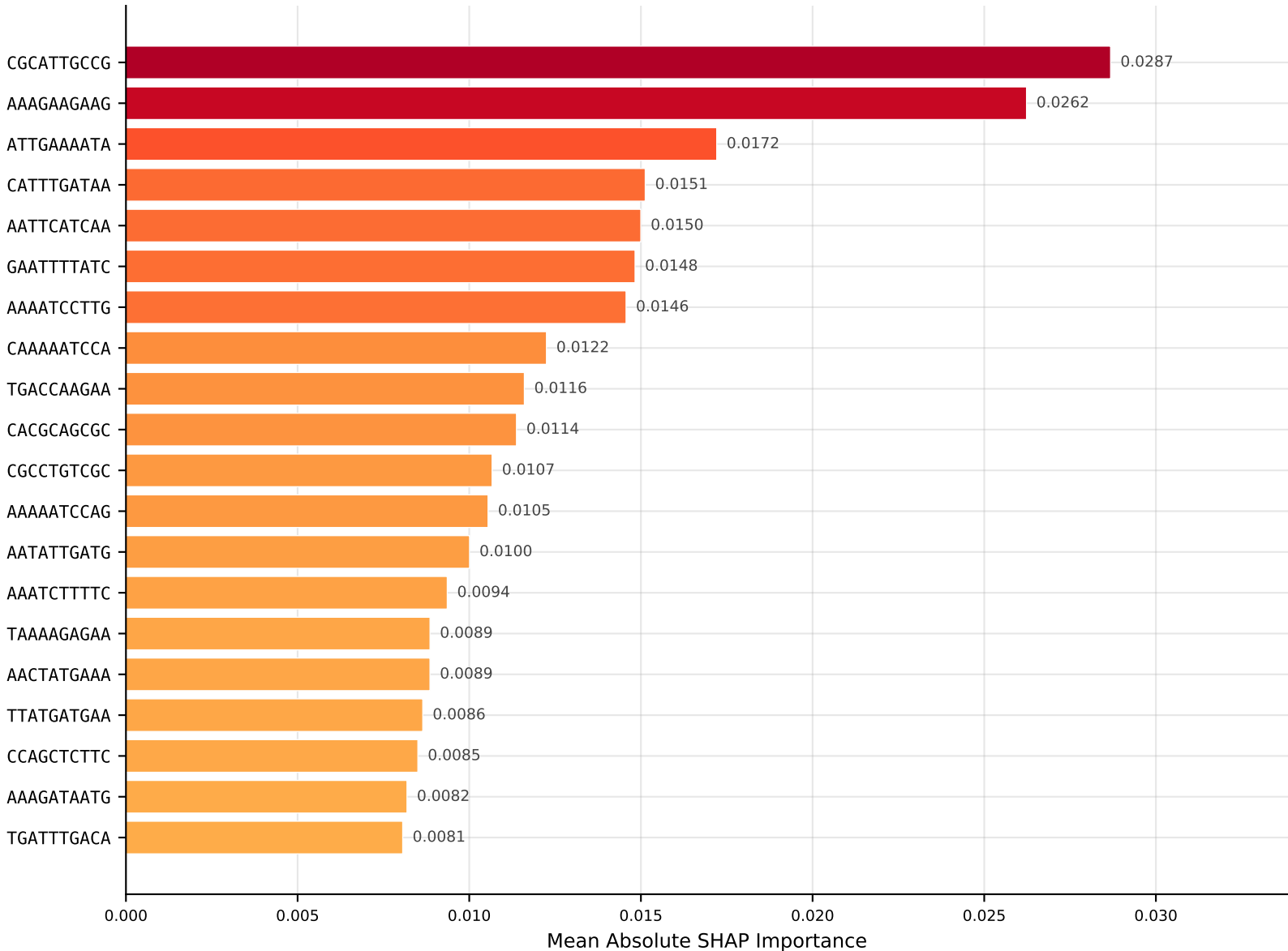


SHAP Feature Importance — Trimethoprim Top 20 k-mer predictors



SHAP Feature Importance — Trimethoprim/Sulfamethoxazole

Top 20 k-mer predictors



SHAP Feature Importance — Vancomycin

Top 20 k-mer predictors

